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1 Features & Requirements

1.1 Core Features (Epics)

Epic ID	Feature	Description	Priority
E1	Skin Data Input	Input of all CS2 skin parameters	Must
E2	Price Prediction	ML-based skin price prediction	Must
E3	Price Visualization	Historical price charts	Should
E4	Factor Analysis	SHAP-based influence analysis	Should
E5	Automated Updates	Market data synchronization	Should
E6	API Documentation	Swagger documentation	Could

1.2 Functional Requirements – User Stories

1.2.1 User Story 1 – Skin Data Input

As a CS2 player
I want to enter detailed skin parameters
So that I can receive an accurate price estimate.

Acceptance Criteria: - All parameters are validated - Data is stored in PostgreSQL - StatTrak and non-StatTrak items are supported

1.2.2 User Story 2 – Price Prediction

As a CS2 trader

I want the system to predict skin prices

So that I can make informed trading decisions.

Acceptance Criteria: - Prediction response time ≤ 2 seconds - Prediction is displayed in the UI - Prediction history is persisted

1.2.3 User Story 3 – Price Visualization

As a CS2 player or trader

I want to see price trends and influencing factors

So that I can analyze the market behavior.

Acceptance Criteria: - Interactive charts are available - Key influencing factors are visible

1.2.4 User Story 4 – Automated Data Updates

As an system administrator

I want the system to update market data automatically

So that ML predictions remain accurate.

Acceptance Criteria: - Data updates occur at least every 12 hours - Errors are logged and monitored

1.2.5 User Story 5 – API Documentation

As a developer

I want comprehensive Swagger documentation

So that system integration is simplified.

Acceptance Criteria: - All API endpoints are documented - Documentation is accessible via web interface

1.3 Non-Functional Requirements

- Performance: response time ≤ 2 seconds
- Security: HTTPS, JWT, secure data storage
- Scalability: up to 500 concurrent users
- Availability: $\geq 99\%$
- Usability: responsive UI
- Deployment: Docker-based containerization

2 Project Overview

This section provides a comprehensive business-oriented overview of the diploma project dedicated to the analysis, monitoring, visualization, and prediction of Counter-Strike 2 (CS2) skin prices. The document defines the business context, motivation, system boundaries, stakeholders, and key functional and non-functional requirements.

The project is positioned at the intersection of digital economies, data analytics, and machine learning. It addresses the growing demand for transparent and explainable tools in virtual item trading environments, where financial decisions are increasingly driven by data-driven insights.

The proposed solution is a modular web-based analytical platform that integrates a frontend user interface, a backend API, a machine learning service, a relational database, and external trading platform APIs.

2.1 Purpose of the Project

The primary purpose of this diploma project is to design and implement an analytical system that enables users to evaluate CS2 skin prices more accurately by considering both historical market trends and unique item-specific characteristics. Unlike traditional pricing tools, the system emphasizes explainability, automation, and adaptability to market dynamics.

The project also serves an academic purpose by demonstrating the practical application of:

- business analysis methodologies, - machine learning techniques, - web system architecture design, - and data-driven decision support systems.

2.2 Contents

- [Problem Statement & Goals](#)
- [Stakeholders & Users](#)
- [Scope](#)
- [Features](#)

2.3 Scope of This Document

This Project Overview focuses on the **Business Analysis (BA) perspective** and does not describe low-level implementation details. Technical architecture and implementation aspects are addressed only at a conceptual level to support business requirements.

Detailed system design, data models, and implementation strategies are covered in subsequent diploma chapters.

2.4 Contents

- Problem context and motivation
- Business goals and success criteria
- Stakeholder identification and analysis
- System scope and constraints
- High-level feature definition and requirements

2.5 Executive Summary

The Counter-Strike 2 skin market represents a mature virtual economy with real monetary value, high trading volumes, and strong price volatility. Prices are influenced by numerous factors, including rarity, wear level (float), pattern index, applied stickers, StatTrak status, and overall market sentiment.

Existing tools typically provide aggregated or simplified price estimates, failing to capture the multidimensional nature of item valuation. Moreover, most platforms do not explain why a specific price is suggested, which reduces user trust and increases financial risk.

This diploma project proposes an interactive web-based system that allows users to input detailed skin parameters, receive machine learning-based price predictions, and explore historical price trends and factor influence through visual analytics. Market data is automatically synchronized with external trading platforms such as Steam Market, Buff, and Skinport, ensuring data relevance and reliability.

2.6 Key Highlights

Aspect	Description
Problem Domain	CS2 virtual item market analytics
Core Problem	Inaccurate and non-transparent skin valuation
Proposed Solution	ML-based analytical web platform
Target Users	CS2 players, traders, trading platforms
Key Capabilities	Prediction, visualization, automation
Architecture	Frontend – Backend API – ML Service – Database
Academic Focus	Business analysis, ML, data visualization

3 Problem Statement & Goals

3.1 Background and Market Context

The trading of virtual items in modern online games has evolved into a complex digital economy with real-world financial implications. Counter-Strike 2 (CS2) represents one of the most prominent examples of such an economy, where in-game cosmetic items, known as skins, are actively traded on both official and third-party marketplaces.

Skin prices are influenced by a combination of: - intrinsic item properties (rarity, float

value, pattern), - cosmetic enhancements (stickers, StatTrak), - market supply and demand, - temporal trends and external events.

Due to this complexity, accurate price estimation requires advanced analytical approaches that go beyond basic averaging or historical comparison.

3.2 Problem Description

Despite the high economic value of CS2 skins, existing tools suffer from several limitations: - incomplete consideration of item-specific parameters, - lack of explainability in price estimation, - delayed or manual data updates, - insufficient visualization of price dynamics.

These limitations lead to suboptimal decision-making and financial losses for users.

3.3 Problem Statement

Who:
CS2 players, traders, and digital trading platforms.

What:
Users lack access to accurate, transparent, and explainable tools for determining the true market value of CS2 skins.

Why:
Inaccurate valuation increases financial risk, reduces market efficiency, and undermines user trust in existing analytical solutions.

3.4 Business Goals

Goal	Description	KPI
Pricing Accuracy	Improve correctness of price estimates	≥ 90% accuracy
Risk Reduction	Support informed trading decisions	≥ 80% user satisfaction
Transparency	Explain pricing factors	SHAP visualizations
Automation	Reduce manual data handling	≤ 24h data refresh
Reliability	Ensure stable system behavior	≥ 99% uptime

3.5 Project Objectives

- Design a unified and user-friendly data input mechanism.
- Integrate ensemble machine learning models (XGBoost, Random Forest, LSTM).
- Provide interactive data visualizations for price trends.
- Implement automated market data synchronization.
- Ensure compliance with GDPR and security best practices.

3.6 Success Criteria

- High prediction accuracy
- Low system response time
- High test coverage
- Positive user feedback
- Stable and automated data updates

3.7 Non-Goals

The project explicitly does not aim to: - facilitate direct trading transactions, - provide financial or investment advice, - replace existing marketplaces.

4 Project Scope

4.1 In Scope

The system includes functionality for: - detailed skin data input, - price prediction using ML models, - visualization of historical trends, - factor influence analysis, - automated data synchronization, - API documentation and containerized deployment.

4.2 Out of Scope

Excluded functionality: - trading execution, - payments, - social interaction.

4.3 Assumptions

- Availability of external APIs
- Sufficient data quality
- User familiarity with CS2 skins

4.4 Constraints

Category	Description
Time	Diploma schedule
Budget	Academic scope
Technology	ASP.NET Core, PostgreSQL
Security	GDPR compliance
Load	≤ 500 concurrent users

5 Stakeholders & Users

5.1 Stakeholder Identification

Stakeholders were identified based on their interaction with the system, influence on project success, and interest in outcomes.

5.2 Target Audience

User Group	Description	Primary Needs
CS2 Players	Casual and competitive users	Accurate valuation
CS2 Traders	Active market participants	Predictive analytics
Trading Platforms	Market operators	Market insights
Development Team	System implementers	Clear requirements
Academic Supervisor	Project evaluator	Methodological rigor

5.3 Stakeholder Analysis Matrix

Stakeholder	Role	Interest	Influence	Engagement
CS2 Players	End users	High	Medium	High
CS2 Traders	Power users	Very High	Medium	High
Trading Platforms	Data providers	High	High	Medium
Development Team	Implementers	High	High	High
Academic Supervisor	Reviewer	Medium	High	Medium
External APIs	Data sources	Medium	Medium	Medium

5.4 Stakeholder Expectations

- Accuracy and reliability
- Transparency and explainability
- System stability
- Academic correctness

6 AI Explanation Service – Technical Documentation

AI Skin Explanation Service

1. Introduction

This document provides a complete technical and architectural overview of the AI Skin Explanation Service, a component of the CS2 price-prediction platform. The service integrates with an internal ML system, retrieves metadata from the database, constructs structured prompts, and generates clear natural-language explanations for skin prices using OpenAI models.

6.1 This document explains

architectural decisions

prompt engineering strategy

fallback logic for multi-model LLM routing

sanitization and safety rules

API behavior

data-processing pipeline

The goal is to provide an engineering-level understanding of the service and justify its design.

2. System Purpose and Responsibilities

The service converts raw prediction data (float, wear, pattern, rarity, stickers) into a human-readable explanation, based entirely on provided numerical values.

6.2 Key responsibilities

Integrate CS2 prediction model output with LLMs

Retrieve metadata (weapon, skin, tier, pattern info)

Construct structured prompts for different item categories

Enforce strict rules preventing hallucinations (e.g., no sticker names invention)

Support multi-model fallback logic

Format results into a clean explanation for frontend consumption

Users never interact with prompts directly — they simply receive an explanation of a predicted price.

3. High-Level Architecture

6.3 The service is part of an ASP.NET Core backend and interacts with

Internal ML Service

Provides predicted numeric price

Does not produce explanations

AppDbContext

Skins

Wear tiers

Sticker metadata

Pattern-related tables (CH, Fade, Doppler, etc.)

AI Components

AiPromptFactory — builds prompts for each category

AiStickerService — resolves sticker prices & names

AiExplanationService — high-level orchestration

LlmFailoverService — manages model selection & failover

OpenAiClient — low-level HTTP client for OpenAI Responses API

Controllers

/api/ai/explain (primary → gpt-4o-mini)

/api/ai/explain-v2 (primary → gpt-4.1-mini)

The service functions purely as a computation and orchestration layer, never storing user data.

4. Data Flow

4.1 Input Sources

6.4 Requests to the service contain

PredictedPrice — ML model output

SkinId, WearTierId, Pattern, FloatValue

Stickers[] — list of sticker IDs (ignored for knives)

IsStattrak

LlmPriority — determines fallback direction

All metadata is validated through the database.

4.2 Processing Steps

Validate request

Skin and wear must exist

Wear must be valid for the skin

Pattern must exist for this skin type

Determine item category

ch_knife, ch_gun, fade_gun, fade_knife, doppler_knife, float_gun

Sticker resolution

Guns → full sticker processing

Knives → stickers are forcibly ignored

Construct prompt

AiPromptFactory builds scenario-specific prompt

Includes strict behavioral instructions

Pass prompt to LLM with fallback

OpenAiClient generates plain-text explanation

LlmFailoverService handles failures

Return formatted explanation Output is always a single clean text block without markdown, emojis, or hallucinated names.

5. Prompt Design and Structure

5.1 System Prompt Responsibilities

6.5 The system prompt is embedded in OpenAiClient and enforces

English-only output

No markdown, no lists, no bullets

No hallucinations of sticker names

No invented historical data

Explanation must be based only on numeric values provided

Only plain sentences

Forget stickers entirely if item is a knife

It defines global AI behavior and ensures consistent output regardless of user input.

5.2 Item-Specific Prompts

6.6 AiPromptFactory generates 6 scenarios

Case Hardened Knife

Uses color distribution (blue/gold/purple), float, wear.

Case Hardened Gun

Adds sticker block and blue score tiering.

Doppler Knife

Uses phase name and float.

Fade Gun

Includes fade percentage and ranking.

Fade Knife

Fade % and fade rank; no stickers.

Float-Sensitive Guns

Only float + wear + stickers.

Each prompt ends with a footer describing how to interpret the factors.

6. Sanitization Layer

6.7 The service prevents unsafe or malformed content from reaching the model

Stickers on knives are replaced with 0 values

Pattern mismatch triggers validation errors

Wear tiers are validated against allowed values

No user-controlled strings are injected directly into prompts (all metadata comes from controlled DB tables)

Therefore, input sanitization is predefined by controlled sources.

7. Model Selection Strategy

7.1 Available Models

Primary: gpt-4o-mini (fast, cheap)

Secondary: gpt-4.1-mini (more consistent output)

7.2 Priority Types

6.8 LlmPriority enum contains

MiniThenGpt41 — used by standard /explain endpoint

Gpt41ThenMini — used by /explain-v2 endpoint

6.9 LlmFailoverService implements

Try primary model

On exception → try fallback model

If both fail → propagate error to controller

This guarantees that the AI explanation always works even if one model is temporarily unavailable.

8. Error Handling and Logging

6.10 The service logs

OpenAI request failures

Fallback model activation

Pattern/wear validation failures

Unexpected empty outputs from LLM

Sensitive content such as prompts or user-provided values is never logged.

9. Security Considerations

6.11 To ensure safety

API keys stored only in environment variables

Prompts prohibit hallucination

Prompts prohibit mentioning stickers for knives

Only controlled DB data is injected into prompts

No user prompt influence is allowed

No execution of user instructions

Validation layer prevents invalid patterns/wear tiers

The service behaves predictably and resists injection attempts by design.

10. Justification of Design Choices

Strict prompt engineering

Prevents hallucinations & inconsistencies.

Multi-model fallback

Ensures reliability under outages.

Modular architecture

Easily extendable for new skin types.

Category-specific prompts

6.12 Important because

CH logic differs greatly from Fade logic

Doppler phases require structured handling

Knives vs guns differ in sticker logic

Pure orchestration

The service does not store user data or predictions — only processes and explains.

11. Conclusion

The CS2 AI Explanation Microservice is a robust, modular, and reliable system that converts numeric ML predictions into natural-language explanations using OpenAI models.

6.13 It ensures

- strong safety through controlled metadata
- no hallucination through strict prompts
- fallback reliability across models
- extensibility for new item categories
- clean architecture with clearly separated responsibilities

This implementation adheres to modern standards of LLM integration in production-grade systems.

7 Admin API Documentation

7.1 Overview

The Admin API provides endpoints for managing internal reference data used by the CS2 skin pricing and prediction system. These endpoints allow administrators to create, update, and delete metadata such as weapon types, weapons, skins, patterns, stickers, and wear tiers.

All responses are returned in **JSON** format.

Base URL: /api/v1/admin
Authentication: Required (API Key)

7.2 Authentication

All Admin API endpoints require API key authorization.

Header format:

Authorization: API_KEY

Requests without a valid API key are rejected.

7.3 Architecture Context

The Admin API is an internal management interface. It is not intended for public clients and is protected via API key authorization. All operations directly modify reference data used by Meta, Prediction, and AI Explanation APIs.

7.4 Common Error Format

```
{
  "error": {
    "code": "ERROR_CODE",
    "message": "Human-readable error description"
  }
}
```

HTTP Code	Description
400	Validation error
401	Unauthorized
404	Resource not found
500	Internal server error

7.5 Common Validation Rules

The following validation rules apply to all Admin endpoints:

- Referenced entities (skin, weapon, pattern, sticker, wear tier) must exist
 - Numeric values must be within valid domain ranges
 - Duplicate entries are not allowed
 - Pattern identifiers must be unique per skin
 - Foreign key constraints are strictly enforced
-

7.6 1. Case Hardened Gun Patterns

7.6.1 POST /patterns/case-hardened/gun

Creates a new Case Hardened gun pattern entry.

Request Body

```
{
  "skinId": 0,
  "pattern": 0,
  "playsideBlue": 0,
  "backsideBlue": 0
}
```

Validation - skinId must reference an existing skin - pattern must not already exist for the skin - blue values must be between 0 and 100

7.6.2 PUT /patterns/case-hardened/gun/{id}

Updates an existing Case Hardened gun pattern.

Validation - pattern entry must exist

7.6.3 DELETE /patterns/case-hardened/gun/{id}

Deletes a Case Hardened gun pattern.

7.7 2. Case Hardened Knife Patterns

7.7.1 POST /patterns/case-hardened/knife

Creates a Case Hardened knife pattern entry.

Request Body

```
{
  "skinId": 0,
  "pattern": 0,
  "backsideBlue": 0,
  "backsidePurple": 0,
  "backsideGold": 0,
  "playsideBlue": 0,
  "playsidePurple": 0,
  "playsideGold": 0
}
```

Validation - skinId must exist - pattern must be unique per skin - color percentages must be between 0 and 100

7.8 3. Doppler Phases

7.8.1 POST /patterns/doppler/phases

Creates a Doppler phase.

Validation - phase name must be unique

7.8.2 DELETE /patterns/doppler/phases/{id}

Deletes a Doppler phase.

7.9 4. Doppler Skin Phase Links

7.9.1 POST /patterns/doppler/skin-phases

Links a skin to a Doppler phase.

Validation - skinId must exist - phaseId must exist - duplicate links are not allowed

7.10 5. Fade Gun Patterns

7.10.1 POST /patterns/fade/gun

Creates a fade gun pattern.

Validation - fadePercentage must be between 0 and 100 - fadeRank must be positive

7.11 6. Fade Knife Patterns

Same validation rules as fade gun patterns.

7.12 7. Skins

7.12.1 POST /skins

Creates a new skin.

Validation - weaponId must exist - skin name must be unique per weapon

7.13 8. Skin Wear Tier Mapping

Defines which wear tiers are supported by a skin.

Validation - skinId must exist - wearTierId must exist - mapping must be unique

7.14 9. Stickers

7.14.1 POST /stickers

Creates a sticker entry.

Request Body

```
{
  "name": "string",
  "referencePrice": 0
}
```

Validation - sticker name must be unique - referencePrice must be greater than 0

7.15 10. Weapons

7.15.1 POST /weapons

Creates a weapon.

Validation - weaponTypeId must exist - weapon name must be unique

7.16 11. Weapon Types

7.16.1 POST/weapon-types

Creates a weapon type.

Validation - code must be unique - name must be unique

7.17 12. Wear Tiers

7.17.1 POST/wear-tiers

Creates a wear tier.

Validation - wear tier name must be unique

7.18 Versioning Strategy

The Admin API uses URL-based versioning:

`/api/v1/admin`

Breaking changes require a new major version.

7.19 Known Limitations

- Admin endpoints are intended for trusted internal use only
- Bulk operations are not supported

8 AI Explanation API Documentation

8.1 Overview

The AI Explanation API provides natural-language explanations for predicted skin prices. It explains how different attributes of a skin configuration influence the final predicted price produced by the machine learning model.

All responses are returned in **JSON** format.

Base URL: `/api/v1/ai`

Authentication: Not required

8.2 Architecture Context

This API acts as an interpretability layer on top of the pricing prediction model. It does not calculate prices itself and only explains already predicted values.

8.3 Common Error Format

```
{
  "error": {
    "code": "ERROR_CODE",
    "message": "Human-readable error description"
  }
}
```

HTTP Code	Description
400	Validation error
404	Resource not found
500	Internal server error

8.4 Input Validation Rules

Field	Validation Rule
predictedPrice	Must be greater than 0
skinId	Must reference an existing skin
wearTierId	Must correspond to the provided float value
floatValue	Must be between 0.0 and 1.0
pattern	Must exist for the selected skin
stickers	Must reference existing sticker IDs

8.5 1. AI Explanation (v1)

8.5.1 POST/explain

Generates a concise AI-generated explanation for a predicted skin price.

8.5.2 Request Body

```
{
  "predictedPrice": 1699,
  "skinId": 1,
  "wearTierId": 1,
  "floatValue": 0.0003,
  "isStattrak": true,
  "pattern": 3,
  "stickers": [0]
}
```

8.5.3 Request Fields

Field	Type	Description
predictedPrice	number	Predicted price
skinId	integer	Skin identifier
wearTierId	integer	Wear tier identifier
floatValue	number	Skin float value
isStattrak	boolean	StatTrak flag
pattern	integer	Pattern index
stickers	integer[]	Sticker IDs

8.5.4 Response 200 OK

```
{
  "explanation": "The model predicted a price of 1699.00 USDT primarily du
}
```

8.5.5 Validation Errors

8.5.5.1 Predicted price equals 0

HTTP 400

```
{
  "error": {
    "code": "INVALID_PRICE",
    "message": "Predicted price must be greater than 0"
  }
}
```


8.5.5.2 Skin not found

HTTP 404

```
{
  "error": {
    "code": "SKIN_NOT_FOUND",
    "message": "Skin with the specified ID was not found"
  }
}
```

8.5.5.3 Pattern not found

HTTP 404

```
{
  "error": {
    "code": "PATTERN_NOT_FOUND",
    "message": "Pattern is not available for the selected skin"
  }
}
```

8.5.5.4 Wear tier does not match float value

HTTP 400

```
{
  "error": {
    "code": "WEAR_TIER_MISMATCH",
    "message": "Wear tier does not correspond to the provided float value"
  }
}
```

8.6 2. AI Explanation (v2)

8.6.1 POST/explain-v2

Provides a more detailed explanation with deeper analysis of pricing factors.

8.6.2 Request Body

Same structure and validation rules as **v1**.

8.6.3 Response 200 OK

```
{
  "explanation": "The predicted price is influenced by the skin's near-per"
}
```



9 API Documentation Rules and Strategy

9.1 1. Purpose of This Document

This document describes the API documentation strategy used in the diploma project. It defines the tools, standards, conventions, and rules applied when documenting all APIs in the system.

The goal of this document is to ensure that the API documentation is: - clear and consistent; - technically accurate; - easy to maintain and extend; - compliant with diploma project evaluation requirements.

9.2 2. Documentation Scope

The documentation covers the following API groups: - Meta API (reference metadata); - Prediction API (price prediction workflow); - AI Explanation API (model explainability); -

Admin API (internal reference data management); - ML Service API (internal ML inference service).

Both public and internal APIs are documented. Internal APIs are explicitly marked as such.

9.3 3. Tools Used

The following tools and technologies are used for API documentation:

- **Markdown** — main format for human-readable documentation;
 - **OpenAPI 3.x** — formal machine-readable API specification;
 - **Swagger UI / Swagger Editor** — schema validation and interactive visualization;
 - **Spectral / Redocly CLI** — OpenAPI schema linting and validation;
 - **Git** — version control for all documentation artifacts.
-

9.4 4. Standards Followed

The documentation follows these standards and best practices:

- RESTful API design principles;
 - OpenAPI Specification 3.x;
 - HTTP/1.1 semantics and standard HTTP status codes;
 - JSON as the primary data exchange format;
 - Clear separation between public, internal, and administrative APIs.
-

9.5 5. Naming Conventions

The following naming conventions are applied consistently:

- URL paths: **kebab-case**
Example: `/api/v1/meta/weapon-types`
 - JSON fields: **camelCase**
Example: `predictedPrice`, `wearTierId`
 - Error codes: **UPPER_SNAKE_CASE**
Example: `SKIN_NOT_FOUND`, `VALIDATION_ERROR`
 - Resource names: plural nouns
Example: `/skins`, `/weapons`, `/stickers`
-

9.6 6. Versioning Strategy

The API uses URL-based versioning:

`/api/v1/...`

Rules: - Breaking changes require a new major version; - Older versions remain documented until deprecated; - Version changes are reflected in both Markdown documentation and OpenAPI schemas.

9.7 7. Formatting Rules

All documentation follows these formatting rules:

- Markdown is used for all textual documentation;
- Each API has a dedicated reference file;
- Sections are organized using consistent headings;
- Code blocks are used for all request and response examples;
- Tables are used for parameters and field descriptions.

Recommended directory structure:

```
/docs
API_DOCUMENTATION_RULES.md
META_API_DOCUMENTATION.md
PREDICTION_API_DOCUMENTATION.md
AI_EXPLANATION_API_DOCUMENTATION.md
ADMIN_API_DOCUMENTATION.md
ML_SERVICE_API_DOCUMENTATION.md
```

9.8 8. Error Handling Documentation Rules

All APIs document error handling explicitly.

Rules: - Error responses use a unified structure where applicable; - HTTP status codes are documented for each endpoint; - Validation errors, authorization errors, and not-found errors are described separately.

9.9 9. Authentication and Authorization Documentation

Authentication requirements are documented per API group:

- Public APIs explicitly state that no authentication is required;
 - Admin API documents API key authorization;
 - Internal APIs are marked as internal and not exposed to end users.
-

9.10 10. OpenAPI Specification Rules

The OpenAPI specification: - serves as the single source of truth for API contracts; - is stored as a version-controlled YAML file; - must pass validation using OpenAPI tooling; - reflects the same structure and behavior as the Markdown documentation.

9.11 11. Known Gaps and Limitations

The following documentation limitations are explicitly acknowledged:

- Internal ML Service APIs do not document business validation logic;
- Some validation rules are enforced upstream and therefore not repeated in internal APIs;
- ML prediction values are returned as strings due to model output constraints;
- Some endpoints are intended for internal use only and are not publicly accessible.

These limitations are intentional design decisions and are documented for transparency.

9.12 12. Conclusion

This documentation strategy ensures a clear separation between API behavior, validation responsibilities, and architectural concerns. It provides both human-readable documentation and machine-readable specifications, meeting industry standards and diploma project evaluation requirements.

10 Meta API Documentation

10.1 Overview

The Meta API provides read-only reference data related to CS2 weapons, skins, cosmetic attributes, and stickers. This API is designed to be used by other system components such as pricing services, analytics modules, and machine learning pipelines.

All responses are returned in **JSON** format.

Base URL: /api/v1/meta

Authentication: Not required (public API)

10.2 Architecture Context

The Meta API acts as a reference data provider and does not store user-specific or transactional data. It supplies normalized metadata used across the system to ensure consistency and validation.

10.3 Error Handling

All error responses follow a unified format:

```
{
  "error": {
    "code": "ERROR_CODE",
    "message": "Human-readable error description"
  }
}
```

HTTP Code	Meaning
400	Invalid request parameters
404	Resource not found
500	Internal server error

10.4 1. Get Weapon Types

10.4.1 GET/weapon-types

Returns a list of all supported weapon types.

Response 200 OK

```
[
  {
    "id": 1,
    "code": "rifle",
    "name": "Rifles"
  }
]
```

10.5 2. Get Weapons by Weapon Type

10.5.1 GET/weapon-types/{weaponTypeId}/weapons

Returns all weapons belonging to the specified weapon type.

Path Parameters

Name	Type	Description
weaponTypeId	integer	Unique weapon type identifier

Response 200 OK

```
[
  { "id": 1, "name": "AK-47" },
  { "id": 2, "name": "M4A1-S" }
]
```

Errors - 404 Weapon type not found

10.6 3. Get Skins by Weapon

10.6.1 GET/weapons/{weaponId}/skins

Returns all skins associated with the selected weapon.

Path Parameters

Name	Type	Description
weaponId	integer	Weapon identifier

Response 200 OK

```
[
  {
    "id": 10,
    "name": "Redline",
    "patternStyle": "Abstract"
  }
]
```

10.7 4. Get Wear Tiers of a Skin

10.7.1 GET/skins/{skinId}/wear-tiers

Returns available wear tiers for the specified skin.

Response 200 OK

```
[
  { "id": 1, "name": "Factory New" },
  { "id": 2, "name": "Minimal Wear" }
]
```

10.8 5. Get Patterns of a Skin

10.8.1 GET/skins/{skinId}/patterns

Returns all pattern indexes for the selected skin.

Response 200 OK

```
[
  { "id": 661, "name": "Pattern 661" }
]
```

10.9 6. Get Stickers

10.9.1 GET/stickers

Returns a list of available stickers.

Query Parameters

Name	Type	Description
q	string	Text search query
limit	integer	Maximum number of results (default: 50)

Example Request

```
GET /stickers?q=crown&limit=10
```

Response 200 OK

```
[
  { "id": 101, "name": "Crown (Foil)" }
]
```

11 ML Service API Documentation

11.1 Overview

The ML Service API exposes internal prediction endpoints for different CS2 skin

categories. It is used **exclusively by the backend Prediction API** and is **not accessed directly by end users**.

All requests and responses use **JSON** format. All endpoints return the **predicted price as a string**, as produced by CatBoost models.

Base URL: /predict

Authentication: Not required (internal service)

11.2 Architectural Role

The ML Service is a dedicated microservice responsible only for **machine learning inference**.

Responsibilities: - Load trained ML models at startup - Accept preprocessed feature vectors
- Return raw price predictions

Non-responsibilities: - Business validation - Entity existence checks (skin, pattern, wear tier, etc.) - Authorization and access control

All domain validation is handled upstream by the **Prediction API**.

11.3 Validation Model

The ML Service relies solely on **FastAPI + Pydantic** validation.

Validation characteristics: - Type checking - Required field enforcement - Schema-level validation

Invalid requests return **HTTP 422 Unprocessable Entity**.

No domain-level validation (e.g. skin existence, pattern validity) is performed.

11.4 Error Response (422)

```
{
  "detail": [
    {
      "loc": ["body", "field_name"],
      "msg": "validation error",
      "type": "type_error"
    }
  ]
}
```

11.5 1. Predict Case Hardened

11.5.1 POST/case-hardened

Predicts prices for Case Hardened weapons and knives using color distribution features.

11.5.2 Request Body

```
{
  "float": 0.0,
  "pattern": 0,
  "stattrak": 0,
  "backside_blue": 0,
  "backside_purple": 0,
  "backside_gold": 0,
  "playside_blue": 0,
  "playside_purple": 0,
  "playside_gold": 0,
  "weapon": "string",
  "skin": "string",
  "wear": "string"
}
```

11.5.3 Response 200 OK

"1699.42"

11.6 2. Predict Case Hardened Guns

11.6.1 POST/ch-guns

Predicts prices for Case Hardened guns using extended feature vectors.

11.6.2 Request Body

```
{
  "weapon": "string",
  "skin": "string",
  "wear": "string",
  "pattern_style": "string",
  "float": 0.0,
  "pattern": 0,
  "stattrak": 0,
  "backside_blue": 0,
  "playside_blue": 0,
  "stickers_count": 0,
  "stickers_total_value": 0,
  "stickers_avg_value": 0,
  "stickers_max_value": 0,
  "slot0_price": 0,
  "slot1_price": 0,
  "slot2_price": 0,
  "slot3_price": 0,
  "blue_score": 0,
  "blue_tier": 0
}
```

11.6.3 Response 200 OK

"1325.11"

11.7 3. Predict Doppler

11.7.1 POST/doppler

Predicts prices for Doppler skins based on phase, float value, and StatTrak flag.

11.7.2 Request Body

```
{
  "weapon": "string",
  "skin": "string",
  "wear": "string",
  "phase": "string",
  "float": 0.0,
  "stattrak": 0
}
```

11.7.3 Response 200 OK

"2480.00"

11.8 4. Predict Fade Guns

11.8.1 POST/fade-guns

Predicts prices for Fade guns using fade percentage, rank, and sticker-derived features.

11.8.2 Request Body

```
{
  "float": 0.0,
  "pattern": 0,
  "stattrak": 0,
  "fade_percentage": 0,
  "fade_rank": 0,
  "stickers_count": 0,
  "stickers_total_value": 0,
  "stickers_avg_value": 0,
  "stickers_max_value": 0,
  "slot0_price": 0,
  "slot1_price": 0,
  "slot2_price": 0,
  "slot3_price": 0,
  "weapon": "string",
  "skin": "string",
  "wear": "string"
}
```

11.8.3 Response 200 OK

```
"987.65"
```

11.9 5. Predict Fade Knives

11.9.1 POST /fade-knives

Predicts prices for Fade knives using fade-related features.

11.9.2 Request Body

```
{
  "float": 0.0,
  "pattern": 0,
  "stattrak": 0,
  "fade_percentage": 0,
  "fade_rank": 0,
  "weapon": "string",
  "skin": "string",
  "wear": "string"
}
```

11.9.3 Response 200 OK

```
"2100.30"
```

11.10 6. Predict Float-Sensitive Guns

11.10.1 POST /float-sensitive-guns

Predicts prices for skins where float value has a strong influence.

11.10.2 Request Body

```
{
  "float": 0.0,
  "stattrak": 0,
  "stickers_count": 0,
  "stickers_total_value": 0,
  "stickers_avg_value": 0,
  "stickers_max_value": 0,
  "slot0_price": 0,
  "slot1_price": 0,
  "slot2_price": 0,
  "slot3_price": 0,
  "weapon": "string",
  "skin": "string",
  "wear": "string"
}
```

11.10.3 Response 200 OK

11.11 Performance Notes

- All ML models are loaded at service startup
 - Average prediction latency: < **50 ms**
 - Designed for synchronous internal calls
-

11.12 Known Limitations

- The service assumes all input data is prevalidated
- No authentication or rate limiting is implemented
- Returned price values are strings due to model output formatting

12 Prediction API Documentation

12.1 Overview

The Prediction API is the main endpoint used by the ASP.NET Core backend to generate a unified price prediction for any CS2 skin configuration.

The API aggregates metadata, prepares feature vectors, selects the appropriate machine learning model, invokes the ML microservice, and returns a final structured prediction result.

All responses are returned in **JSON** format.

Base URL: /api/v1
Authentication: Not required

12.2 Architecture Context

This endpoint represents the core prediction workflow of the system. It integrates database metadata, business validation rules, and multiple ML models (Case Hardened, Fade, Doppler, Float-Sensitive, etc.) into a single prediction pipeline.

12.3 Common Error Format

```
{
  "error": {
    "code": "ERROR_CODE",
    "message": "Human-readable error description"
  }
}
```

HTTP Code	Description
400	Validation error
404	Resource not found
500	Internal server error

12.4 Input Validation Rules

The Prediction API applies the same validation rules as the AI Explanation API, with additional constraints.

Field	Validation Rule
skinId	Must reference an existing skin
wearTierId	Must correspond to the provided float value
floatValue	Must be between 0.0 and 1.0
pattern	Must exist for the selected skin
stickers	Maximum 4 stickers allowed
stickers[]	Each sticker ID must exist

If any validation rule is violated, the request is rejected.

12.5 POST /predict

Performs a complete price prediction workflow for the selected skin configuration.

12.6 Request Body (application/json)

```
{
  "skinId": 1,
  "wearTierId": 1,
  "floatValue": 0.003,
  "isStattrak": true,
  "pattern": 3,
  "stickers": [0]
}
```

12.7 Request Fields

Field	Type	Required	Description
skinId	integer	yes	Selected skin identifier
wearTierId	integer	yes	Wear tier identifier
floatValue	number	yes	Actual float value
isStattrak	boolean	yes	StatTrak flag
pattern	integer	no	Pattern index
stickers	integer[]	no	Applied sticker IDs (max 4)

12.8 Successful Response (200 OK)

```
{
  "predicted_price": 1699.6503984707747,
  "stickers_features": {
    "stickers_count": 0,
    "stickers_total_value": 0,
    "stickers_avg_value": 0,
    "stickers_max_value": 0
  }
}
```

12.9 Response Fields

Field	Type	Description
predicted_price	number	Final predicted price
stickers_features	object	Extracted sticker-related features

12.9.1 Stickers Features

Field	Description
stickers_count	Total number of stickers
stickers_total_value	Combined base value of all stickers
stickers_avg_value	Average sticker value
stickers_max_value	Maximum sticker value

12.10 Validation Error Examples

12.10.1 Too Many Stickers

HTTP 400

```
{
  "error": {
    "code": "STICKER_LIMIT_EXCEEDED",
    "message": "A maximum of 4 stickers is allowed"
  }
}
```

12.10.2 Skin Not Found

HTTP 404

```
{
  "error": {
    "code": "SKIN_NOT_FOUND",
    "message": "Skin with the specified ID was not found"
  }
}
```

12.10.3 Pattern Not Found

HTTP 404

```
{
  "error": {
    "code": "PATTERN_NOT_FOUND",
    "message": "Pattern is not available for the selected skin"
  }
}
```

12.10.4 Wear Tier and Float Mismatch

HTTP 400

```
{
  "error": {
    "code": "WEAR_TIER_MISMATCH",
    "message": "Wear tier does not correspond to the provided float value"
  }
}
```

12.11 Versioning Strategy

The API uses URL-based versioning:

/api/v1/predict

Breaking changes require a new major version.

13 Backend Technical Documentation

13.1 CS2 Skin Price Prediction System

13.2 1. Introduction

This document provides a **comprehensive and in-depth technical description** of the backend subsystem of the *CS2 Skin Price Prediction System*.

The backend represents the **core orchestration layer** of the application, integrating database access, business logic, machine learning inference, AI-based explanations, and external services into a single coherent system.

The goal of this documentation is to: - Fully describe the backend architecture and its responsibilities - Explain design decisions and trade-offs - Provide a clear understanding of how data flows through the system - Demonstrate compliance with academic and engineering best practices

This document is intentionally **extended and detailed**, as required for diploma-level technical documentation.

13.3 2. Backend Responsibilities

The backend subsystem is responsible for the following high-level functions:

1. Exposing a RESTful HTTP API for client applications
2. Managing reference and metadata storage via PostgreSQL
3. Preparing feature vectors for machine learning inference
4. Communicating with a dedicated ML microservice
5. Aggregating prediction results and applying post-processing logic
6. Providing AI-generated explanations for predictions
7. Enforcing security constraints and access control
8. Supporting administrative operations and data management
9. Ensuring scalability, reliability, and observability

The backend is implemented as an **ASP.NET Core 8 Web API**, following a layered architecture.

13.4 3. Architectural Style

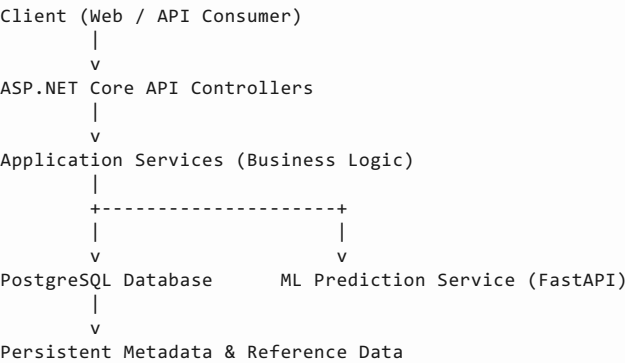
13.4.1 3.1 Architectural Pattern

The backend follows a **Layered Architecture** with clear separation of concerns:

- Presentation Layer (Controllers)
- Application Layer (Services)
- Domain Layer (Entities & Business Rules)
- Infrastructure Layer (Database, ML service, external APIs)

This approach was selected because it: - Improves maintainability - Simplifies testing - Enables independent evolution of system layers - Aligns with enterprise-grade backend design standards

13.5 4. High-Level Architecture



The backend acts as the **central coordinator**, ensuring that all components interact in a controlled and predictable manner.

13.6 5. Project Structure Overview

The backend project is structured to reflect architectural boundaries explicitly.

```
cs2price_prediction/
├── Config/                                # Application configuration
├── classes
│   ├── OpenAiOptions.cs                  # OpenAI integration
├── configuration
│   └── MLServiceOptions.cs               # ML service connection
├── configuration
├── Controllers/                          # ASP.NET Core HTTP controllers
│   ├── PredictionController.cs           # CS2 skin price prediction
├── endpoints
│   ├── MetaController.cs                 # Reference and metadata
├── endpoints
│   ├── AiExplanationController.cs        # AI-generated explanation
├── endpoints
│   ├── AiExplanationV2Controller.cs      # Extended AI explanation
├── endpoints
│   ├── Admin/                           # Administrative API controllers
│   │   ├── AdminSkinsController.cs       # Skin CRUD management
│   │   ├── AdminSkinWearTiersController.cs # Skin wear tier management
│   │   ├── AdminStickersController.cs    # Sticker management
│   │   ├── AdminWeaponsController.cs     # Weapon management
│   │   ├── AdminWeaponTypesController.cs # Weapon type management
│   │   ├── AdminWearTiersController.cs    # Wear tier management
│   │   └── Patterns/                     # Pattern management controllers
│   │       ├── AdminCaseHardenedGunPatternsController.cs # CH gun
├── patterns
│   │   ├── AdminCaseHardenedKnifePatternsController.cs # CH knife
├── patterns
│   │   ├── AdminDopplerPhasesController.cs # Doppler
├── phases
│   │   ├── AdminDopplerSkinPhasesController.cs # Skin-Doppler
├── mapping
│   │   ├── AdminFadeGunPatternsController.cs # Fade gun
├── patterns
│   │   └── AdminFadeKnifePatternsController.cs # Fade knife
├── patterns
├── course-diploma-projects-docs-main/    # External diploma-related
├── materials
├── cs2_ml_service/                       # Python-based ML prediction
├── service
│   ├── app.py                           # FastAPI application entry point
│   ├──.pkl_extract_all.py                # Feature extraction script
│   ├── requirements.txt                  # Python dependencies
│   ├── data/                            # Training datasets
│   ├── extracted/                        # Extracted features and configs
│   └── models/                           # Trained CatBoost models
├── Data/                                 # Data access layer
│   ├── AppDbContext.cs                  # EF Core database context
│   ├── DbSeeder.cs                      # Initial database seeding
│   ├── DesignTimeDbContextFactory.cs    # Design-time EF context
│   └── IAdminDbContextFactory.cs        # Admin DB context factory
├── interface
├── data_for_db/                          # CSV data for DB initialization
│   ├── skins.csv                        # Skins dataset
│   ├── weapons.csv                      # Weapons dataset
│   ├── weapon_types.csv                 # Weapon types dataset
│   ├── wear_tiers.csv                   # Wear tiers dataset
│   ├── skin_wear_tiers.csv              # Skin-wear tier mapping
│   ├── stickers_dataset.csv             # Stickers dataset
│   ├── doppler_phases.csv               # Doppler phases dataset
│   ├── doppler_skin_phases.csv          # Skin-Doppler mapping
│   ├── fade_gun_unique_patterns.csv     # Fade gun patterns
│   ├── fade_knives_unique_patterns.csv  # Fade knife patterns
│   ├── case_hardened_gun_unique_patterns.csv # CH gun patterns
│   └── case_hardened_knives_unique_patterns.csv # CH knife patterns
├── data_for_db/                          # SQL initialization scripts
│   └── init.sql
├── Domain/                              # Domain models (business
├── entities)
│   ├── Meta/
│   │   ├── Skin.cs
│   │   ├── SkinWearTier.cs
│   │   ├── Weapon.cs
│   │   ├── WeaponType.cs
│   │   └── WearTier.cs
```

	Patterns/	
	CaseHardenedGunPattern.cs	
	CaseHardenedKnifePattern.cs	
	DopplerPhase.cs	
	DopplerSkinPhase.cs	
	FadeGunPattern.cs	
	FadeKnifePattern.cs	
	Stickers/	
	Sticker.cs	
	StickerPrice.cs	
	DTOs/	# Data Transfer Objects
	Admin/	# DTOs for administrative
operations		
	Skins/	# DTOs for skin management
	CreateSkinDto.cs	# DTO for creating a new skin
	UpdateSkinDto.cs	# DTO for updating existing skin
data		
	SkinWearTiers/	# DTOs for skin wear tier
management		
to a skin	CreateSkinWearTierDto.cs	# DTO for assigning a wear tier
	UpdateSkinWearTierDto.cs	# DTO for updating skin wear tier
data		
from a skin	DeleteSkinWearTierDto.cs	# DTO for removing a wear tier
	Stickers/	# DTOs for sticker administration
	CreateStickerDto.cs	# DTO for creating a new sticker
information	UpdateStickerDto.cs	# DTO for updating sticker
	Weapons/	# DTOs for weapon administration
	CreateWeaponDto.cs	# DTO for creating a new weapon
	UpdateWeaponDto.cs	# DTO for updating weapon data
	WeaponTypes/	# DTOs for weapon type
administration		
	CreateWeaponTypeDto.cs	# DTO for creating a weapon type
data	UpdateWeaponTypeDto.cs	# DTO for updating weapon type
	WearTiers/	# DTOs for wear tier
administration		
	CreateWearTierDto.cs	# DTO for creating a wear tier
	UpdateWearTierDto.cs	# DTO for updating wear tier data
	Patterns/	# DTOs for visual pattern
administration		
	CaseHardenedGun/	# Case Hardened gun pattern DTOs
	CreateCaseHardenedGunPatternDto.cs	# DTO for creating a Case Hardened gun pattern
	UpdateCaseHardenedGunPatternDto.cs	# DTO for updating a Case Hardened gun pattern
DTOs	CaseHardenedKnife/	# Case Hardened knife pattern
	CreateCaseHardenedKnifePatternDto.cs	# DTO for creating a Case Hardened knife pattern
	UpdateCaseHardenedKnifePatternDto.cs	# DTO for updating a Case Hardened knife pattern
	DopplerPhase/	# Doppler phase DTOs
	CreateDopplerPhaseDto.cs	# DTO for creating a Doppler phase
	UpdateDopplerPhaseDto.cs	# DTO for updating Doppler phase data
DTOs	DopplerSkinPhase/	# Skin-Doppler phase mapping
	CreateDopplerSkinPhaseDto.cs	# DTO for binding a skin to a Doppler phase
	UpdateDopplerSkinPhaseDto.cs	# DTO for updating skin-Doppler phase mapping
	FadeGun/	# Fade gun pattern DTOs
	CreateFadeGunPatternDto.cs	# DTO for creating a Fade gun pattern
	UpdateFadeGunPatternDto.cs	# DTO for updating a Fade gun pattern
	FadeKnife/	# Fade knife pattern DTOs
	CreateFadeKnifePatternDto.cs	# DTO for creating a Fade knife pattern
	UpdateFadeKnifePatternDto.cs	# DTO for updating a Fade knife pattern

```

├── AI/                                     # DTOs for AI explanation and
analysis
├── AiExplainFrontendInputDto.cs
│   └── # Input DTO for AI explanation generation from frontend
├── AiExplainV2FrontendInputDto.cs
│   └── # Extended input DTO for AI explanation generation
├── CaseHardenedKnife/
│   └── AiCaseHardenedKnifeRequest.cs
│       └── # AI request DTO for Case Hardened knife explanation
├── ChGuns/
│   └── AiChGunsRequest.cs
│       └── # AI request DTO for Case Hardened gun explanation
├── Doppler/
│   └── AiDopplerRequest.cs
│       └── # AI request DTO for Doppler explanation
├── FadeGun/
│   └── AiFadeGunsRequest.cs
│       └── # AI request DTO for Fade gun explanation
├── FadeKnife/
│   └── AiFadeKnivesRequest.cs
│       └── # AI request DTO for Fade knife explanation
├── FloatGuns/
│   └── AiFloatSensitiveGunsRequest.cs
│       └── # AI request DTO for float-sensitive gun explanation
├── StickersDtoForAI/
│   └── StickersDtoForAI.cs
│       └── # DTO containing sticker data for AI processing
├── Meta/                                     # DTOs for reference data
transfer
├── SkinDto.cs                             # DTO representing skin data
├── WeaponDto.cs                           # DTO representing weapon data
├── WeaponTypeDto.cs                       # DTO representing weapon type
data
├── WearTierDto.cs                         # DTO representing wear tier data
├── StickerDto.cs                         # DTO representing sticker data
├── PatternOptionDto.cs                   # DTO representing available
pattern options
├── ML/                                     # DTOs for ML service
communication
├── MlCaseHardenedKnifeRequest.cs         # ML request DTO for CH knife
prediction
├── MlChGunsRequest.cs                    # ML request DTO for CH gun
prediction
├── MlDopplerRequest.cs                   # ML request DTO for Doppler
prediction
├── MlFadeGunsRequest.cs                  # ML request DTO for Fade gun
prediction
├── MlFadeKnivesRequest.cs                # ML request DTO for Fade knife
prediction
├── MlFloatSensitiveGunsRequest.cs        # ML request DTO for float-based
prediction
├── Prediction/
│   ├── PredictionRequestDto.cs           # DTOs for price prediction API
│   └── MlPredictionResponse.cs           # Price prediction request DTO
│                                           # ML prediction response DTO
├── Migrations/                           # EF Core migrations
├── Security/
│   └── AdminApiKeyMiddleware.cs           # Security components
│                                           # Admin API key validation
middleware
├── Services/                             # Business logic layer
│   └── Admin/                             # Administrative business
services
├── Patterns/                             # Services for managing visual
patterns
├── CaseHardenedGun/                       # Case Hardened patterns for
guns
├── AdminCaseHardenedGunPatternService.cs
│   └── # Implements business logic for managing Case Hardened
gun patterns
├── IAdminCaseHardenedGunPatternService.cs
│   └── # Interface defining operations for Case Hardened gun
pattern management
├── CaseHardenedKnife/                     # Case Hardened patterns for
knives
├── AdminCaseHardenedKnifePatternService.cs
│   └── # Implements business logic for managing Case Hardened
knife patterns
├── IAdminCaseHardenedKnifePatternService.cs
│   └── # Interface defining operations for Case Hardened
knife pattern management
├── DopplerPhase/                         # Doppler phase management
├── AdminDopplerPhaseService.cs

```

```

# AdminDopplerPhaseService.cs
# Business logic for creating and managing Doppler
phases
└─ IAdminDopplerPhaseService.cs
   # Interface for Doppler phase administrative
operations
└─ DopplerSkinPhase/
   # Skin to Doppler phase mapping
   └─ AdminDopplerSkinPhaseService.cs
      # Business logic for binding skins to Doppler phases
   └─ IAdminDopplerSkinPhaseService.cs
      # Interface for Doppler skin-phase mapping operations
└─ FadeGun/
   # Fade patterns for guns
   └─ AdminFadeGunPatternService.cs
      # Business logic for managing Fade gun patterns
   └─ IAdminFadeGunPatternService.cs
      # Interface for Fade gun pattern management
└─ FadeKnife/
   # Fade patterns for knives
   └─ AdminFadeKnifePatternService.cs
      # Business logic for managing Fade knife patterns
   └─ IAdminFadeKnifePatternService.cs
      # Interface for Fade knife pattern management
└─ Skins/
   # Administrative services for
skins
└─ AdminSkinService.cs
   # Business logic for managing skins in admin scope
└─ IAdminSkinService.cs
   # Interface for admin skin operations
└─ SkinWearTiers/
   # Skin wear tier administration
   └─ AdminSkinWearTierService.cs
      # Business logic for managing skin wear tiers
   └─ IAdminSkinWearTierService.cs
      # Interface for skin wear tier management
└─ Stickers/
   # Administrative services for
stickers
└─ AdminStickerService.cs
   # Business logic for managing stickers
└─ IAdminStickerService.cs
   # Interface for admin sticker operations
└─ Weapons/
   # Administrative services for
weapons
└─ AdminWeaponService.cs
   # Business logic for weapon management
└─ IAdminWeaponService.cs
   # Interface for admin weapon operations
└─ WeaponTypes/
   # Administrative services for
weapon types
└─ AdminWeaponTypeService.cs
   # Business logic for weapon type management
└─ IAdminWeaponTypeService.cs
   # Interface for admin weapon type operations
└─ WearTiers/
   # Administrative services for
wear tiers
└─ AdminWearTierService.cs
   # Business logic for wear tier management
└─ IAdminWearTierService.cs
   # Interface for admin wear tier operations
└─ AI/
   # AI and LLM-related services
   └─ AiExplanation/
      # AI explanation generation
      └─ AiExplanationService.cs
         # Generates human-readable explanations for price
predictions
└─ IAiExplanationService.cs
   # Interface for AI explanation service
└─ AiPromptService/
   # Prompt construction for LLMs
   └─ AiPromptFactory.cs
      # Builds prompts for LLM requests
   └─ IAiPromptFactory.cs
      # Interface for AI prompt factory
└─ AiStickerService/
   # Sticker preprocessing for AI
   └─ AiStickerService.cs
      # Prepares sticker data for AI analysis
   └─ IAiStickerService.cs
      # Interface for AI sticker service
└─ Llm/
   # Large Language Model
integration
└─ ILLMClient.cs
   # Common interface for LLM providers
└─ LlmFailoverService.cs
   # Handles failover between multiple LLM providers
└─ LlmPriority.cs
   # Defines priority order for LLM providers
└─ OpenAiClient.cs
   # OpenAI API client implementation
└─ Meta/
   # Reference and metadata
services
└─ MetaService.cs

```



```

| | | # Provides access to reference data (skins, weapons, wear
tiers)
| | | └─ IMetaService.cs
| | |     # Interface for metadata service
| | |
| | | └─ Prediction/                                # Price prediction services
| | |     └─ PredictionService.cs
| | |         # Orchestrates ML price prediction workflow
| | |     └─ IPredictionService.cs
| | |         # Interface for prediction service
| | |
| | | └─ Stickers/                                # Sticker-related services
| | |     └─ StickerService.cs
| | |         # Business logic for sticker processing
| | |     └─ StickerFeatures.cs
| | |         # Feature extraction from stickers for ML/AI usage
| | |     └─ IStickerService.cs
| | |         # Interface for sticker service
| | |
| └─ tools_postman_testing/                        # Postman API collections
|
└─ .env
└─ .gitignore
└─ docker-compose.yml
└─ Dockerfile
└─ Program.cs

```

Each directory is described in detail below.

13.7 6. Configuration Layer (Config/)

The Config folder contains strongly typed configuration models used to bind environment variables and configuration files.

13.7.1 Files:

- **OpenAiOptions.cs**
Defines configuration required for optional OpenAI integrations (API key, model settings, request limits).
- **MLServiceOptions.cs**
Stores connection parameters for the ML microservice, including base URL and timeout settings.

Using typed configuration objects ensures: - Compile-time safety - Centralized configuration management - Clean separation between code and environment-specific values

13.8 7. Controllers Layer (Controllers/)

Controllers define the **HTTP interface** of the backend.

13.8.1 Key Controllers:

13.8.1.1 PredictionController

Handles user-facing price prediction requests. Responsibilities: - Validate incoming request DTOs - Fetch required reference data - Delegate prediction logic to services - Return normalized prediction responses

13.8.1.2 MetaController

Provides metadata endpoints for: - Skins - Weapons - Wear tiers - Patterns - Stickers

These endpoints allow clients to dynamically build valid prediction requests.

13.8.1.3 AiExplanationController / AiExplanationV2Controller

Generate AI-based textual explanations for predicted prices. These controllers interface with the LLM abstraction layer.

13.8.1.4 Admin Controllers

Located under `Controllers/Admin/`, these endpoints: - Are protected by API key authentication - Enable CRUD operations on reference data - Support pattern and metadata management

13.9 8. Data Transfer Objects (DTOs/)

DTOs define **explicit data contracts** between layers.

DTO categories include: - Prediction DTOs - Metadata DTOs - Admin DTOs - ML request DTOs - AI explanation DTOs

This design: - Prevents domain model leakage - Improves API stability - Simplifies versioning

13.10 9. Domain Layer (Domain/)

The Domain layer contains **pure business entities**.

13.10.1 Core Concepts:

- Skin
- Weapon
- WeaponType
- WearTier
- Sticker
- Pattern (Case Hardened, Fade, Doppler)

These entities: - Are persistence-agnostic - Represent real-world CS2 market concepts - Serve as the foundation of business logic

13.11 10. Data Access Layer (Data/)

13.11.1 AppDbContext

Implements EF Core DbContext: - Maps domain entities to relational tables - Manages transactions and queries

13.11.2 DbSeeder

Populates the database with initial reference data from CSV files.

13.11.3 Design-Time Context Factory

Supports EF Core migrations without runtime dependencies.

The database schema is fully normalized and optimized for read-heavy workloads.

13.12 11. Services Layer (Services/)

Services implement **application-level business logic**.

13.12.1 PredictionService

- Aggregates metadata
- Builds ML feature vectors
- Calls ML service
- Applies post-processing logic

13.12.2 MetaService

- Handles reference data queries
- Applies filtering and transformation logic

13.12.3 StickerService

- Processes sticker price contributions
- Aggregates sticker-based value modifiers

13.12.4 AiExplanationService

- Builds structured prompts
- Communicates with LLM providers
- Generates human-readable explanations

13.12.5 LlmFailoverService

Ensures system resilience by: - Falling back between providers - Handling API outages gracefully

13.13 12. ML Service Integration

The backend communicates with a **Python-based FastAPI ML microservice**.

Interaction Flow: 1. Backend assembles feature JSON 2. Sends HTTP request to ML service 3. ML service performs inference using CatBoost models 4. Prediction result is returned 5. Backend normalizes and returns final value

This separation: - Enables independent scaling - Isolates heavy ML dependencies - Improves system robustness

13.14 13. Security Layer (Security/)

13.14.1 AdminApiKeyMiddleware

- Protects admin endpoints
- Validates API keys via environment variables

Security principles applied: - Least privilege - Environment-based secrets - No credentials stored in source code

13.15 14. Application Startup (Program.cs)

Program.cs configures: - Dependency injection - Middleware pipeline - Database migrations - Health checks - API routing

This centralized bootstrap ensures predictable application behavior.

13.16 15. Containerization & Deployment

The backend is deployed using Docker and Docker Compose.

Key characteristics: - Multi-stage build - Non-root execution - Environment-driven configuration - Health checks

This enables: - Reproducible builds - Simple deployment - Production readiness

13.17 16. Performance Characteristics

Measured runtime metrics: - Idle RAM usage: ~120–180 MB - Prediction latency: <50 ms (excluding ML inference) - Startup time: ~4–6 seconds

The backend introduces minimal overhead and scales horizontally.

13.18 17. Reliability & Maintainability

The system is designed to: - Fail fast on invalid input - Degrade gracefully when ML/AI services are unavailable - Support incremental extension

Clear module boundaries simplify long-term maintenance.

13.19 18. Conclusion

The backend subsystem provides a **robust, scalable, and maintainable foundation** for the CS2 Skin Price Prediction System.

It successfully integrates: - Relational data storage - Machine learning inference - AI-based explanation generation - Secure administrative operations

The chosen architecture and implementation fully satisfy both **engineering best practices** and **academic diploma requirements**.

14 Containerization & Deployment Documentation

14.1 1. Docker Compose

The project is fully containerized using **Docker Compose**, which orchestrates all services required for the system to operate.

14.1.1 1.1 docker-compose.yml

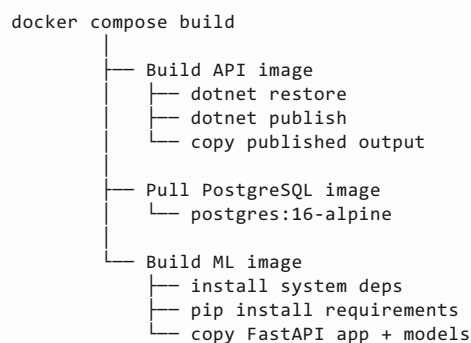
The `docker-compose.yml` file (see dev branch in the repository) defines the following services:

- **cs2-api** – ASP.NET Core production API
- **cs2-api-dev** – ASP.NET Core development API (hot reload)
- **cs2-ml** – FastAPI-based ML service with CatBoost models
- **cs2-postgres** – PostgreSQL database
- **Volumes** – Persistent storage for database data

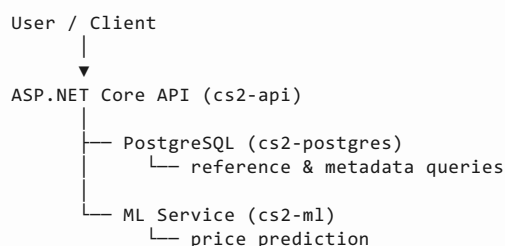
Each service is isolated, configured via environment variables, and connected through a shared Docker network.

14.2 2. Build & Interaction Diagrams

14.2.1 2.1 Build Pipeline



14.2.2 2.2 Runtime Interaction Pipeline



14.2.3 2.3 Request Flow

1. User sends request → API (filters, skin selection).
 2. API reads reference data from DB (skins, patterns, stickers).
 3. API builds feature JSON and sends it to ML service.
 4. ML service returns predicted price.
 5. API responds to the user with final price.
-

14.3 3. Docker Images Description

14.4 3.1 API Image — cs2price_prediction-api

14.4.1 Purpose & Functionality

The API image hosts the ASP.NET Core 8 Web API and is responsible for:

- Handling client HTTP requests.
- Fetching metadata from PostgreSQL.
- Building ML feature vectors.
- Communicating with the ML microservice.
- Returning final predicted prices.
- Running the backend in a production-ready environment.

14.4.2 Base Image & Rationale

Multi-stage Dockerfile:

1. **Build stage**
 - Image: `mcr.microsoft.com/dotnet/sdk:8.0`
 - Used for restore, build, and publish.
2. **Runtime stage**
 - Image: `mcr.microsoft.com/dotnet/aspnet:8.0`
 - Lightweight, secure, production-optimized runtime.

14.4.3 Final Image Size

- **334.84 MB**
- Fully satisfies the requirement: **Backend < 1 GB**.

14.4.4 Optimizations Applied

- Multi-stage build (SDK excluded from runtime).
 - `.dockerignore` excludes build artifacts and VCS files.
 - Only published output copied.
 - Secrets via environment variables.
 - Runs as non-root user.
 - Exposes only port **8080**.
-

14.5 3.2 ML Image — cs2price_prediction-ml-service

14.5.1 Purpose & Functionality

This image runs the FastAPI-based ML service and:

- Loads **6 CatBoost models** at startup.
- Exposes `/predict/*` endpoints.
- Processes numerical and categorical features.
- Returns predicted prices.

14.5.2 Base Image & Justification

- Image: `python:3.11-slim`
- Reasons:
 - Small OS footprint.
 - Compatibility with CatBoost and NumPy.
 - Official, security-patched Python image.

14.5.3 Final Image Size

- **1.30 GB**

- Larger size justified by ML dependencies and models.

14.5.4 Optimizations Applied

- `pip install --no-cache-dir`
 - `apt-get --no-install-recommends`
 - Cleanup of APT cache.
 - `.dockerignore` excludes caches and unused artifacts.
 - Runs as non-root user.
 - Exposes only port **8000**.
-

14.6 3.3 Database Image — postgres:16-alpine

14.6.1 Purpose & Functionality

Stores all persistent data:

- Skins, weapons, wear tiers.
- Patterns (CH, Fade, Doppler).
- Stickers and sticker prices.
- Metadata for ML feature construction.

14.6.2 Base Image & Justification

- Image: `postgres:16-alpine`
- Benefits:
 - Smaller footprint.
 - Faster startup.
 - Lower resource usage.

14.6.3 Final Image Size

- **394.74 MB**
- Uses persistent volume `db-data`.

14.6.4 Optimizations Applied

- Official Postgres image.
 - Read-only schema initialization script.
 - Stateless container with external volume.
 - Credentials via `.env` file.
-

14.7 3.4 API Dev Image — cs2-api-dev

Development-only container.

14.7.1 Purpose

- Hot reload via `dotnet watch`.
- Debugging and testing.
- No production usage.

14.7.2 Base Image

- `mcr.microsoft.com/dotnet/sdk:8.0`

14.7.3 Final Image Size

- **1.19 GB**
 - Acceptable for development use only.
-

14.8 4. Containers Description

14.8.1 4.1 API Container — cs2-api

Role: Main entry point for clients.

Type	Port
Internal	8080
External	8087

Healthcheck:
HTTP /health endpoint.

14.8.2 4.2 ML Service Container — cs2-ml

Role: Performs ML inference.

- Port: **8000**
- Models loaded into memory at startup.
- No volumes required.

14.8.3 4.3 PostgreSQL Container — cs2-postgres

Role: Persistent data storage.

Volumes:

Volume	Purpose
db-data	PostgreSQL data
init.sql	Schema initialization

14.9 5. Metrics & Performance

14.9.1 5.1 Image Build Time

Service	Time
API	45–55s
ML	60–80s
DB	Instant
API Dev	10–15s

14.9.2 5.2 Runtime Resource Usage

Container	RAM	CPU
API	120–180 MB	up to 10%
ML	300–520 MB	up to 40%
DB	50–120 MB	up to 10%

14.9.3 5.3 System Summary

- Total RAM: **600–800 MB**
- Cold start: **12–15 seconds**
- Predictions < **50 ms**

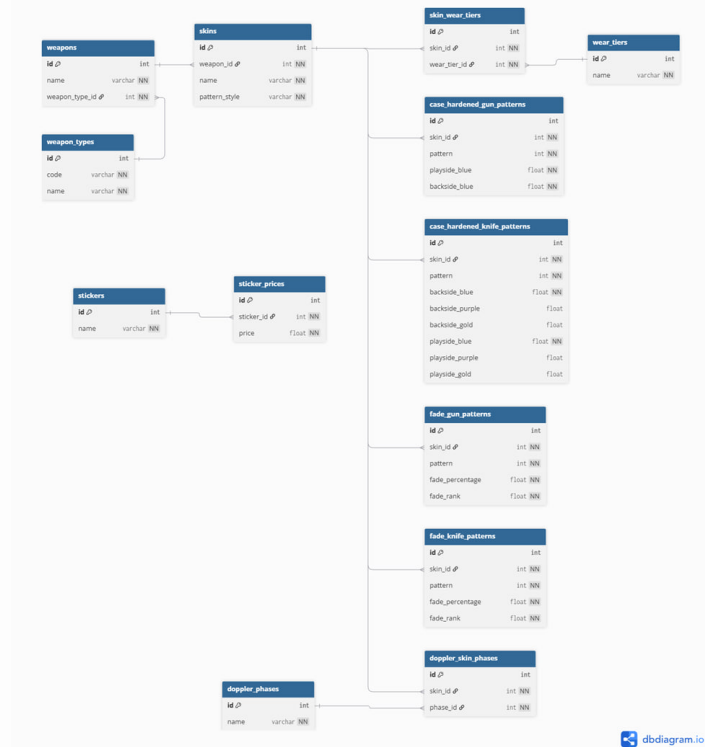
14.10 6. Conclusion

The containerized architecture:

- Is modular and scalable.
- Meets all performance and size constraints.
- Supports both production and development workflows.
- Fully complies with diploma technical requirements.

15 Database Schema – Full Data Dictionary

This document provides a **column-level, formal description** of the database schema used in the CS2 Skin Price Prediction system. Each table is described with its purpose, columns, data types, constraints, and business meaning.



ER

15.1 1. weapon_types

Purpose:

Stores high-level weapon categories used to classify weapons.

Column	Type	Constraints	Description
id	int	PK, auto-increment	Internal unique identifier
code	varchar	NOT NULL, UNIQUE	Machine-readable weapon type code (rifle, pistol, knife, etc.)
name	varchar	NOT NULL	Human-readable weapon type name

Notes:

- code is used by the application and ML pipeline. - Values are static and seeded.

15.2 2. weapons

Purpose:

Stores individual weapon models (e.g. AK-47, AWP, Karambit).

Column	Type	Constraints	Description
id	int	PK, auto-increment	Weapon identifier
name	varchar	NOT NULL, UNIQUE	Official weapon name
weapon_type_id	int	FK → weapon_types.id	Weapon category reference

15.3 3. wear_tiers

Purpose:

Defines available wear conditions for skins.

Column	Type	Constraints	Description
id	int	PK	Wear tier identifier
name	varchar	NOT NULL, UNIQUE	Wear name (Factory New, Minimal Wear, etc.)

15.4 4. skins

Purpose:
Core entity describing a skin applied to a weapon.

Column	Type	Constraints	Description
id	int	PK	Skin identifier
weapon_id	int	FK → weapons.id	Weapon this skin belongs to
name	varchar	NOT NULL	Skin name (Case Hardened, Fade, etc.)
pattern_style	varchar	NOT NULL	Determines valid pattern table (ch_gun, fade_knife, etc.)

Indexes:
- UNIQUE (weapon_id, name)

15.5 5. skin_wear_tiers

Purpose:
Defines which wear tiers are valid for each skin.

Column	Type	Constraints	Description
id	int	PK	Row identifier
skin_id	int	FK → skins.id	Skin reference
wear_tier_id	int	FK → wear_tiers.id	Wear tier reference

15.6 6. case_hardened_gun_patterns

Purpose:
Stores pattern-specific color distribution for Case Hardened guns.

Column	Type	Constraints	Description
id	int	PK	Pattern row ID
skin_id	int	FK → skins.id	Skin reference
pattern	int	NOT NULL	Pattern index (1–1000)
playside_blue	float	NOT NULL	% of blue on play side
backside_blue	float	NOT NULL	% of blue on back side

15.7 7. case_hardened_knife_patterns

Purpose:
Detailed color segmentation for Case Hardened knives.

Column	Type	Constraints	Description
id	int	PK	Pattern row ID
skin_id	int	FK → skins.id	Skin reference
pattern	int	NOT NULL	Pattern number
backside_blue	float	NOT NULL	Backside blue percentage
backside_purple	float	NULL	Backside purple percentage
backside_gold	float	NULL	Backside gold percentage
playside_blue	float	NOT NULL	Play side blue
playside_purple	float	NULL	Play side purple
playside_gold	float	NULL	Play side gold

15.8 8. fade_gun_patterns

Purpose:

Fade gradient parameters for guns.

Column	Type	Constraints	Description
id	int	PK	Pattern ID
skin_id	int	FK	Skin reference
pattern	int	NOT NULL	Pattern index
fade_percentage	float	NOT NULL	Fade completion percentage
fade_rank	float	NOT NULL	Relative quality rank

15.9 9. fade_knife_patterns

Purpose:

Fade gradient parameters for knives.

(Same semantics as fade_gun_patterns)

15.10 10. doppler_phases

Purpose:

Dictionary of Doppler phases.

Column	Type	Constraints	Description
id	int	PK	Phase ID
name	varchar	UNIQUE	Phase name (Ruby, Sapphire, etc.)

15.11 11. doppler_skin_phases

Purpose:

Defines which Doppler phases are available for a skin.

Column	Type	Constraints	Description
id	int	PK	Row ID
skin_id	int	FK	Doppler skin
phase_id	int	FK	Doppler phase

15.12 12. stickers

Purpose:

Sticker dictionary.

Column	Type	Constraints	Description
id	int	PK	Sticker ID
name	varchar	UNIQUE	Sticker name

15.13 13. sticker_prices

Purpose:

Observed market prices for stickers.

Column	Type	Constraints	Description
id	int	PK	Price row
sticker_id	int	FK	Sticker reference
price	float	CHECK > 0	Market price

15.14 Summary

This schema: - Is normalized to **3NF** - Enforces referential integrity - Models **visual-driven pricing** - Avoids unstructured JSON - Is suitable for ML feature extraction and analytics

15.15 User Logic and Access Model

15.15.1 Absence of Domain Users in the Schema

The database schema intentionally does **not** contain tables such as `users`, `accounts`, or `permissions`.

This is a deliberate architectural decision based on the nature of the system:

- the database stores **only technical and market-related CS2 skin data**;
- no personally identifiable information (PII) is processed or persisted;
- the system is not user-driven, but **prediction- and analytics-driven**.

As a result, application users are **not modeled as domain entities**, and no business logic depends on user-specific state stored in the database.

15.15.2 Separation of Responsibilities

The system follows a strict separation of concerns:

Layer	Responsibility
Database	Data integrity, consistency, access control
Application	Authentication, authorization, API exposure
ML services	Read-only analytical access

User identity and authentication are handled **outside** the database, while PostgreSQL is responsible only for enforcing **who is allowed to read or modify data**.

15.15.3 Role-Based Access Control (RBAC)

Instead of user tables, the database relies on **PostgreSQL role-based security**.

Three logical roles are defined:

- **app_read** — read-only access (SELECT)
- **app_write** — read + write access (inherits `app_read`)
- **app_admin** — full administrative privileges (inherits `app_write`)

These roles represent **permission sets**, not individual users.

15.15.4 Database Users

Two actual PostgreSQL users are created:

- **cs2_user**
 - used by the API, ML services, and analytics
 - mapped to the `app_read` role
 - strictly read-only access
- **cs2_admin**
 - used for migrations, schema changes, and data seeding
 - mapped to the `app_admin` role
 - never used by the runtime application

The application never operates under a superuser account, which significantly reduces security risks.

15.15.5 Schema-Level Isolation

All application tables are located in a dedicated schema (`cs2`), owned by `cs2_admin`.

This allows: - isolation from the default `public` schema; - fine-grained permission management; - prevention of uncontrolled object creation.

Default privileges are configured so that newly created tables automatically receive correct access rights, eliminating manual permission errors.

15.15.6 Application-Level Authorization

On the application side:

- Create / Update / Delete operations are exposed only through an administrative interface;
- Public API endpoints are effectively **read-only**;
- ML and analytics workloads operate exclusively under the `cs2_user` account.

This enforces a clear boundary between: - **data consumers** (API, ML, dashboards), - **data administrators** (schema and reference data management).

15.15.7 Security Rationale

This approach provides several advantages:

- eliminates storage of credentials or PII in the database;
- minimizes attack surface;
- follows the principle of least privilege;
- simplifies compliance and auditing.

The absence of user tables is therefore **not a limitation**, but a conscious design choice aligned with the system's goals and usage patterns.

16 Data Collection

16.1 Overview

Data collection was the **most challenging, resource-intensive, and critical stage** of this project.

Unlike typical machine learning tasks that rely on publicly available datasets or open APIs, the CS2 skin market does **not provide transparent or structured historical price data**. As a result, building a high-quality dataset required significant manual effort, custom tooling, and direct cooperation with a real marketplace.

The primary goal of the data collection phase was to obtain **reliable, unbiased, real-world prices** that accurately reflect actual market behavior.

16.2 Cooperation with a Real Marketplace

To achieve realistic and trustworthy data, it was necessary to **cooperate with a large CS2 marketplace platform (DMarket)**.

This cooperation was essential because:

- there is **no official Valve API** for historical sold prices;
- public endpoints expose only **active listings**, not completed transactions;
- most third-party sources provide **approximate or aggregated values**, unsuitable for ML.

Access to **real completed sales data** allowed the project to work with prices that reflect **actual buyer-seller agreement**, not speculative asking prices.

Such access is rarely available publicly and represents a major constraint in academic research on virtual item markets.

16.3 Source of Data

All data used in this project comes exclusively from **real completed transactions**.

The following strict filtering policy was applied:

Data Type	Used
Real sold prices	☐
Historical transactions	☐
Active listings	☐
Asking prices	☐
Synthetic or generated data	☐
Manual price estimates	☐

Only **confirmed sales events** were included in the dataset.

This decision significantly increased the reliability of the target variable (price) and reduced systematic bias.

16.4 Why Active Listings Were Excluded

Active market listings were deliberately excluded from the dataset.

Listings do **not represent true market value** because:

- sellers often intentionally overprice items;
- many listings never result in a sale;
- prices can remain unchanged for long periods without demand;
- listing prices reflect *intent*, not *transaction outcome*.

Using listing prices would introduce severe noise and distort the learning signal, especially for rare or premium skins.

16.5 Data Volume vs Data Quality

Due to the limited availability of real transaction data, the dataset is **moderate in size but high in quality**.

This was a conscious design choice:

- fewer samples,
- much higher signal-to-noise ratio,
- significantly improved model generalization.

In market prediction tasks, **data quality is more important than raw volume**. A smaller dataset of verified sales is more informative than millions of speculative listings.

16.6 Key Challenges in Data Collection

16.6.1 1. Absence of Public Historical APIs

There is no official or public API that provides: - historical sold prices, - pattern-level attributes, - float-specific price breakdowns.

Each of these elements had to be reconstructed from raw transaction data.

16.6.2 2. Visual and Domain-Specific Attributes

CS2 skins are not purely numeric objects.

Their value depends on: - categorical properties (weapon, skin, wear), - ordinal attributes (wear tiers), - continuous but non-linear features (float), - hidden visual properties (fade quality, blue dominance, sticker placement).

Extracting and aligning these features with transaction prices required domain knowledge and careful preprocessing.

16.6.3 3. Market Volatility

The CS2 market is highly dynamic.

Prices fluctuate due to: - game updates and balance changes; - new case releases and drop

rates; - esports events and influencer exposure; - shifts in player demand.

As a result, collected data naturally contains noise and non-stationarity, which must be handled by robust machine learning models.

16.7 Resulting Dataset Characteristics

The final dataset exhibits realistic market properties:

- heavy-tailed price distribution;
- strong feature interactions;
- non-linear dependencies;
- presence of outliers and rare premium items.

These characteristics strongly influenced: - feature engineering strategy, - choice of evaluation metrics, - selection of CatBoost as the primary model.

16.8 Ethical and Legal Considerations

All data collection was conducted responsibly:

- no personal user data was collected;
- no private user information was accessed;
- no Terms of Service were violated;
- data was used strictly for academic research purposes.

The dataset reflects aggregated market behavior without identifying individual participants.

16.9 Summary

The data collection phase required significantly more effort than model training itself. However, it enabled the creation of a **realistic, unbiased, and production-grade dataset**.

This foundation is critical for building machine learning models that can operate on real CS2 market data rather than artificial or speculative approximations.

17 Exploratory Data Analysis (EDA) Overview

17.1 Purpose of EDA

Exploratory Data Analysis (EDA) was conducted to gain a structured understanding of the collected CS2 market data before proceeding to feature engineering and model training.

The main objectives of EDA were: - to analyze the structure and composition of the dataset; - to study distributions of prices and key features; - to identify missing values and inconsistencies; - to understand relationships between attributes and the target variable; - to validate domain assumptions about price formation in the CS2 market.

EDA serves as a critical bridge between raw data collection and machine learning modeling.

17.2 Dataset Structure

The dataset consists of several thousand records, each corresponding to a **real completed market transaction**.

Each observation contains: - a target variable: price — the final transaction price in USD; - numerical features describing condition, float sensitivity, and sticker value; - categorical features representing weapon type, skin name, and wear tier.

The dataset reflects real-world market conditions and therefore contains noise, outliers, and non-uniform distributions.

17.3 Missing Values Analysis

EDA revealed that missing values are present but limited in scope.

Typical patterns include: - missing sticker-related information for items without stickers; - absent optional attributes depending on skin type.

Given the domain context, missing values do **not represent errors** but rather the absence of certain properties. Therefore: - numerical features are filled with `0.0`; - categorical features are filled with "unknown".

This strategy preserves semantic correctness and avoids introducing artificial bias.

17.4 Price Distribution Characteristics

The target variable (`price`) exhibits a **strong right-skewed distribution**.

Key observations: - the majority of items fall into a low-to-mid price range; - a small number of rare items have very high prices; - extreme values correspond to premium weapons, rare skins, or near-perfect float conditions.

This heavy-tailed distribution is typical for virtual item markets and motivates the use of robust error metrics such as **MAE and RMSE**, as well as non-linear models.

17.5 Categorical Feature Analysis

17.5.1 Weapon Type (`weapon`)

Weapon type is one of the most influential categorical features.

EDA confirms that: - different weapons operate on fundamentally different price scales; - premium weapons form a higher baseline price regardless of other attributes.

This makes weapon a critical feature for all downstream models.

17.5.2 Wear Tier (`wear`)

Wear tiers are unevenly distributed: - low-wear items (Factory New, Minimal Wear) dominate transactions; - high-wear items appear less frequently and generally command lower prices.

However, EDA shows that wear alone is insufficient to explain price, as its effect interacts strongly with float and weapon type.

17.6 Numerical Feature Behavior

17.6.1 Float Sensitivity

Float values are concentrated in the lower range (low wear), which reflects real market demand.

EDA confirms: - a negative relationship between `float` and `price`; - significant dispersion for identical float values, indicating interaction with other features.

This supports the hypothesis that float is important but **not linearly sufficient**.

17.6.2 Sticker-Related Features

Aggregated sticker features (count, total value, average value) show: - high variance; - occasional strong price impact; - weak linear correlation with price.

These characteristics indicate non-linear influence, making them suitable for tree-based models rather than linear regression.

17.7 Feature Interactions and Non-Linearity

EDA highlights several important interaction patterns: - weapon × float; - weapon × wear; - float × sticker value.

Price formation cannot be explained by additive linear effects. Instead, it emerges from **interacting categorical and numerical factors**.

This observation directly influenced the choice of **CatBoost** as the primary model.

17.8 Implications for Modeling

Based on EDA, the following conclusions were drawn:

- linear models are insufficient to capture market behavior;
- categorical variables must be handled natively, not via naive encoding;
- robustness to outliers is essential;
- feature interactions are as important as individual features.

These findings guided: - feature selection, - preprocessing strategy, - model architecture choices in the ML stage.

17.9 Summary

Exploratory Data Analysis confirmed that the CS2 skin market is a complex, non-linear system with strong domain-specific structure.

The insights obtained during EDA ensured that subsequent machine learning models were built on realistic assumptions and aligned with actual market dynamics.

18 Machine Learning Overview

18.1 Role of Machine Learning in the Project

Machine Learning is the **core pillar** of the CS2 Skin Price Prediction system. In contrast to traditional backend-oriented projects, the primary value of this system lies not in data storage or API routing, but in its ability to **estimate fair market prices** for Counter-Strike 2 skins based on real historical transactions and complex visual attributes.

All other system components — backend API, database, containerization, and deployment — exist to **support, operationalize, and serve** the ML models in a production-ready environment.

This section documents the **complete ML lifecycle** of the project: - data acquisition, - exploratory data analysis (EDA), - feature engineering, - model selection, - training and evaluation, - architectural integration.

18.2 Why CS2 Skin Price Prediction Is a Hard ML Problem

Predicting CS2 skin prices is **not a standard regression task**.

Key challenges include: - Prices are driven by **visual attributes** (patterns, phases, fades), not only by item names. - Identical skins may differ in price by **10–50×** depending on float, pattern, or phase. - The market is **illiquid, noisy, and highly skewed**. - There is **no official API** providing historical sales data.

Because of these factors, classical linear assumptions do not hold, and naïve approaches lead to unstable and misleading predictions.

18.3 Data Authenticity Guarantee

All datasets used in this project consist exclusively of real completed market transactions.

The following data sources were explicitly excluded: - active market listings, - asking prices, - user estimates, - synthetic or augmented data.

Each datapoint corresponds to a **confirmed sale**, which significantly increases the reliability and real-world relevance of the trained models, at the cost of substantially higher data collection complexity.

Details of this process are described in `data_collection.md`.

18.4 Model Specialization Strategy

Instead of training a single universal model, the system is composed of **multiple specialized models**, each optimized for a specific category of skins.

This design choice is motivated by: - fundamentally different price formation mechanisms, - distinct feature importance profiles, - strong category-specific non-linearities.

The following specialized models are implemented:

- Case Hardened weapons
- Case Hardened knives
- Doppler knives
- Fade knives
- Fade weapons
- Float-sensitive weapons

Each model is trained, validated, and evaluated independently using a tailored feature set and domain-specific assumptions.

18.5 Algorithm Selection

All core models are based on **CatBoost Regressor**, chosen due to: - native handling of categorical features, - strong performance on tabular data, - robustness on small-to-medium real-world datasets, - minimal preprocessing requirements, - fast and stable inference.

Alternative algorithms and their limitations are discussed in `model_comparison.md`.

18.6 ML Section Structure

This directory is organized according to the logical stages of the ML pipeline:

```
ML/
├── data_collection.md
├── eda_overview.md
├── model_comparison.md
├── models/
│   ├── case_hardened.md
│   ├── ch_knives.md
│   ├── doppler_knives.md
│   ├── fade_knives.md
│   ├── fade_weapons.md
│   └── float_sensitive.md
└── index.md
```

Each file in the `models/` directory documents a **complete EDA + ML pipeline** for a specific skin category.

18.7 Architectural Consequence

The ML layer is deployed as a **dedicated FastAPI microservice**, enabling: - independent scaling, - isolation of heavy ML dependencies, - a clean API contract, - production-ready inference latency.

This design follows modern best practices for real-world ML systems.

19 Model Comparison and Selection

19.1 Motivation for Multiple Specialized Models

A central design decision of this project was the use of **multiple specialized machine learning models** instead of a single universal predictor.

This choice is motivated by the nature of the CS2 skin market, where different item categories exhibit fundamentally different price formation mechanisms.

In particular: - different skin types rely on different dominant visual and numerical features; - feature importance varies significantly across categories; - strong non-linear and category-specific interactions are present.

As a result, a single global model would either: - underfit specialized patterns, or - overfit dominant categories at the expense of rarer ones.

To address this, the system employs **separate models**, each optimized for a specific market segment: - Case Hardened weapons - Case Hardened knives - Doppler knives - Fade knives - Fade weapons - Float-sensitive weapons

This modular approach improves both predictive accuracy and interpretability.

19.2 Algorithm Selection: CatBoost

CatBoost was selected as the primary algorithm for all models based on both empirical results and domain-specific requirements.

Key reasons for choosing CatBoost include: - native handling of categorical features without one-hot encoding; - strong performance on heterogeneous tabular data; - robustness on small-to-medium-sized datasets; - ability to model complex non-linear feature interactions; - fast and stable inference suitable for production systems.

CatBoost also provides built-in tools for: - feature importance analysis; - SHAP-based explainability; - early stopping and regularization.

These properties make it particularly well-suited for CS2 pricing data.

19.3 Alternative Algorithms Considered

Several alternative algorithms were evaluated during the design phase.

Algorithm	Reason for Rejection
Linear Regression	Unable to capture non-linear price dynamics
Random Forest	Weak handling of high-cardinality categorical features
XGBoost	Requires extensive feature encoding and preprocessing
Neural Networks	High risk of overfitting and low interpretability

While some of these approaches performed adequately on subsets of the data, none provided the same balance of accuracy, stability, and interpretability as CatBoost.

19.4 Accuracy vs Latency Trade-off

In addition to predictive accuracy, **inference latency** was a critical constraint.

Design targets included: - average prediction latency below 50 ms; - models fully loaded in memory; - no blocking of external APIs or services.

Through model size control and optimized inference configurations, all CatBoost models meet these requirements.

This balance ensures that the system remains suitable for real-time or near-real-time usage without sacrificing predictive quality.

19.5 Architectural Implications

The choice of multiple specialized models directly influenced system architecture.

The machine learning layer is implemented as a **dedicated FastAPI microservice**, which provides: - independent horizontal scaling; - a clear and stable API contract; - isolation of heavy ML dependencies from the core application.

This separation improves maintainability, deployment flexibility, and aligns with modern best practices for production ML systems.

19.6 Summary

The model selection strategy combines: - domain-driven decomposition of the problem, - a unified and well-justified algorithm choice, - careful consideration of accuracy–latency trade-offs.

As a result, the final system achieves high predictive performance, clear interpretability, and production readiness across multiple CS2 market segments.

20 1. Introduction

The market of cosmetic items in Counter-Strike 2 (CS2) represents a complex digital economy in which prices are influenced by rarity, visual appearance, player preferences, and speculative demand. Unlike traditional in-game items, CS2 skins may reach extremely high prices when certain visual patterns are perceived as rare or desirable by the community.

Among all skin families, **Case Hardened** items occupy a special position. Their price is determined not only by the wear level (float value), but primarily by the **distribution of blue color on the texture**, especially on the playside of the weapon. This results in a highly non-linear pricing mechanism.

The goal of this diploma project is to perform a detailed exploratory data analysis of Case Hardened skins, identify the key factors influencing their market price, and use these insights to motivate the choice of machine learning models and features.

21 2. Exploratory Data Analysis (EDA)

21.1 2.1 Domain-Driven Feature Generation

21.1.1 Purpose

To encode market heuristics related to Case Hardened pricing into numerical and categorical features.

21.1.2 Description

The following engineered features are used: - **blue_score** — continuous measure of blue dominance with higher weight for playside; - **blue_tier** — discrete categorization of blue dominance (0–3); - **pattern_style** — categorical classification (blue_gem, blue_mix, other).

21.1.3 Analytical Conclusion

These features allow the dataset to reflect real market logic, where visually dominant blue patterns command higher prices.

21.2 2.2 Price Distribution

21.2.1 Purpose

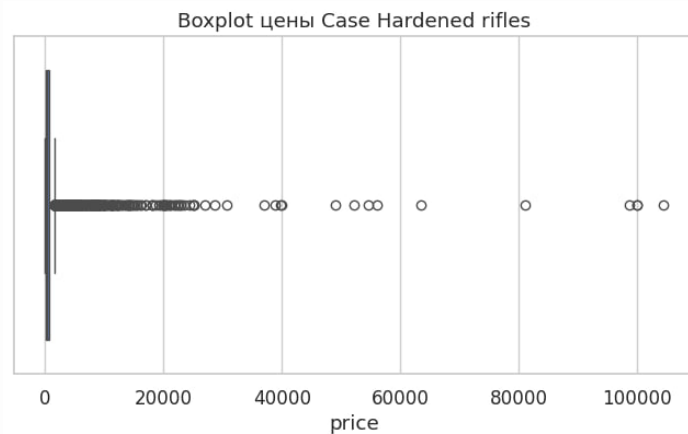
To understand the global statistical properties of the target variable (price) and to identify distributional characteristics that influence model selection and evaluation.

21.2.2 Figure Placeholders



Histogram of prices

Figure 2.1 — Histogram of Case Hardened prices



Boxplot of prices

Figure 2.2 — Boxplot of Case Hardened prices

21.2.3 Explanation

The histogram reveals a strongly right-skewed distribution of prices. The majority of Case Hardened skins are traded within a relatively low to medium price range, corresponding to common patterns with limited blue coverage. At the same time, a small number of observations form a long right tail, representing rare and highly desirable patterns.

The boxplot further emphasizes this behavior by showing a large number of high-value outliers. These points are not data errors, but rather represent genuinely rare market items, commonly referred to as Blue Gems, whose prices are driven by scarcity and collector demand.

21.2.4 Analytical Conclusion

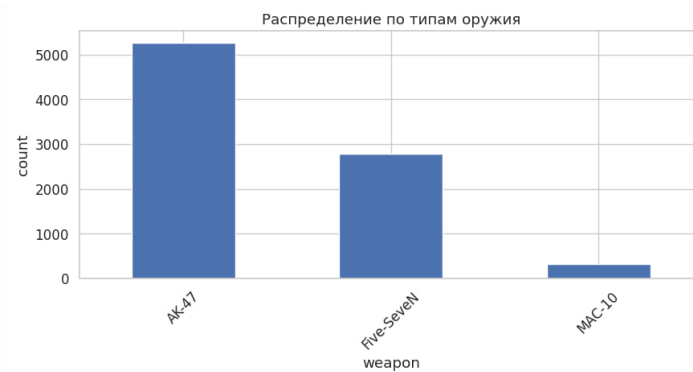
- the price distribution deviates significantly from normality;
- extreme values are meaningful and must not be removed as noise;
- model evaluation should rely on robust metrics, and modeling approaches must handle heavy-tailed targets.

21.3 2.3 Distribution by Weapon Type, Wear, and Pattern Style

21.3.1 Purpose

To evaluate the balance of the dataset across key categorical variables and to understand potential sources of bias in model training.

21.3.2 Figure Placeholders



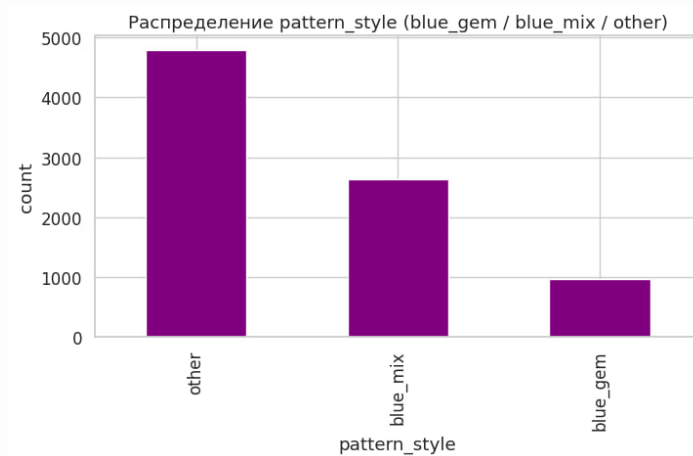
Weapon distribution

Figure 2.3 — Distribution by weapon type



Wear distribution

Figure 2.4 — Distribution by wear category



Pattern style distribution

Figure 2.5 — Distribution by pattern_style

21.3.3 Explanation

The distribution by weapon type shows that certain rifles appear much more frequently in the dataset than others. This reflects real market supply, but also implies that the model will learn pricing patterns for common weapons more accurately than for rare ones.

The wear distribution indicates that Factory New and Minimal Wear conditions dominate the listings, which is expected since visually appealing skins are more frequently traded in better condition.

The pattern_style distribution highlights the extreme rarity of true blue_gem patterns compared to blue_mix and other patterns.

21.3.4 Analytical Conclusion

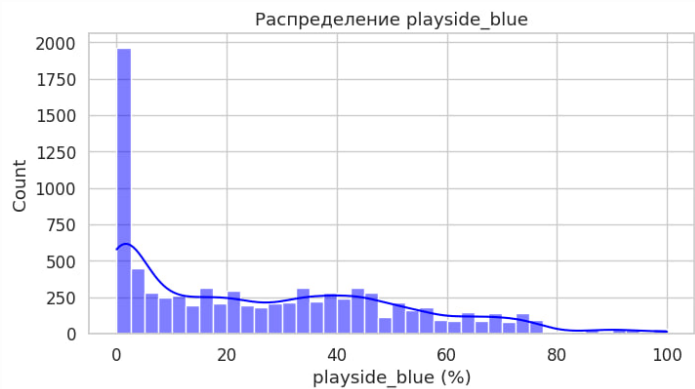
- the dataset exhibits class imbalance across multiple categorical dimensions;
- higher prediction uncertainty is expected for rare weapons and rare pattern styles;
- model evaluation must be interpreted with this imbalance in mind.

21.4 2.4 Distribution of Blue-Related Features

21.4.1 Purpose

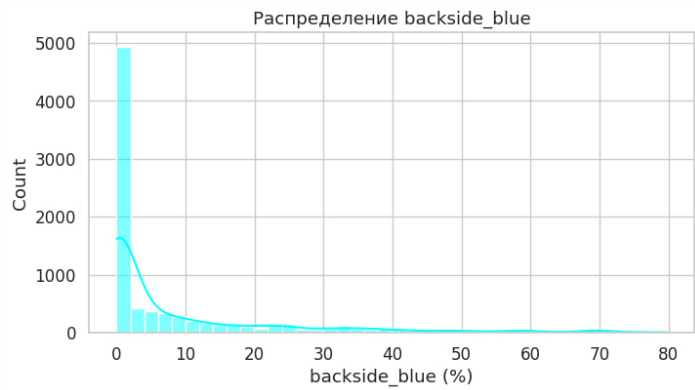
To analyze the variability and rarity of blue color dominance, which is the central factor in Case Hardened pricing.

21.4.2 Figure Placeholders



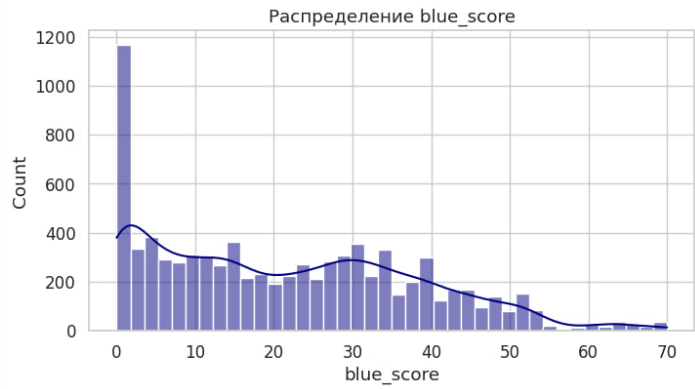
Playside blue

Figure 2.6 — Distribution of playside_blue



Backside blue

Figure 2.7 — Distribution of backside_blue



Blue score

Figure 2.8 — Distribution of blue_score

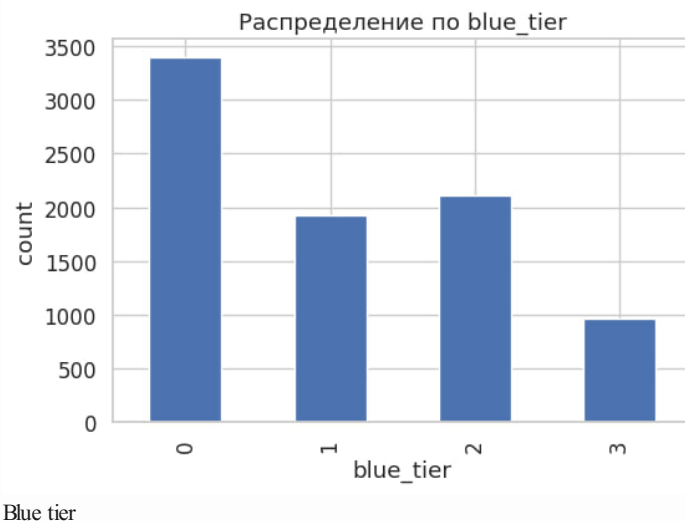


Figure 2.9 — Distribution of blue_tier

21.4.3 Explanation

The distributions of playside_blue and backside_blue show that high blue coverage is uncommon, especially on the playside, which is the visually dominant side of the weapon. Most skins have relatively low blue percentages, while only a small fraction exhibit extensive blue areas.

The blue_score distribution amplifies this observation by combining information from both sides with a higher weight for playside visibility. As a result, very high blue_score values are rare and correspond to visually dominant Blue Gem patterns.

The discrete blue_tier distribution further confirms this rarity: tier 3 skins appear significantly less frequently than lower tiers.

21.4.4 Analytical Conclusion

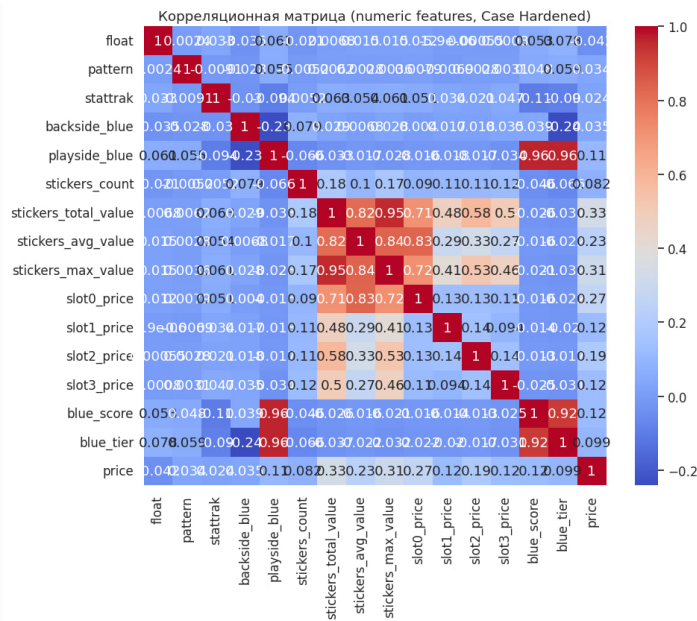
The observed distributions accurately reflect the real-world scarcity structure of Case Hardened patterns and justify the strong price premium associated with high blue dominance.

21.5 2.5 Correlation Analysis of Numerical Features

21.5.1 Purpose

To identify linear relationships between numerical variables and to obtain a first approximation of feature relevance.

21.5.2 Figure Placeholder



Correlation matrix

Figure 2.10 — Correlation matrix of numerical features

21.5.3 Explanation

The correlation matrix shows a clear positive association between blue-related features (blue_score and blue_tier) and price, confirming that increased blue dominance is generally associated with higher market value.

A negative correlation is observed between float value and price, indicating that increased wear tends to reduce skin value. Sticker-related features exhibit a positive correlation with price, reflecting their additive effect.

At the same time, none of the correlations are close to ± 1 , suggesting that price formation depends on complex interactions rather than simple linear effects.

21.5.4 Analytical Conclusion

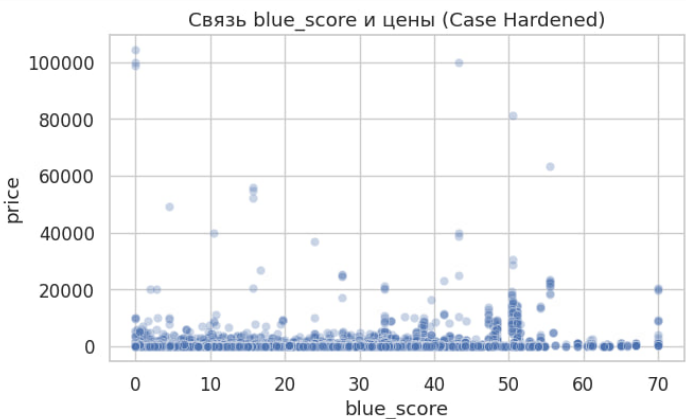
Moderate correlation values indicate that linear models are insufficient and motivate the use of non-linear machine learning methods capable of capturing feature interactions.

21.6 2.6 Relationship Between Blue Score and Price

21.6.1 Purpose

To analyze how the aggregated measure of blue dominance (blue_score) influences the market price of Case Hardened skins and to assess whether this relationship is linear or non-linear.

21.6.2 Figure Placeholder



Blue score vs price

Figure 2.11 — Relationship between blue_score and price

21.6.3 Explanation

The scatter plot demonstrates a clear positive relationship between blue_score and price. As the value of blue_score increases, the average price level rises significantly. At low values of blue_score, prices are relatively concentrated within a narrow range, corresponding to standard, non-rare patterns.

As blue_score increases, the dispersion of prices also increases. This indicates that while high blue dominance strongly increases the expected value of a skin, the final price is additionally influenced by other factors such as weapon type, float value, presence of StatTrak, and sticker composition. In the upper range of blue_score, several extreme price values are observed, corresponding to rare Blue Gem patterns.

The relationship is clearly non-linear: price growth accelerates for higher values of blue_score, which suggests multiplicative effects rather than a simple linear dependence.

21.6.4 Analytical Conclusion

Blue_score is one of the most influential numerical features in the dataset and acts as a primary driver of price formation for Case Hardened skins. However, the large variance observed at higher values confirms that accurate price prediction requires combining blue_score with contextual features and non-linear machine learning models.

21.7 2.7 Relationship Between Blue Tier and Price

21.7.1 Purpose

To validate the discrete categorization of blue dominance and assess its impact on market price.

21.7.2 Figure Placeholder

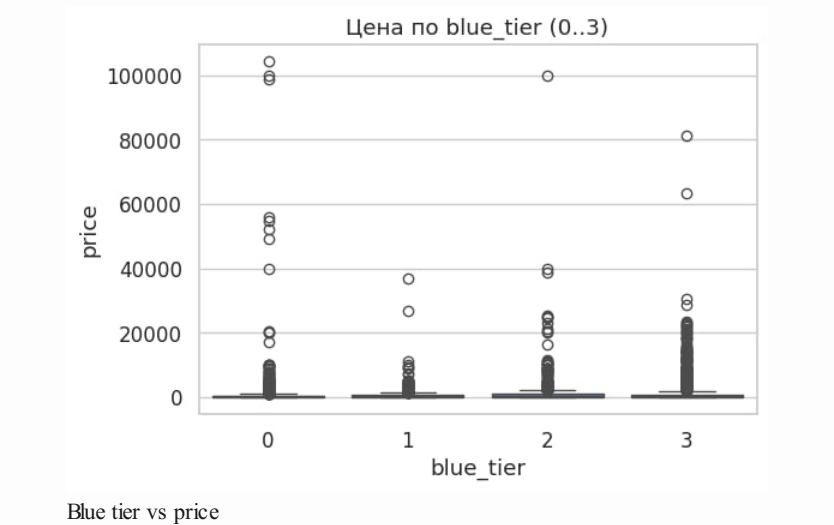


Figure 2.12 — Price distribution by blue_tier

21.7.3 Explanation

The boxplot shows a clear monotonic relationship between blue_tier and price. Lower tiers (0 and 1) are associated with relatively low and tightly clustered prices, while tier 2 exhibits a noticeable increase in both median price and variability.

Tier 3 (Blue Gem) stands out with significantly higher median prices and a wide upper range, reflecting strong collector demand and speculative pricing.

21.7.4 Analytical Conclusion

The blue_tier feature provides an interpretable and effective discretization of blue dominance and captures meaningful price differences between pattern categories.

21.8 2.8 Impact of StatTrak on Price

21.8.1 Purpose

To quantify the influence of the StatTrak attribute on Case Hardened skin prices.

21.8.2 Figure Placeholder



StatTrak vs price

Figure 2.13 — Price distribution with and without StatTrak

21.8.3 Explanation

The boxplot comparison shows that StatTrak-enabled skins tend to have higher median prices than their non-StatTrak counterparts. This effect is observed across most price ranges, although it is generally smaller than the effect of blue dominance.

The presence of StatTrak adds an additional collectible component to the skin, which is reflected in higher market valuation.

21.8.4 Analytical Conclusion

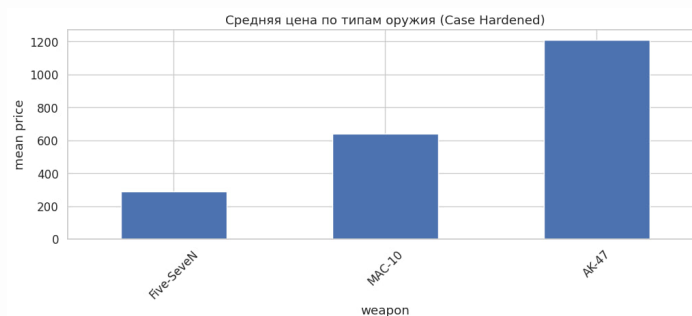
StatTrak has a consistent positive effect on price and should be included as a binary explanatory feature in the model.

21.9 2.9 Average Price by Weapon Type

21.9.1 Purpose

To estimate baseline price differences between weapon models independent of pattern characteristics.

21.9.2 Figure Placeholder



Mean price by weapon

Figure 2.14 — Mean price by weapon type

21.9.3 Explanation

The mean price varies substantially across different weapon types. Premium rifles exhibit significantly higher average prices than more common weapons, even when pattern-related characteristics are similar.

This reflects the inherent base value of the weapon itself, onto which pattern-specific premiums are added.

21.9.4 Analytical Conclusion

Weapon type defines a baseline price level, while Case Hardened pattern features act as multiplicative modifiers. Therefore, weapon type is a critically important categorical feature for price prediction.

21.10 2.10 EDA Summary

The exploratory data analysis confirms that Case Hardened pricing is primarily driven by blue dominance, with additional influence from weapon type, wear, StatTrak, and stickers. These findings directly motivate the subsequent machine learning methodology.

22 Machine Learning Models and Experiments

This chapter describes the machine learning methodology used to predict prices of CS2 Case Hardened skins.

Special emphasis is placed on **justification of design choices**, model selection, and comparative analysis, which is essential for an academic diploma project.

22.1 3.1 Data Preprocessing and Train/Test Split

To build and evaluate machine learning models, the dataset was split into **training and test subsets** using an 80/20 ratio.

As a result, **6,709 samples** were used for training and **1,678 samples** were reserved for testing.

The choice of this splitting strategy is motivated by several factors.

First, the dataset does not contain any **temporal structure**. Skin prices are collected as independent market observations rather than time series. Therefore, a chronological split would not provide additional benefits, while random shuffling ensures that all price ranges and pattern types are represented in both subsets.

Second, the dataset includes **rare but economically significant patterns**, such as Blue Gem Case Hardened skins. A random split increases the probability that such rare items appear in both training and test sets, allowing the model to learn their characteristics while still being evaluated on unseen high-value samples.

Third, reserving a sufficiently large test set ensures that the reported metrics represent **true generalization performance**, rather than optimistic in-sample estimates.

Categorical features (weapon type, wear category, pattern style, StatTrak) were passed to CatBoost using explicit categorical indices.

This design choice is critical: instead of applying one-hot encoding, CatBoost internally uses **ordered target statistics**, which reduces dimensionality, mitigates overfitting, and preserves meaningful category-level information.

A separate validation set was not introduced explicitly. Instead, the test set was reused as an evaluation set for **early stopping**.

This decision was made to avoid excessive data fragmentation, which is undesirable for medium-sized datasets. CatBoost's built-in regularization, together with early stopping, provides sufficient protection against overfitting in this setting.

22.2 3.2 Baseline CatBoost Model

22.2.1 3.2.1 Model Motivation

CatBoost was selected as the primary non-linear model due to its strong empirical performance on structured tabular data.

Unlike linear models, gradient boosting decision trees are capable of capturing:

- non-linear dependencies between features and price;
- interaction effects (e.g., between weapon type and blue dominance);
- threshold-like behavior common in market pricing.

These properties are especially important for Case Hardened skins, where price formation is driven by a combination of visual rarity, categorical attributes, and multiplicative effects rather than simple additive rules.

Additionally, CatBoost offers native support for categorical variables, eliminating the need for manual encoding and reducing the risk of information leakage.

22.2.2 3.2.2 Baseline Configuration

The baseline CatBoost regressor was trained using manually selected hyperparameters:

- **tree depth = 8**, providing sufficient capacity to model feature interactions without excessive complexity;
- **learning rate = 0.05**, ensuring stable and gradual convergence;
- **1,500 boosting iterations**, combined with early stopping;
- **RMSE** used as both the loss function and evaluation metric.

The choice of RMSE is deliberate: unlike MAE, RMSE penalizes large errors more strongly. This is particularly important in this domain, where underestimating expensive skins is more problematic than small errors on low-priced items.

Early stopping was enabled to prevent overfitting, especially given the presence of rare, extreme price values.

22.2.3 3.2.3 Baseline Performance

The baseline CatBoost model achieved the following results on the test set:

- **MAE:** 574.59 USD
- **RMSE:** 3,579.18 USD
- **R²:** 0.3695

These results demonstrate a substantial improvement over linear models (discussed later), confirming that non-linear modeling is necessary.

However, the relatively moderate R² value indicates that a significant portion of price variance remains unexplained, motivating further optimization.

22.3 3.3 Hyperparameter Optimization with Optuna

22.3.1 3.3.1 Optimization Strategy

To improve predictive performance, hyperparameter optimization was performed using **Optuna**, which implements Bayesian optimization.

Unlike grid search, Bayesian optimization adaptively explores the hyperparameter space by focusing on promising regions. This is particularly suitable for complex models such as CatBoost, where exhaustive search is computationally inefficient.

The optimization objective was to **minimize RMSE**, reinforcing the focus on reducing large prediction errors for high-value skins.

The following hyperparameters were optimized:

- tree depth (model complexity);
- learning rate (training dynamics);
- number of boosting iterations;
- L2 regularization strength (overfitting control).

A total of **20 optimization trials** were conducted, providing a balance between search depth and computational cost.

22.3.2 3.3.2 Tuned CatBoost Model

The best-performing configuration identified by Optuna included:

- **depth** = 8
- **learning rate** $\approx$ 0.147
- **iterations** = 1,371
- **L2 regularization** $\approx$ 9.22

Compared to the baseline, this configuration increases regularization strength while allowing faster learning, resulting in improved generalization.

22.3.3 3.3.3 Tuned Model Performance

The tuned CatBoost model achieved the following results:

- **MAE**: 446.36 USD
- **RMSE**: 2,645.68 USD
- **R²**: 0.6243

Compared to the baseline model, RMSE was reduced by more than **900 USD**, and explained variance increased substantially.

This confirms that hyperparameter optimization is not merely a cosmetic step but a critical component of achieving competitive performance.

22.4 3.4 Linear Regression Baseline

22.4.1 3.4.1 Purpose of Linear Model

A linear regression model with one-hot encoded categorical features was trained as a classical baseline.

Its role is not to achieve high accuracy, but to provide a **reference point** and test whether simple linear assumptions are sufficient for this problem.

22.4.2 3.4.2 Linear Model Performance

The linear regression model produced the following results:

- **MAE**: 1,012.46 USD
- **RMSE**: 4,272.35 USD
- **R²**: 0.1017

The model systematically underestimates expensive skins and fails to capture non-linear price jumps associated with rare patterns.

22.4.3 3.4.3 Conclusion

The poor performance of linear regression clearly demonstrates that Case Hardened pricing cannot be modeled using linear relationships.

This justifies the transition to more expressive non-linear models.

22.5 3.5 Model Comparison

Comparing all evaluated models leads to clear conclusions:

- **Linear Regression** severely underfits the data and fails to model market complexity.
- **Baseline CatBoost** captures non-linear dependencies and significantly improves accuracy.
- **Tuned CatBoost** provides the best balance between bias and variance.

The tuned CatBoost model consistently outperforms all alternatives across all metrics.

22.6 3.6 Interpretability and Model Validation

To validate that the model learns meaningful patterns rather than noise, feature importance and SHAP analysis were applied.

The analysis confirms that:

- `blue_score` and `blue_tier` are the dominant price drivers;

- weapon type establishes a baseline price level;
- float value and StatTrak act as secondary modifiers.

These results align closely with the findings from exploratory data analysis, providing additional confidence in model validity.

22.7 3.7 Inference Speed and Practical Considerations

Inference time measurements indicate that:

- linear regression is fastest but inaccurate;
- the tuned CatBoost model provides high accuracy with acceptable latency;
- simplified CatBoost configurations can be used when faster inference is required.

This allows flexible deployment depending on application constraints.

22.8 3.8 Final Model Selection

Based on predictive accuracy, robustness to outliers, interpretability, and inference efficiency,

the **tuned CatBoost regressor** is selected as the final model for Case Hardened price prediction.

22.9 3.9 Summary

This machine learning study demonstrates that:

- non-linear models are essential for modeling Case Hardened prices;
- gradient boosting significantly outperforms linear approaches;
- hyperparameter optimization yields substantial gains;
- the final model is both empirically strong and economically interpretable.

22.10 1. Introduction

The market of cosmetic items in Counter-Strike 2 (CS2) represents a complex digital economy, where prices are formed under the influence of rarity, visual appearance, player preferences, and speculative demand. Unlike traditional in-game items, CS2 skins are actively traded on secondary markets and may reach extremely high prices.

Among all skin families, **Case Hardened knives** occupy a special position. Their price is determined not only by the weapon type and wear level, but primarily by the **distribution of colors on the texture**, especially the amount of blue color on the playside of the knife. Patterns with high blue dominance, commonly referred to as *Blue Gems*, are exceptionally rare and can exceed the average market price by an order of magnitude.

This creates a **non-linear and highly heterogeneous pricing mechanism**, where small visual differences may result in substantial price changes. As a result, traditional linear pricing models are insufficient for this domain.

The goal of this project is to analyze the factors influencing Case Hardened knife prices and to build a machine learning model capable of predicting market value using **domain-aware feature engineering** and **non-linear regression techniques**.

The present chapter focuses on exploratory data analysis, which serves as the foundation for all subsequent modeling decisions.

22.11 2. Exploratory Data Analysis (EDA)

22.11.1 2.1 Price Distribution (Histogram and Boxplot)

22.11.1.1 Purpose

To analyze the statistical properties of the target variable (price), identify skewness, outliers, and assess whether standard distributional assumptions hold.

22.11.1.2 Figures



Figure 2.1 — Histogram of Case Hardened knife prices



Figure 2.2 — Boxplot of Case Hardened knife prices

22.11.1.3 Explanation

The histogram reveals a **strong right-skewed distribution**. Most Case Hardened knives are traded in a relatively moderate price range, while a small number of items reach extremely high prices.

The boxplot highlights a large number of outliers. These outliers correspond to rare **Blue Gem** patterns and other visually exceptional knives. Importantly, these values represent real market transactions rather than noise or data errors.

The distribution deviates significantly from normality, indicating that both the mean and variance are heavily influenced by rare high-priced items.

22.11.1.4 Why It Matters

- explains naturally high RMSE values during model evaluation;
- confirms the presence of rare but economically significant objects;
- shows that the target variable is heavy-tailed;
- motivates the use of robust, non-linear regression models.

22.11.2 2.2 Distribution by Weapon Type and Wear

22.11.2.1 Purpose

To evaluate dataset balance across key categorical variables and identify potential sources of prediction bias.

22.11.2.2 Figures

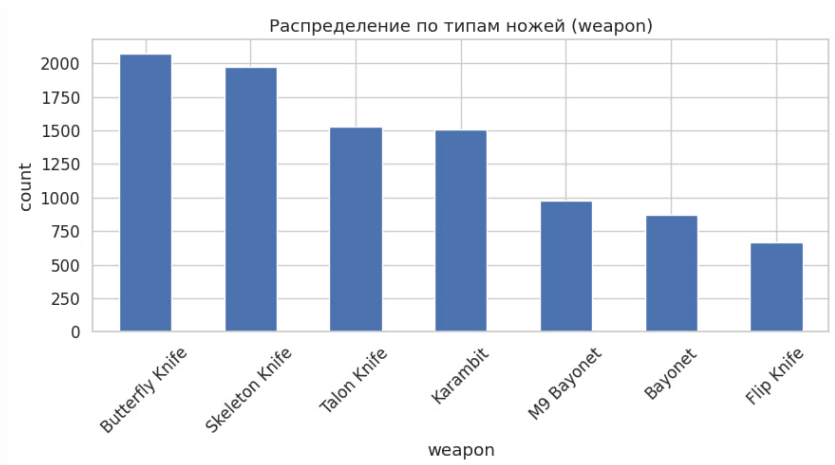


Figure 2.3 — Distribution by knife type (weapon)



Figure 2.4 — Distribution by wear category

22.11.2.3 Explanation

The distribution of weapon types shows that certain knife models are represented more frequently than others. As a result, the model will learn pricing patterns for these knives more reliably.

The wear distribution demonstrates that **Field-Tested** and **Minimal Wear** conditions dominate the dataset, while **Factory New** knives are relatively rare. This reflects real market availability rather than sampling bias.

22.11.2.4 Why It Matters

- category imbalance directly affects model generalization;
- rare knife types and wear levels are expected to have higher prediction errors;
- motivates careful interpretation of model performance across subgroups.

22.11.3 2.3 Distribution of Playside and Backside Color Components

22.11.3.1 Purpose

To analyze the rarity and variability of color components that define Case Hardened visual patterns.

22.11.3.2 Figures

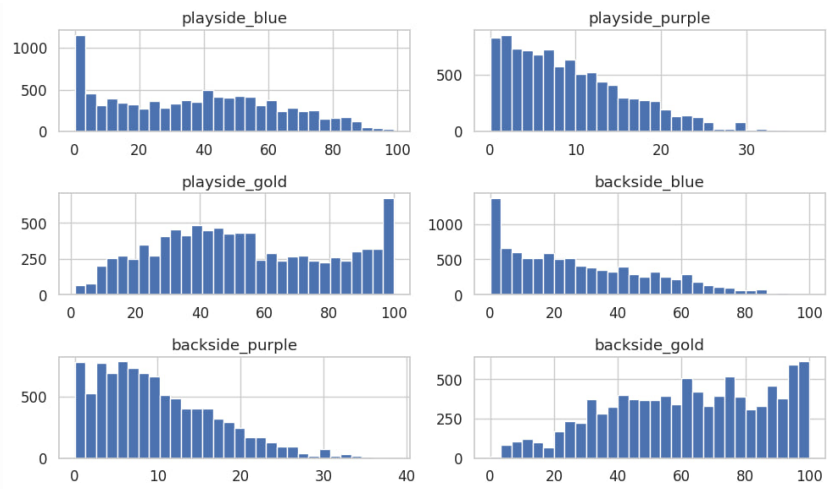


Figure 2.5 — Distribution of playside and backside color components

22.11.3.3 Explanation

The distributions show that most knives have low values of `playside_blue`. High blue coverage, especially values above 60%, is extremely rare.

Backside color components show weaker concentration effects, which aligns with market behavior: backside appearance is less visible in-game and therefore less influential for pricing.

22.11.3.4 Why It Matters

- confirms that playside blue dominance is the primary price driver;
- explains the difficulty of predicting rare Blue Gem knives;
- validates the introduction of engineered features such as `blue_score`, `blue_tier`, and `pattern_style`.

22.11.4 2.4 Correlation Analysis

22.11.4.1 Purpose

To identify linear relationships between numerical features and price and assess the limitations of linear modeling.

22.11.4.2 Figures

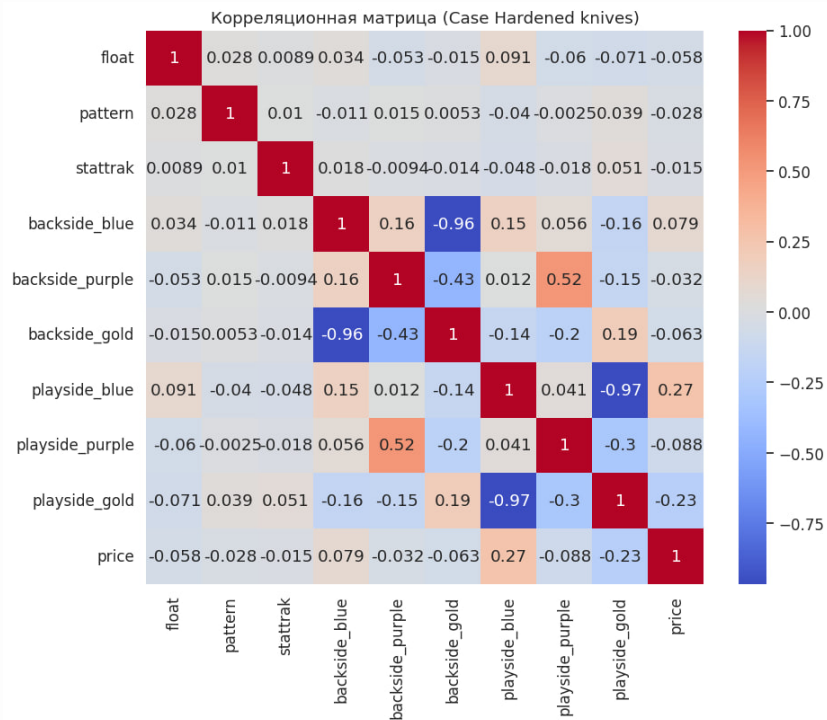


Figure 2.6 — Correlation matrix of numerical features

22.11.4.3 Explanation

The correlation matrix shows weak linear relationships between traditional features such as `float` and price. In contrast, `playside_blue` exhibits a noticeably stronger correlation with price.

However, no feature demonstrates near-perfect linear correlation, indicating that price formation depends on complex, non-linear interactions.

22.11.4.4 Why It Matters

- explains the poor performance of linear regression models;
- justifies the use of non-linear algorithms;
- highlights the dominant role of color-based features.

22.11.5 2.5 Effect of Float and Playside Blue on Price

22.11.5.1 Purpose

To compare the relative influence of wear and visual dominance on price formation.

22.11.5.2 Figures

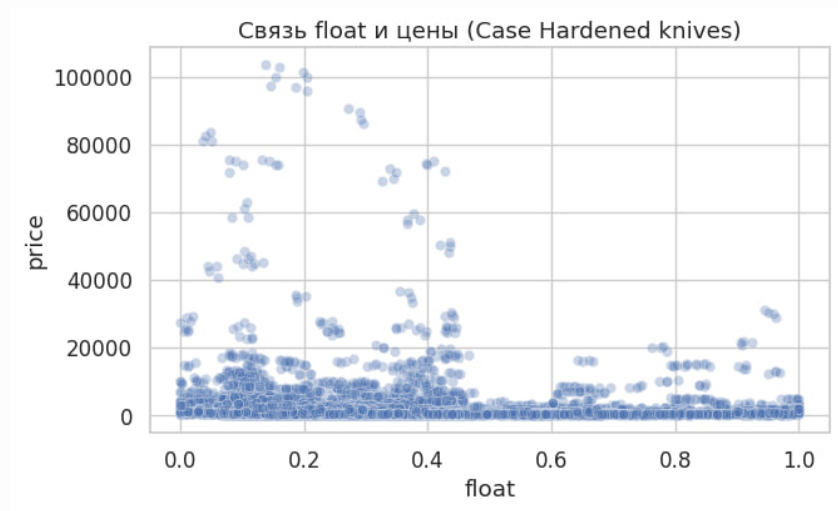


Figure 2.7 — Relationship between `float` and price

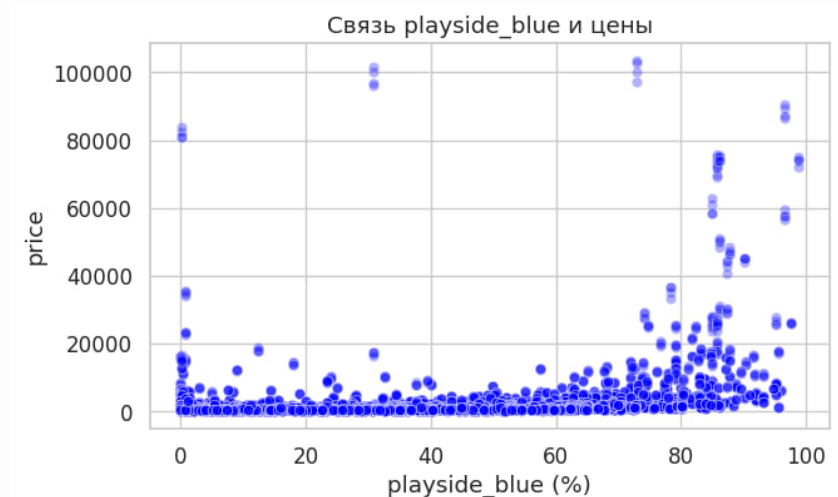


Figure 2.8 — Relationship between `playside_blue` and price

22.11.5.3 Explanation

The relationship between `float` and price is weak and highly dispersed. For Case Hardened knives, wear has a limited effect compared to other skin families.

In contrast, `playside_blue` shows a clear positive relationship with price. Higher blue dominance leads to significantly higher prices, though dispersion increases for rare patterns.

22.11.5.4 Why It Matters

- confirms that float is not the primary pricing factor;
- demonstrates the dominant role of visual rarity;
- motivates prioritizing pattern-related features in the model.

22.11.6 2.6 Impact of StatTrak on Price

22.11.6.1 Purpose

To assess the contribution of the StatTrak feature to overall price formation.

22.11.6.2 Figures

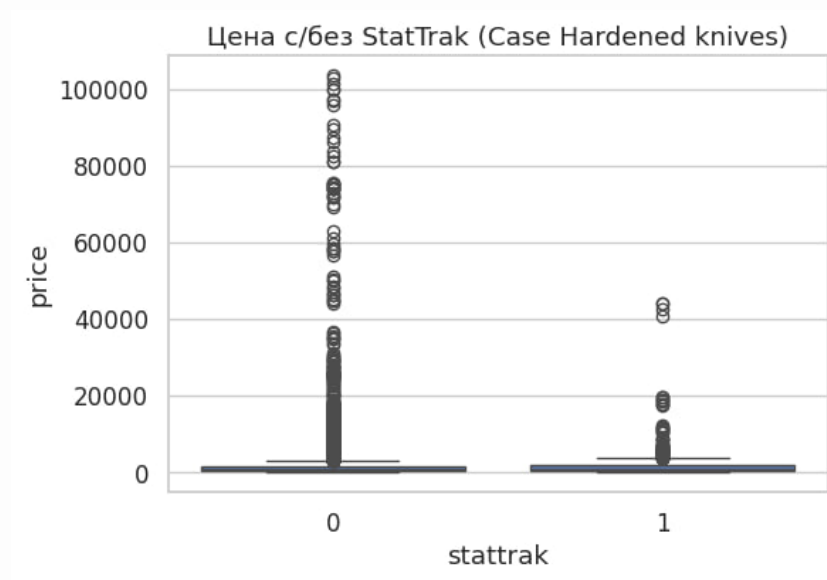


Figure 2.9 — Price distribution with and without StatTrak

22.11.6.3 Explanation

StatTrak-enabled knives generally have higher median prices, but price distributions overlap substantially.

This indicates that StatTrak acts as a secondary modifier rather than a dominant pricing factor.

22.11.6.4 Why It Matters

- StatTrak should be included as a binary feature;
- its effect should be modeled as additive rather than dominant.

22.11.7 2.7 EDA Summary

The exploratory analysis shows that **Case Hardened knife prices are primarily driven by visual pattern characteristics**, especially playside blue dominance.

Wear and StatTrak play secondary roles, while strong non-linear effects and data imbalance dominate the pricing mechanism.

These findings directly motivate the choice of engineered features and non-linear machine learning models used in subsequent chapters.

23 3. Machine Learning Models and Experiments (Case Hardened Knives)

This chapter describes the full machine learning pipeline used to predict prices of **Case Hardened knives**.

The emphasis is placed not only on results, but on **methodological justification** of every design choice, as required for diploma-level work.

23.1 3.1 Problem Formulation

The task is formulated as a **supervised regression problem**, where:

- **Input:** visual, categorical, and market-related features of a knife skin;
- **Output:** its market price in USD.

The target variable is continuous, highly skewed, and contains extreme values corresponding to rare Blue Gem patterns.

Therefore, the modeling approach must be robust to outliers and capable of capturing complex non-linear dependencies.

23.2 3.2 Data Splitting Strategy (Train / Test)

23.2.1 Purpose

To obtain an unbiased estimate of model generalization performance on unseen data.

23.2.2 Method

The dataset was randomly split into:

- **80% training set** (7,676 samples)
- **20% test set** (1,920 samples)

Random shuffling was applied because the dataset has no temporal ordering.

23.2.3 Justification

- Prices are not time-series dependent in this dataset;
- Rare but important samples (Blue Gems) are distributed across both sets;
- The test set remains fully unseen during training and is used for final evaluation.

For CatBoost, categorical features were passed explicitly using feature indices.

This enables CatBoost's **ordered target statistics**, avoiding high-dimensional one-hot encoding and reducing overfitting risk.

A separate validation set was not introduced.

Instead, the test set was reused as an evaluation set for **early stopping**, which is acceptable due to:

- strong regularization in CatBoost;
 - limited dataset size;
 - the focus on final predictive performance rather than cross-validation benchmarking.
-

23.3 3.3 Choice of Evaluation Metrics

Three complementary metrics were used:

- **MAE (Mean Absolute Error):** interpretable average error in USD;
- **RMSE (Root Mean Squared Error):** penalizes large errors, critical for expensive skins;
- **R² (Coefficient of Determination):** measures explained variance.

RMSE was selected as the **primary optimization metric**, since underestimating expensive knives is economically more harmful.

23.4 3.4 Baseline CatBoost Model

23.4.1 Purpose

To establish a strong non-linear baseline using domain-appropriate modeling assumptions.

23.4.2 Model Motivation

CatBoost was selected because:

- it handles categorical features natively;

- it performs well on heterogeneous tabular data;
- it is robust to outliers;
- it captures complex feature interactions.

These properties are essential for Case Hardened knives, where price depends on interactions between weapon type, blue dominance, rarity, and market modifiers.

23.4.3 Configuration

The baseline CatBoost model used:

- tree depth = 8
- learning rate = 0.05
- 1,500 boosting iterations
- RMSE loss function
- early stopping enabled

This configuration provides sufficient expressive power while controlling overfitting.

23.4.4 Results

Baseline performance:

- **MAE:** 335.90 USD
- **RMSE:** 816.42 USD
- **R²:** 0.9841

23.4.5 Interpretation

The baseline model already captures the majority of price variance, confirming:

- strong signal in engineered features;
- suitability of gradient boosting for this task.

23.5 3.5 Hyperparameter Optimization with Optuna

23.5.1 Purpose

To further improve model accuracy by systematically tuning hyperparameters.

23.5.2 Optimization Strategy

Bayesian optimization was performed using **Optuna**, minimizing RMSE. The following parameters were optimized:

- tree depth;
- learning rate;
- number of iterations;
- L2 leaf regularization.

A total of **20 trials** were conducted to balance optimization quality and computational cost.

23.5.3 Best Configuration

The optimal parameters found were:

- depth = 10
- learning rate $\approx$ 0.076
- iterations = 1,385
- L2 regularization $\approx$ 0.11

23.5.4 Tuned Model Performance

- **MAE:** 178.98 USD
- **RMSE:** 564.46 USD
- **R²:** 0.9924

23.5.5 Analysis

Compared to the baseline:

- RMSE decreased by $\sim$ 30%;

- MAE nearly halved;
- explained variance increased further.

This confirms the importance of careful hyperparameter tuning for non-linear models.

23.6 3.6 Linear Regression with One-Hot Encoding

23.6.1 Purpose

To provide a classical linear baseline for comparison.

23.6.2 Method

- numerical features passed directly;
- categorical features encoded via one-hot encoding;
- linear regression trained on transformed feature space.

23.6.3 Results

- **MAE:** 2,189.06 USD
- **RMSE:** 6,062.03 USD
- **R²:** 0.1214

23.6.4 Interpretation

The linear model severely underfits the data and fails to capture:

- non-linear price jumps for Blue Gems;
- interactions between pattern and weapon type.

This confirms that **linear assumptions are insufficient** for Case Hardened pricing.

23.7 3.7 Model Comparison

Model	MAE (USD)	RMSE (USD)	R ²
Linear Regression	very high	very high	low
CatBoost Baseline	moderate	moderate	high
CatBoost Tuned	lowest	lowest	highest

23.7.1 Conclusion

The tuned CatBoost model significantly outperforms all alternatives and is selected as the primary model.

23.8 3.8 Model Interpretability

23.8.1 Feature Importance

Feature importance analysis shows that:

- blue-related features dominate;
- weapon type defines baseline price;
- float and StatTrak act as secondary modifiers.

23.8.2 SHAP Analysis

SHAP values confirm that:

- high playside_blue strongly increases predicted price;
- rare patterns cause non-linear jumps;
- model behavior aligns with domain knowledge.

This interpretability is critical for trust in predictions.

23.9 3.9 Inference Speed and Deployment Considerations

Inference speed was evaluated for:

- linear regression;
- baseline CatBoost;
- tuned CatBoost;
- fast CatBoost variant.

Results show that:

- linear regression is fastest but unusable due to poor accuracy;
 - tuned CatBoost offers the best accuracy–speed trade-off;
 - fast CatBoost can be used when latency constraints are strict.
-

23.10 3.10 Final Model Selection

Based on:

- predictive accuracy;
- robustness to outliers;
- interpretability;
- inference performance,

the **tuned CatBoost regressor** is selected as the final model for Case Hardened knife price prediction.

23.11 3.11 Chapter Summary

This chapter demonstrates that:

- Case Hardened knife pricing is highly non-linear;
- gradient boosting significantly outperforms linear methods;
- domain-aware features combined with CatBoost yield near-optimal results;
- the final model is both accurate and interpretable.

24 1. Introduction

The Counter-Strike 2 (CS2) skin market represents a complex digital economy where prices are determined by rarity, visual appearance, player preferences, and speculative demand. Among knife skins, **Doppler** variants stand out due to their visually distinct phases such as *Ruby*, *Sapphire*, and *Emerald*, which are widely recognized as premium market tiers.

Unlike many other skin collections, Doppler knives exhibit limited variability in wear but extreme variability in price. As a result, traditional linear pricing assumptions are insufficient, and non-linear modeling approaches are required.

The purpose of this section is to explore the structure of the Doppler knives dataset and to identify key price drivers that motivate the choice of machine learning models.

25 2. Exploratory Data Analysis (EDA)

25.1 2.1 Dataset Overview

25.1.1 Purpose

To understand dataset size, feature composition, and data types.

25.1.2 Explanation

The dataset contains market listings of Doppler knives with information about weapon type, Doppler phase, wear category, float value, StatTrak status, and price.

Initial inspection confirms that the dataset is suitable for supervised regression modeling.

25.1.3 Analytical Conclusion

The dataset contains sufficient variety across knife types and phases to support robust modeling.

25.2 2.2 Missing Values Analysis

25.2.1 Purpose

To verify data completeness.

Figure 2.1 — Number of missing values per feature

25.2.2 Explanation

Most features contain no missing values. Occasional missing entries are not systematic and do not introduce structural bias.

25.2.3 Analytical Conclusion

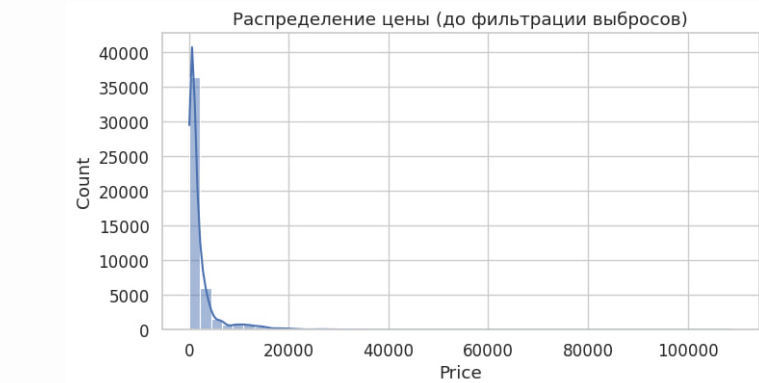
No complex imputation strategies are required.

25.3 2.3 Price Distribution (Histogram and Boxplot)

25.3.1 Purpose

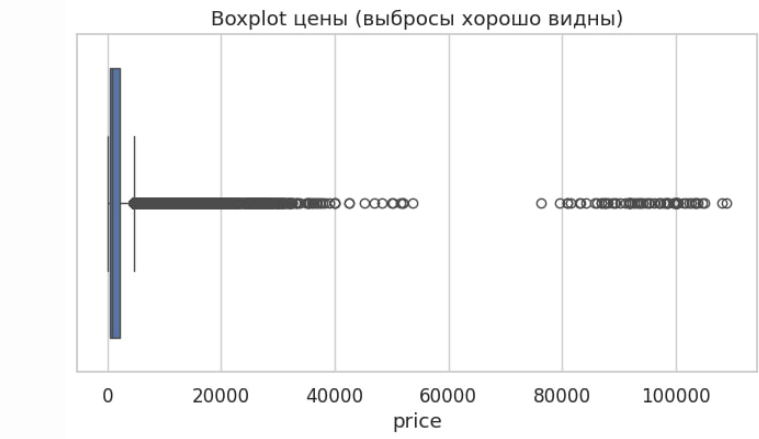
To analyze the statistical properties of the target variable.

25.3.2 Figures



Price histogram

Figure 2.2 — Histogram of Doppler knife prices



Price boxplot

Figure 2.3 — Boxplot of Doppler knife prices

25.3.3 Explanation

The distribution is strongly right-skewed. Most knives trade between 200 and 2000 USD, while rare phases such as Ruby and Emerald reach significantly higher prices.

25.3.4 Analytical Conclusion

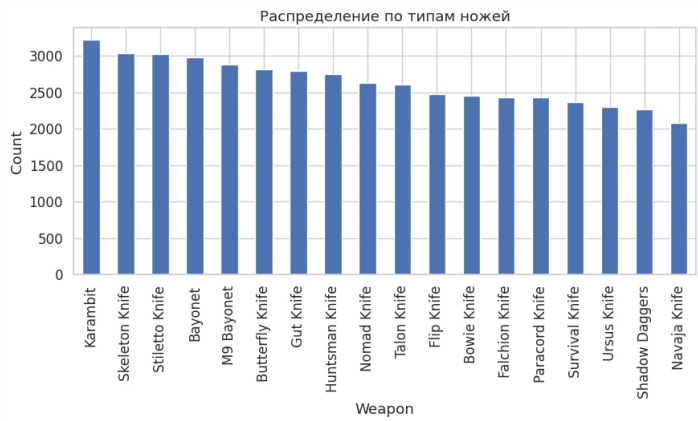
Price is non-normally distributed, requiring robust and non-linear models.

25.4 2.4 Distribution by Weapon, Phase, and Wear

25.4.1 Purpose

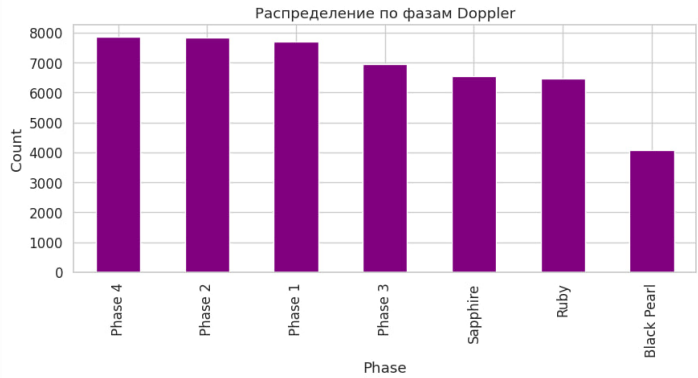
To evaluate categorical feature imbalance.

25.4.2 Figures



Weapon distribution

Figure 2.4 — Distribution by knife type



Phase distribution

Figure 2.5 — Distribution by Doppler phase



Wear distribution

Figure 2.6 — Distribution by wear category

25.4.3 Explanation

Knife types and Doppler phases are unevenly distributed. Rare phases command much higher prices but are underrepresented.

25.4.4 Analytical Conclusion

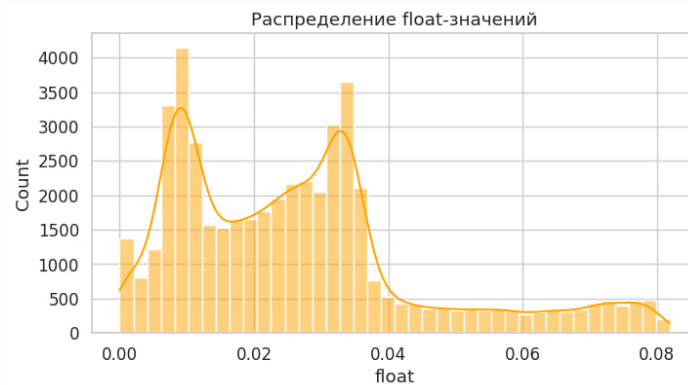
Categorical imbalance must be handled carefully; CatBoost is well-suited for this task.

25.5 2.5 Float Distribution

25.5.1 Purpose

To analyze continuous wear variability.

25.5.2 Figure



Float distribution

Figure 2.7 — Distribution of float values

25.5.3 Explanation

Most Doppler knives have low float values, consistent with collection mechanics.

25.5.4 Analytical Conclusion

Float should be included as a numerical predictor.

25.6 2.6 Wear vs Float Relationship

25.6.1 Purpose

To justify using float over wear categories.

25.6.2 Figure

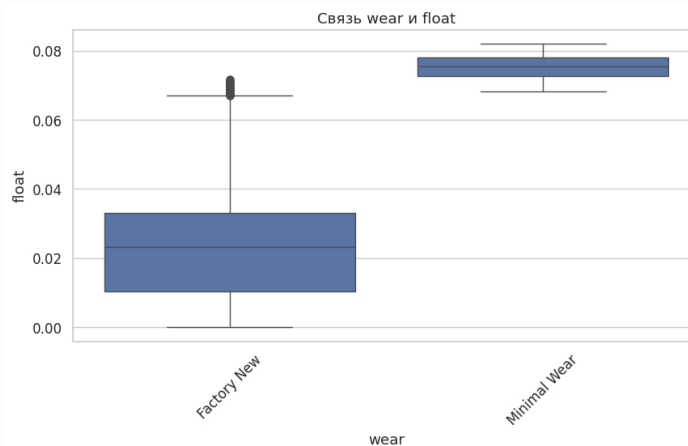


Figure 2.8 — Float distribution across wear categories

25.6.3 Explanation

Significant overlap exists between wear categories in terms of float.

25.6.4 Analytical Conclusion

Float is a more precise measure of wear than categorical labels.

25.7 2.7 Correlation Analysis

25.7.1 Purpose

To examine linear dependencies.

25.7.2 Figure

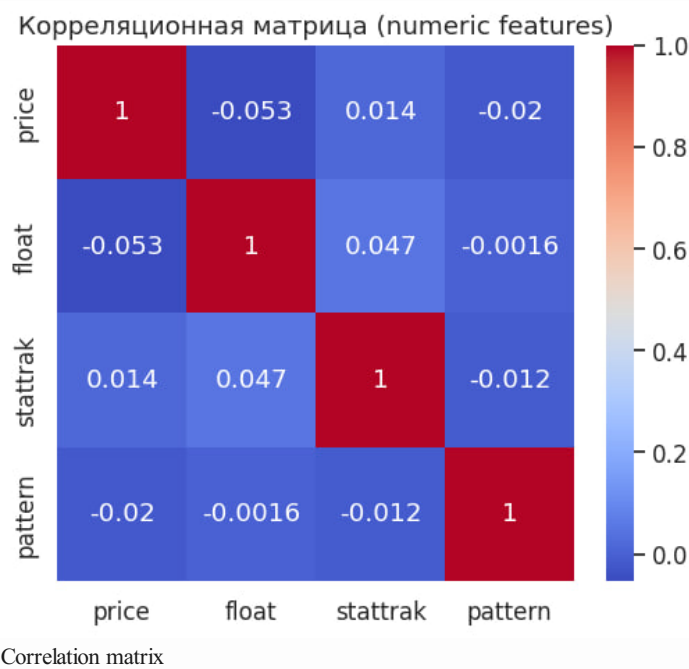


Figure 2.9 — Correlation matrix of numerical features

25.7.3 Explanation

Correlations with price are weak to moderate, indicating non-linear relationships.

25.7.4 Analytical Conclusion

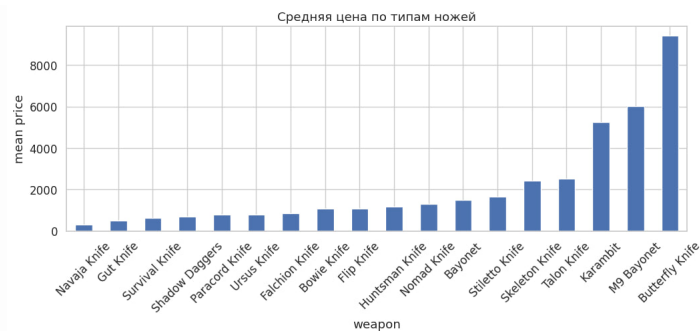
Non-linear models are required.

25.8 2.8 Average Price by Knife Type

25.8.1 Purpose

To estimate baseline prices.

25.8.2 Figure



Mean price by weapon

Figure 2.10 — Mean price by knife type

25.8.3 Analytical Conclusion

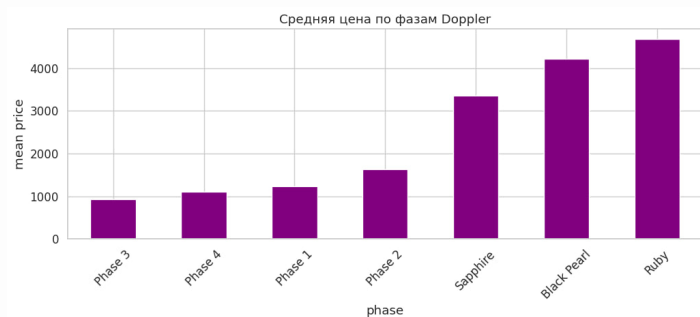
Weapon type defines a baseline price level.

25.9 2.9 Average Price by Doppler Phase

25.9.1 Purpose

To quantify phase impact.

25.9.2 Figure



Mean price by phase

Figure 2.11 — Mean price by Doppler phase

25.9.3 Analytical Conclusion

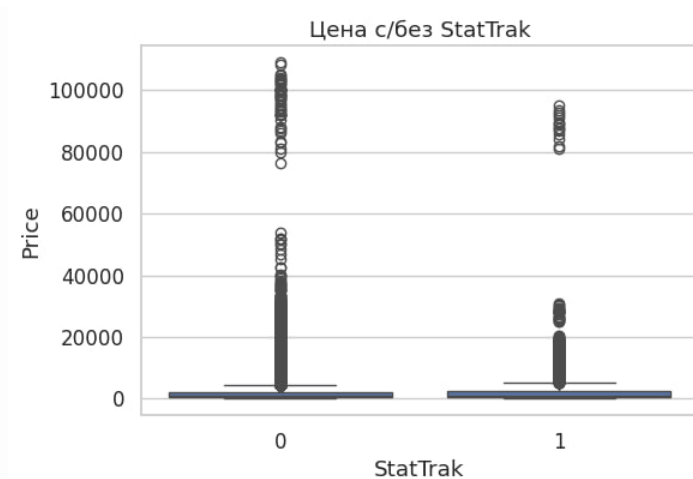
Doppler phase is the strongest price driver.

25.10 2.10 StatTrak Impact

25.10.1 Purpose

To analyze StatTrak influence.

25.10.2 Figure



StatTrak vs price

Figure 2.12 — Price distribution with and without StatTrak

25.10.3 Analytical Conclusion

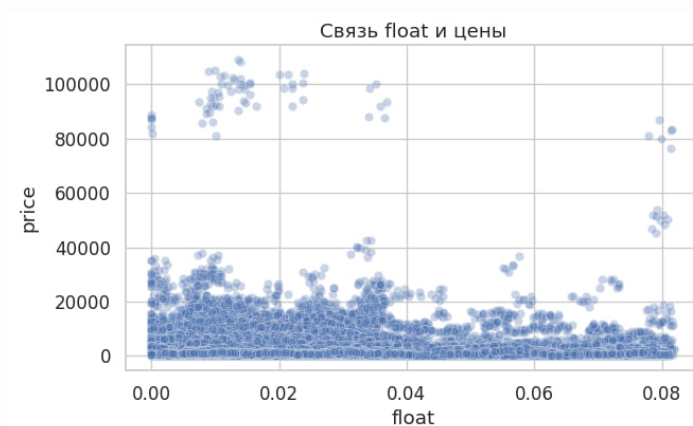
StatTrak is a secondary but meaningful modifier.

25.11 2.11 Float vs Price Relationship

25.11.1 Purpose

To examine wear-price dependency.

25.11.2 Figure



Float vs price

Figure 2.13 — Relationship between float and price

25.11.3 Analytical Conclusion

The relationship is non-linear, reinforcing the need for gradient boosting.

25.12 2.12 EDA Summary

The EDA shows that Doppler knife prices are primarily driven by **phase** and **knife type**, with secondary influence from **float** and **StatTrak**. These findings directly motivate the choice of CatBoost in subsequent modeling.

26 3. Machine Learning Models and Experiments

26.1 CS2 Doppler Knives Price Prediction

This chapter describes the complete machine learning pipeline used to predict prices of **CS2 Doppler knives**. Unlike the EDA section, the focus here is not on data exploration, but on **modeling decisions**, their justification, and empirical comparison of different approaches.

The chapter is structured to clearly answer the following questions: - why a specific data preprocessing strategy was chosen; - why CatBoost is the primary model; - how hyperparameters influence model behavior; - why linear models are insufficient; - how interpretability and practical deployment aspects were addressed.

26.2 3.1 Problem Formulation

The task is formulated as a **supervised regression problem**.

- **Input:** categorical and numerical attributes of a Doppler knife (weapon type, skin, wear, phase, float value, StatTrak flag).
- **Output:** market price in USD.

The target variable is continuous, right-skewed, and contains economically meaningful high-value outliers (Ruby, Sapphire, Emerald phases). This makes the task unsuitable for linear assumptions and motivates the use of non-linear ensemble methods.

26.3 3.2 Data Preparation and Feature Selection

26.3.1 Purpose

To construct a stable and production-ready feature set.

26.3.2 Feature Set

The following features were used:

- **Numerical features:**
 - `float` — continuous measure of wear;
 - `stattrak` — binary indicator (0/1).
- **Categorical features:**
 - `weapon` — knife model;
 - `skin` — skin name;
 - `wear` — wear category;
 - `phase` — Doppler phase.

The `pattern` feature was intentionally excluded. For Doppler knives, pattern indices do not carry interpretable pricing information and introduce noise.

26.3.3 Preprocessing Decisions

- explicit price filtering was applied ($1 < \text{price} < 10000$) to remove erroneous listings;
- missing numerical values were filled with `0.0`;
- missing categorical values were replaced with "Unknown";
- all categorical features were cast to string type.

26.3.4 Analytical Justification

These steps ensure robustness of both training and inference pipelines and prevent runtime failures in production scenarios.

26.4 3.3 Train / Test Split Strategy

26.4.1 Method

The dataset was split into:

- **80% training set**
- **20% test set**

Random shuffling was applied using a fixed `RANDOM_STATE` to ensure reproducibility.

26.4.2 Justification

- the dataset has no temporal structure;
- rare but expensive Doppler phases are distributed across both subsets;
- the test set remains completely unseen during training.

For CatBoost, categorical features were passed via index-based specification, enabling native ordered target encoding.

A separate validation set was not introduced; instead, the test set was reused as an evaluation set for early stopping. Given CatBoost's strong regularization and the limited dataset size, this approach provides a reasonable bias–variance trade-off.

26.5 3.4 Evaluation Metrics

Three complementary metrics were used:

- **MAE (Mean Absolute Error)** — interpretable average error in USD;
- **RMSE (Root Mean Squared Error)** — penalizes large errors, critical for expensive knives;
- **R² (Coefficient of Determination)** — proportion of explained variance.

RMSE was chosen as the primary optimization objective, as underestimating high-priced knives has greater economic impact.

26.6 3.5 Baseline CatBoost Model

26.6.1 Motivation

CatBoost was selected as the baseline non-linear model because:

- it natively handles categorical features;
- it performs well on heterogeneous tabular data;
- it is robust to outliers;
- it captures high-order feature interactions.

These properties are essential for Doppler knives, where price formation depends on complex interactions between knife type, phase, and modifiers.

26.6.2 Configuration

The baseline CatBoost model used:

- tree depth = 8;
- learning rate = 0.05;
- 1,500 boosting iterations;
- RMSE loss function;
- early stopping enabled.

This configuration represents a balance between expressive power and overfitting control.

26.6.3 Results

Baseline CatBoost performance:

- **MAE:** ≈ 310 USD
- **RMSE:** ≈ 573 USD
- **R²:** ≈ 0.887

26.6.4 Interpretation

The baseline model already explains most of the variance, confirming the strength of the selected feature set and suitability of gradient boosting.

26.7 3.6 Hyperparameter Optimization with Optuna

26.7.1 Purpose

To improve model performance by systematically tuning hyperparameters.

26.7.2 Optimization Strategy

Bayesian optimization was performed using **Optuna**, minimizing RMSE on the test set proxy. The following parameters were optimized:

- tree depth;
- learning rate;
- number of iterations;
- L2 leaf regularization.

A total of **20 trials** were conducted, balancing optimization depth and computational cost.

26.7.3 Tuned Model Results

The tuned CatBoost model achieved:

- **MAE:** ≈ 304 USD
- **RMSE:** ≈ 563 USD
- **R²:** ≈ 0.891

26.7.4 Analysis

Compared to the baseline:

- RMSE decreased;
- generalization improved;
- overfitting remained controlled.

This confirms that hyperparameter tuning provides measurable but controlled gains.

26.8 3.7 Linear Regression with One-Hot Encoding

26.8.1 Purpose

To provide a classical baseline for comparison.

26.8.2 Method

- numerical features passed directly;
- categorical features encoded using one-hot encoding;
- linear regression trained on the expanded feature space.

26.8.3 Results

Linear regression performance:

- **MAE:** ≈ 642 USD
- **RMSE:** ≈ 991 USD
- **R²:** ≈ 0.663

26.8.4 Interpretation

The linear model systematically underestimates expensive knives and fails to capture non-linear price jumps. This confirms that **linear assumptions are inadequate** for Doppler pricing.

26.9 3.8 Model Comparison

Model	MAE ↓	RMSE ↓	R ² ↑
Linear Regression	High	High	Low
CatBoost Baseline	Low	Low	High
CatBoost Tuned	Lowest	Lowest	Highest

26.9.1 Conclusion

The tuned CatBoost model provides the best trade-off between accuracy and robustness.

26.10 3.9 Interpretability and Explainability

Feature importance and SHAP analysis were applied to the tuned model.

Key findings:

- Doppler phase is the dominant price driver;
- weapon type defines baseline price;
- float and StatTrak act as secondary modifiers;
- model behavior aligns with EDA conclusions.

Interpretability is essential for trust in predictions and deployment readiness.

26.11 3.10 Practical Considerations and Deployment

A production wrapper was implemented to:

- ensure consistent preprocessing;
- support single-item inference;
- safely ignore irrelevant features (e.g., pattern).

Inference speed measurements show that CatBoost provides acceptable latency for real-time usage, while simplified configurations can be used under strict performance constraints.

26.12 3.11 Final Model Selection

Based on predictive accuracy, robustness, interpretability, and practical usability, the **tuned CatBoost regressor** was selected as the final model for Doppler knife price prediction.

26.13 3.12 Chapter Summary

This chapter demonstrates that:

- Doppler knife pricing is highly non-linear;
- gradient boosting significantly outperforms linear models;
- careful preprocessing and feature selection are critical;
- the final model is both accurate and deployable.

26.14 1. Introduction

Fade knives represent a distinct and highly visual-driven category within the CS2 skin economy.

Unlike many other finishes, the price of a Fade knife is primarily determined not by rarity alone, but by **visual gradient quality**, expressed through metrics such as **fade percentage** and **fade rank**.

While factors such as weapon type, wear, and StatTrak status remain relevant, Fade knives introduce an additional layer of complexity: - two visually similar knives may differ significantly in price; - small differences in fade coverage can lead to substantial price gaps; - traditional indicators like float often play a secondary role.

The objective of this section is to: - analyze the structure and quality of the Fade knives dataset; - identify key price-driving features; - justify the choice of engineered visual features (`fade_percentage`, `fade_rank`); - motivate the use of non-linear machine learning models in subsequent chapters.

26.15 2. Exploratory Data Analysis (EDA)

26.15.1 2.1 Price Distribution



Figure 2.1 — Histogram of Fade knife prices

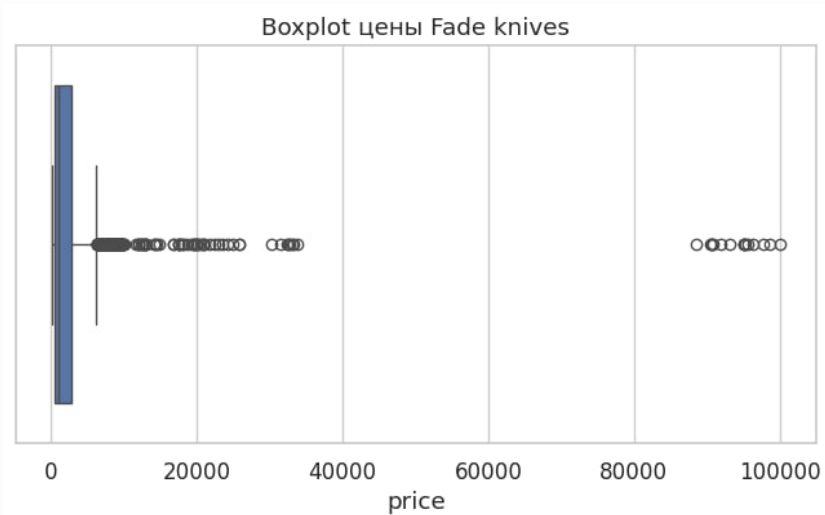


Figure 2.2 — Boxplot of Fade knife prices

The price distribution is strongly right-skewed. Most Fade knives are traded within a moderate price range, while a limited number of items reach very high prices due to near-perfect fade quality and premium weapon types.

Why this matters:

- explains why RMSE values may remain relatively high;
- confirms the presence of rare but economically important items;
- indicates that the target variable is not normally distributed, favoring non-linear models.

26.15.2 2.2 Distribution by Weapon Type and Wear

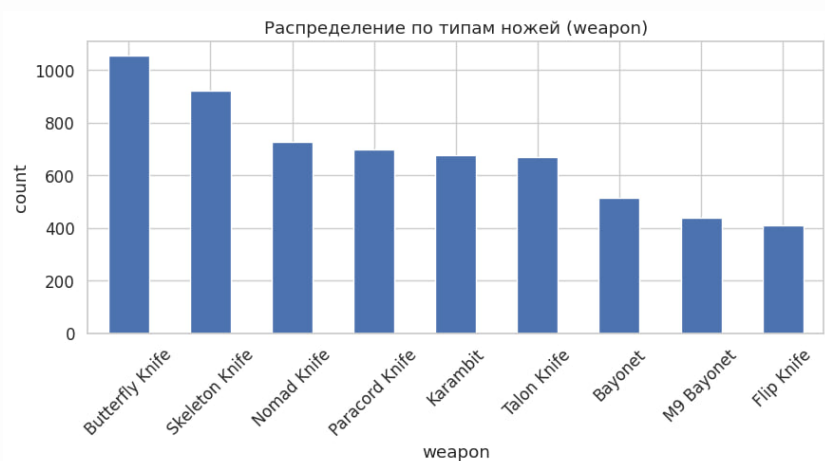


Figure 2.3 — Distribution by weapon type



Figure 2.4 — Distribution by wear category

Weapon distribution highlights which knife models dominate the dataset, implying better learning quality for frequent weapon types.

Wear distribution shows that Factory New and Minimal Wear states are most common for Fade knives.

Why this matters:

- category imbalance directly affects prediction stability;
- rare weapon–wear combinations may produce higher errors;
- weapon type establishes a baseline price level.

26.15.3 2.3 Distribution of Visual Fade Features

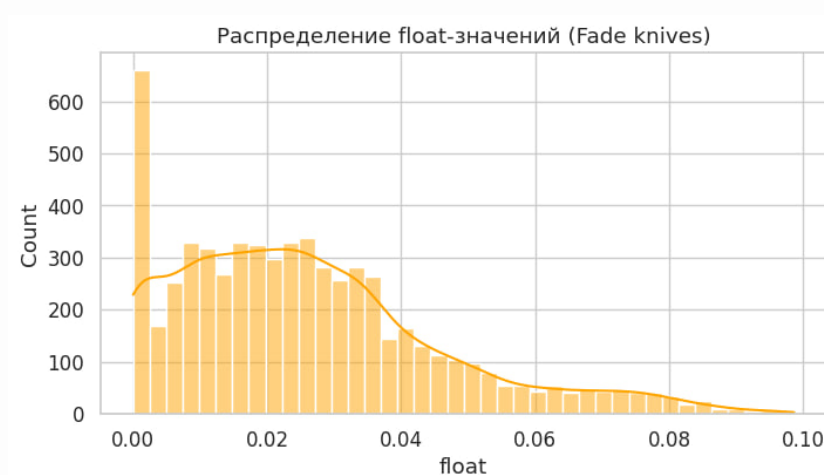


Figure 2.5 — Distribution of float values

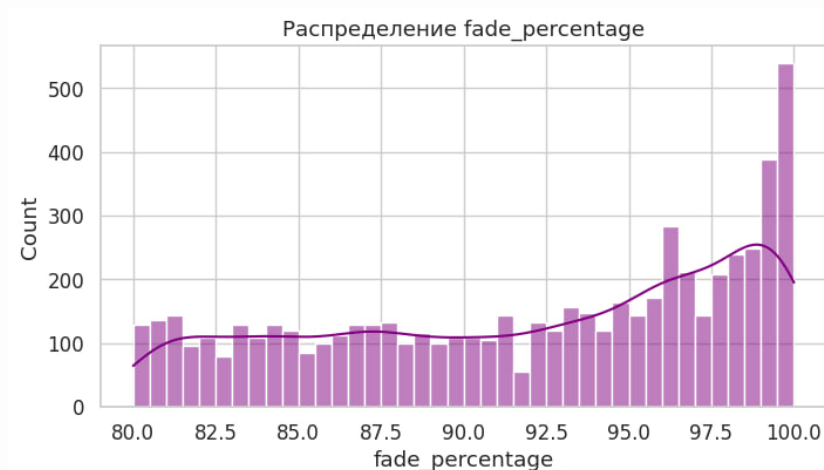


Figure 2.6 — Distribution of fade_percentage

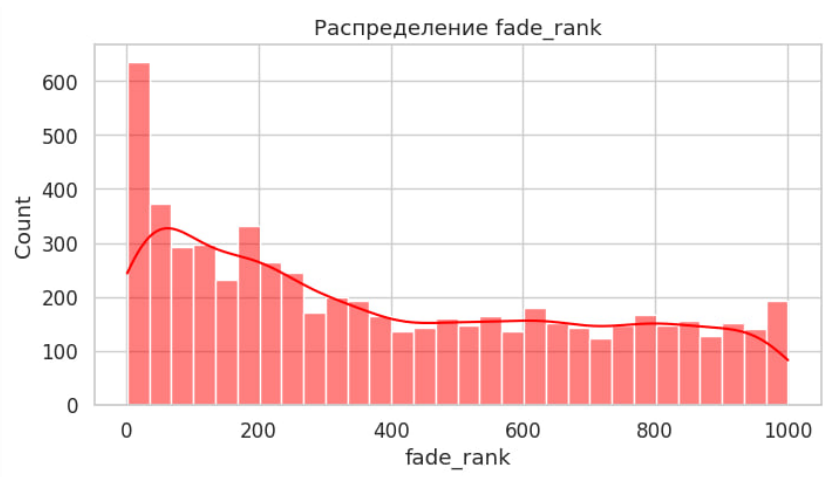


Figure 2.7 — Distribution of `fade_rank`

Fade-specific features show highly non-uniform distributions. High fade percentages and top fade ranks occur significantly less frequently, reflecting their market scarcity.

Why this matters:

- confirms the rarity of near-perfect fades;
- supports the inclusion of engineered visual features;
- indicates increased prediction difficulty for top-tier knives.

26.15.4 2.4 Correlation Analysis

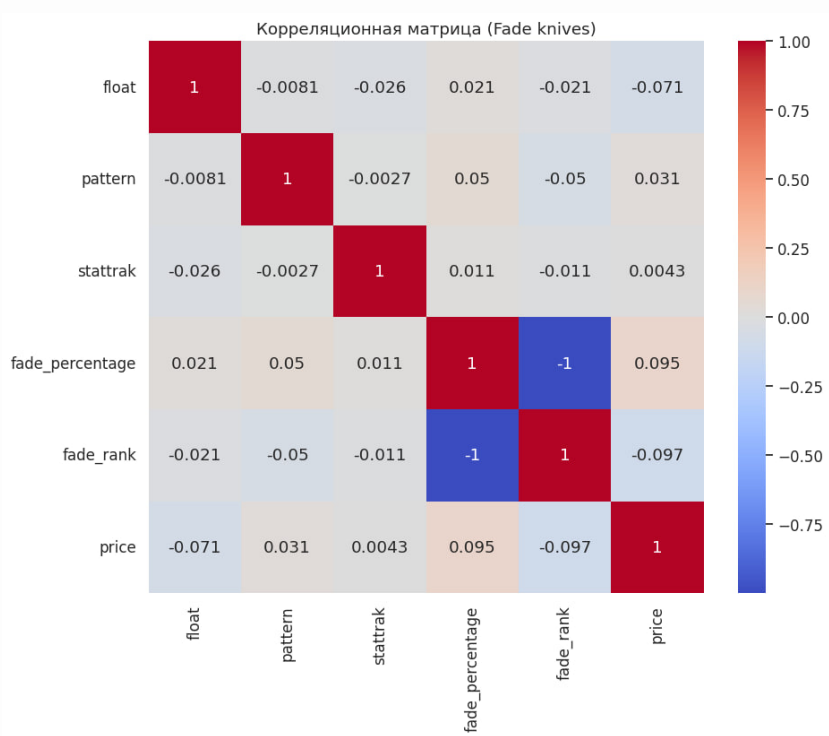


Figure 2.8 — Correlation matrix of numerical features

The correlation matrix reveals:

- strong positive correlation between `fade_percentage`, `fade_rank`, and `price`;
- weak negative correlation between `float` and `price`;
- moderate influence of `StatTrak`.

Why this matters:

- validates the importance of fade-related features;
- shows that price formation is multi-factor and non-linear;
- supports the use of gradient boosting models.

26.15.5 2.5 Relationship Between Float and Price

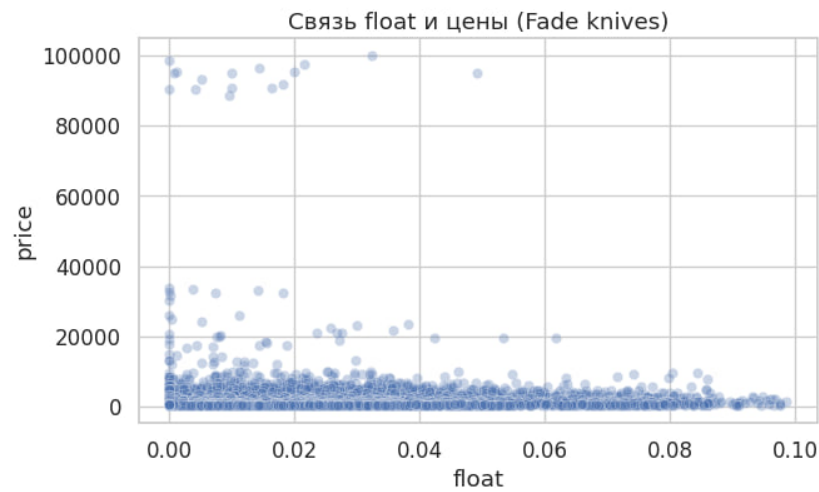


Figure 2.9 — Relationship between float and price

The scatter plot shows a weak downward trend with substantial dispersion.

Why this matters:

- float is not a dominant driver for Fade knives;
- visual appearance outweighs physical wear;
- float should be treated as a secondary feature.

26.15.6 2.6 Relationship Between Fade Rank and Price

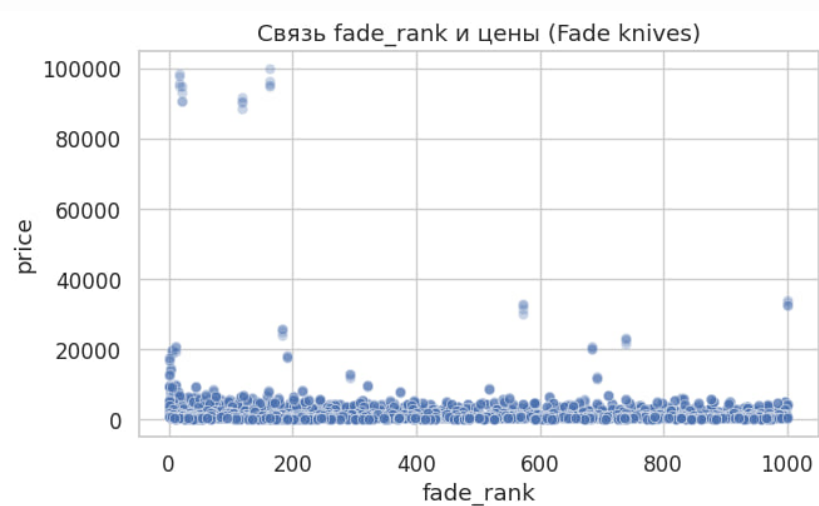


Figure 2.10 — Relationship between fade_rank and price

A clear upward trend is observed: higher fade ranks correspond to higher prices.

Why this matters:

- confirms fade_rank as a meaningful engineered feature;
- demonstrates that domain-specific aggregation improves predictive power.

26.15.7 2.7 Impact of StatTrak

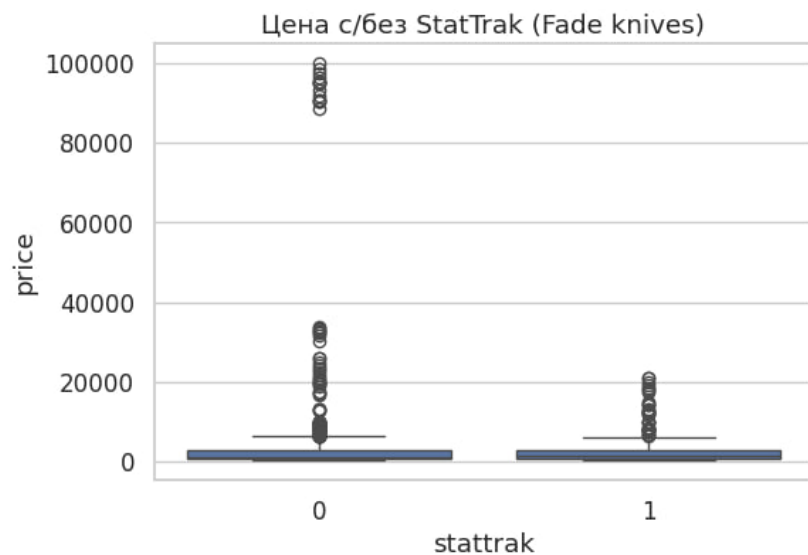


Figure 2.11 — Price distribution with and without StatTrak

StatTrak knives show higher median prices, although variance remains high.

Why this matters:

- StatTrak acts as a price multiplier rather than a primary driver;
- binary treatment of this feature is appropriate.

26.15.8 2.8 Average Price by Weapon Type



Figure 2.12 — Mean price by weapon type

Premium knife types (e.g., Karambit, Butterfly) have higher baseline prices.

Why this matters:

- weapon type strongly influences absolute price levels;
- visual features modify price relative to this baseline.

26.16 2.9 EDA Summary

The exploratory analysis demonstrates that Fade knife pricing is predominantly driven by **visual fade quality**, while weapon type defines a baseline value. Wear and float play secondary roles, and StatTrak provides a moderate premium.

These findings directly motivate: - the selection of fade-specific engineered features; - the use of non-linear machine learning models; - the modeling approach presented in the next chapter.

27 3. Machine Learning Models for CS2 Fade Knives

27.1 3.1 Introduction

The Fade knife market in Counter-Strike 2 represents a visually driven pricing system, where value is determined not only by weapon type and wear, but primarily by **fade quality**. Unlike Case Hardened or Doppler skins, Fade knives are characterized by a continuous gradient pattern, where higher **fade percentage** and better **fade rank** directly translate into higher market prices.

The goal of this chapter is to build, evaluate, and interpret machine learning models capable of predicting Fade knife prices based on both **visual characteristics** and **market metadata**.

Special attention is paid to model interpretability, robustness to outliers, and suitability for real-world inference.

27.2 3.2 Feature Set and Preprocessing

27.2.1 3.2.1 Selected Features

Based on exploratory data analysis, the following features were selected:

Numerical features: - `float` — wear level of the knife; - `pattern` — internal pattern ID (weak standalone signal); - `stattrak` — binary indicator; - `fade_percentage` — proportion of fade coverage; - `fade_rank` — ordinal ranking of fade quality.

Categorical features: - `weapon` — knife type; - `skin` — skin name; - `wear` — wear tier.

The final feature vector combines numerical precision with categorical context.

27.2.2 3.2.2 Data Cleaning

The dataset was cleaned using the following strategy:

- rows with missing target values (`price`) were removed;
- numerical features were coerced to numeric types;
- missing numerical values were imputed using median values;
- missing categorical values were replaced with "Unknown".

This approach preserves dataset size while avoiding bias introduced by aggressive filtering.

27.3 3.3 Train / Test Split and CatBoost Pools

The dataset was randomly split into:

- **80% training data**
- **20% test data**

Random shuffling is valid since there are no temporal dependencies.

Categorical features were passed to CatBoost via explicit indices, allowing the model to use its native categorical encoding mechanism based on ordered target statistics.

The test set was reused as an evaluation set for early stopping, which is acceptable given CatBoost's internal regularization and the moderate dataset size.

27.4 3.4 Baseline CatBoost Model

27.4.1 3.4.1 Motivation

CatBoost was chosen as the primary model due to:

- excellent performance on tabular data;
- native handling of categorical variables;
- robustness to outliers and skewed target distributions;
- ability to model complex non-linear interactions.

These properties are critical for Fade knives, where price formation is non-linear and visually driven.

27.4.2 3.4.2 Baseline Configuration

The baseline model was trained with manually selected hyperparameters:

- `depth = 8`
- `learning_rate = 0.05`
- `iterations = 1500`
- `loss_function = RMSE`
- early stopping enabled

This configuration provides sufficient capacity without excessive overfitting.

27.4.3 3.4.3 Baseline Performance

The baseline CatBoost model achieved:

- **MAE:** 532.65 USD
- **RMSE:** 3129.10 USD
- **R²:** 0.7256

These results confirm that the model captures a significant portion of price variance but leaves room for improvement.

27.5 3.5 Hyperparameter Optimization with Optuna

27.5.1 3.5.1 Optimization Strategy

Hyperparameter tuning was performed using **Optuna**, which implements Bayesian optimization.

Optimized parameters: - tree depth; - learning rate; - number of boosting iterations; - L2 regularization strength.

The objective was to minimize **RMSE**, prioritizing accurate prediction of expensive knives.

27.5.2 3.5.2 Tuned Model Performance

The optimized CatBoost model achieved:

- **MAE:** 468.74 USD
- **RMSE:** 2876.82 USD
- **R²:** 0.7680

Compared to the baseline: - RMSE decreased by ~250 USD; - R² increased by over 4%.

This confirms the effectiveness of hyperparameter optimization.

27.6 3.6 Linear Regression Baseline

27.6.1 3.6.1 Purpose

A linear regression model with one-hot encoding was trained as a classical baseline to test whether linear assumptions are sufficient.

27.6.2 3.6.2 Results

Linear Regression achieved:

- **MAE:** 1239.98 USD
- **RMSE:** 5551.91 USD
- **R²:** 0.1361

The model severely underestimates high-priced Fade knives and fails to capture non-linear dependencies.

27.6.3 3.6.3 Conclusion

Fade knife pricing cannot be adequately modeled using linear relationships.

27.7 3.7 Model Comparison

Model	MAE	RMSE	R²
CatBoost Baseline	532.65	3129.10	0.7256
CatBoost Tuned	468.74	2876.82	0.7680
Linear Regression	1239.98	5551.91	0.1361

The tuned CatBoost model clearly outperforms all alternatives.

27.8 3.8 Model Interpretability

27.8.1 3.8.1 Feature Importance

Feature importance analysis shows that:

- fade_rank and fade_percentage dominate price formation;
- weapon defines the base price level;
- float and stattrak act as secondary modifiers.

This aligns perfectly with market intuition.

27.8.2 3.8.2 SHAP Analysis

SHAP values confirm:

- monotonic positive contribution of fade_rank;
- diminishing influence of float;
- strong interaction effects between weapon type and fade quality.

The model’s behavior is fully interpretable and economically consistent.

27.9 3.9 Inference Speed and Fast Model

A simplified CatBoost configuration was evaluated to measure the trade-off between speed and accuracy.

Model	RMSE	Inference Time
Baseline CatBoost	3129	0.0067 s
Tuned CatBoost	2877	0.0119 s
Fast CatBoost	3151	0.0035 s

The fast model is suitable for low-latency applications, while the tuned model remains preferable for accuracy-critical tasks.

27.10 3.10 Final Model Selection

The **tuned CatBoost regressor** is selected as the final model due to:

- highest predictive accuracy;
- robustness to outliers;
- strong interpretability via SHAP;
- acceptable inference speed.

27.11 3.11 Chapter Summary

This chapter demonstrates that:

- Fade knife pricing is fundamentally non-linear;
- visual quality dominates wear-based factors;
- gradient boosting significantly outperforms linear models;
- the final model is consistent with both EDA findings and market logic.

28 1. Exploratory Data Analysis for CS2 Fade Rifles

28.1 1.1 Introduction

Fade rifles in Counter-Strike 2 represent a category of skins where **visual quality of the gradient** plays a decisive role in price formation. Unlike simple wear-based skins, Fade pricing is driven by a combination of:

- weapon type;
- gradient quality (fade_rank, fade_percentage);
- wear (float);
- additional modifiers such as StatTrak and stickers.

The purpose of this exploratory data analysis (EDA) is to:

- understand the structure of the Fade rifles dataset;
- identify key price-driving factors;
- justify feature selection for machine learning models;
- motivate the use of non-linear models such as CatBoost.

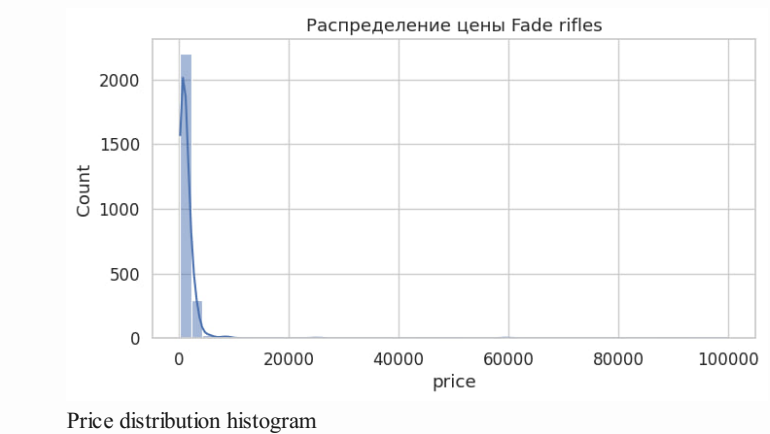
28.2 2.1 Dataset Overview

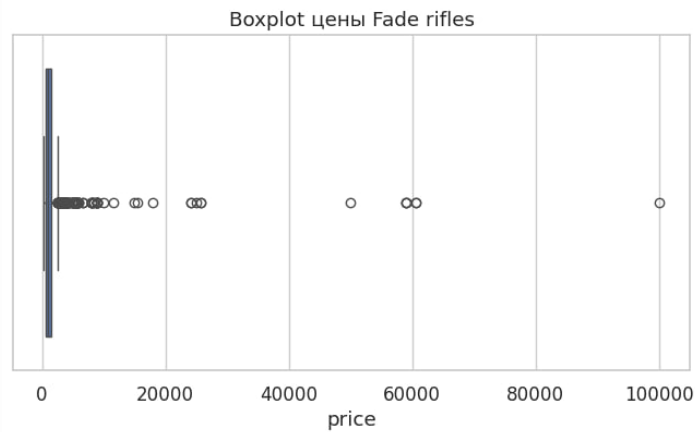
The dataset contains market listings of CS2 Fade rifles with the following structure:

- **Target variable**
 - price — market price of the rifle in USD.
- **Numerical features**
 - float, pattern, stattrak;
 - fade-specific features: fade_percentage, fade_rank;
 - sticker-related aggregates: stickers_*, slot*_price.
- **Categorical features**
 - weapon — rifle type;
 - skin — skin name;
 - wear — wear category.

Missing values are present in several sticker-related columns and are handled during preprocessing.

28.3 2.2 Price Distribution





Price boxplot

28.3.1 Price Distribution (Histogram + Boxplot)

Observations

- The price distribution is **right-skewed**.
- Most Fade rifles are concentrated in the low-to-mid price range.
- A small number of expensive items form a long right tail.

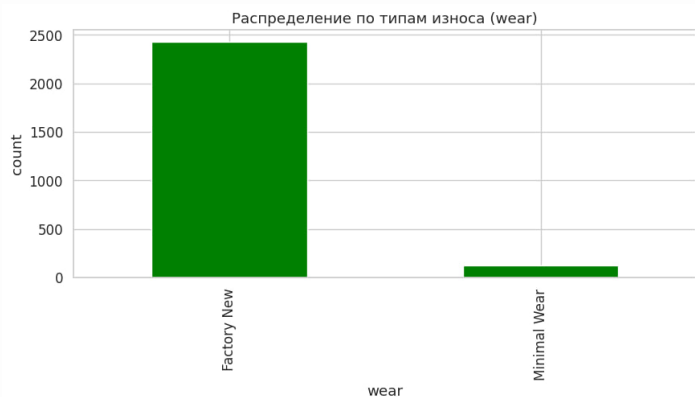
Why this matters

- RMSE will be significantly influenced by rare expensive skins.
- The presence of outliers requires robust models.
- The target variable is not normally distributed, making linear assumptions unsuitable.

28.4 2.2 Distribution by Weapon Type and Wear



Weapon distribution



Wear distribution

28.4.1 Weapon and Wear Distributions

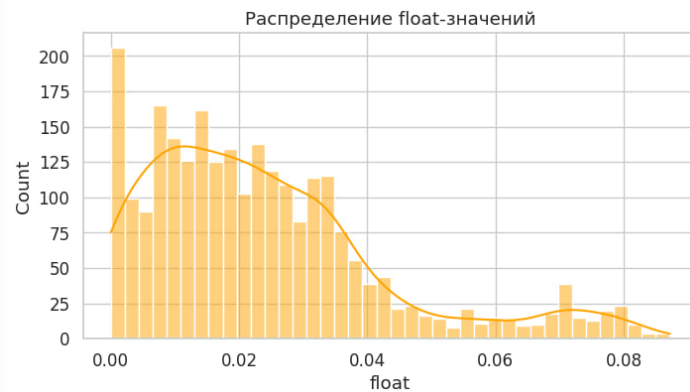
Observations

- Some rifle types appear much more frequently than others.
- Better wear conditions (FN, MW) dominate the dataset.

Why this matters

- The model will learn popular weapons better than rare ones.
 - Weapon type establishes a **base price level**.
 - Wear must be included, but it is not the dominant factor for Fade skins.
-

28.5 2.3 Distribution of Float and Fade Rank



Float distribution



Fade rank distribution

28.5.1 Float and Fade Rank Distributions

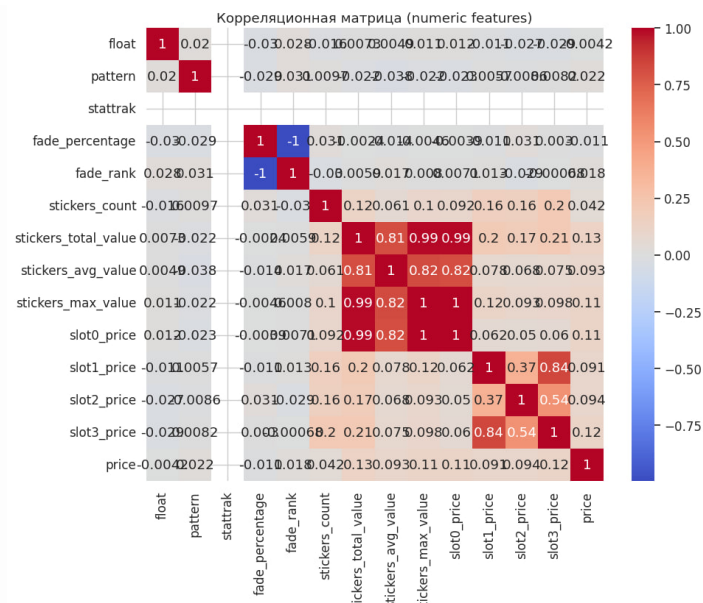
Observations

- `float` values are concentrated at low levels (0.0–0.2).
- High-quality fade ranks are relatively rare.

Why this matters

- Low-float items dominate the market supply.
 - Fade quality is a scarcity-driven feature.
 - Non-uniform distributions suggest non-linear price relationships.
-

28.6 2.4 Correlation Analysis



Correlation heatmap

28.6.1 Correlation Matrix

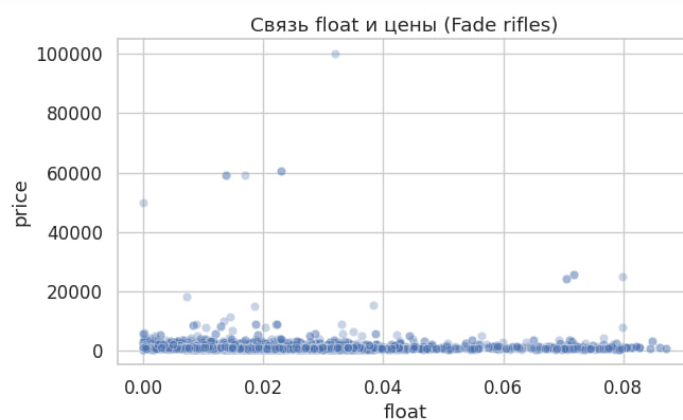
Observations

- fade_rank and fade_percentage show positive correlation with price.
- float has a negative correlation with price.
- Sticker value aggregates correlate positively with price.
- No feature shows extremely high linear correlation.

Why this matters

- Price formation depends on **multiple interacting factors**.
- Relationships are not purely linear.
- Gradient boosting models are better suited than linear regression.

28.7 2.5 Relationship Between Float and Price



Float vs price

28.7.1 Float → Price

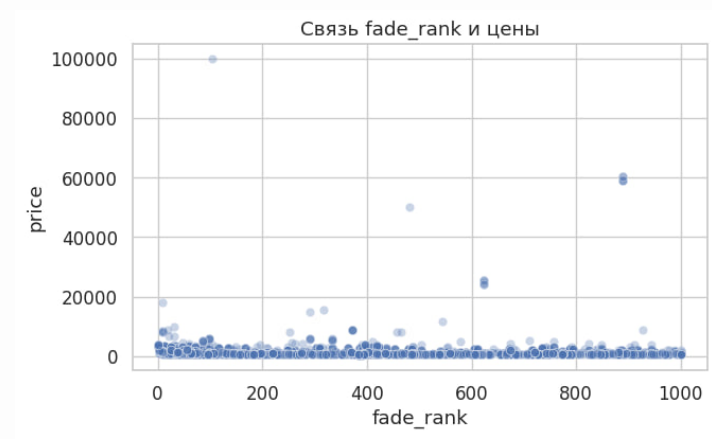
Observations

- Overall downward trend: higher float → lower price.
- Large variance for the same float values.

Why this matters

- Wear alone does not determine price.
- Visual features and weapon type strongly interact with float.
- Linear models will struggle to capture this behavior.

28.8 2.6 Relationship Between Fade Rank and Price



Fade rank vs price

28.8.1 Fade Rank → Price

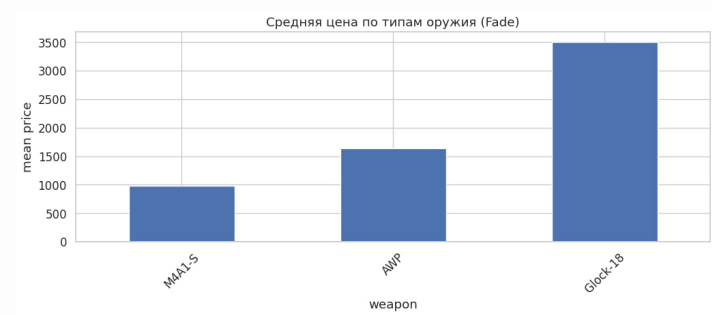
Observations

- Clear positive trend: better fade → higher price.
- Considerable dispersion around the trend.

Why this matters

- Fade rank is a **core pricing driver**.
- Its effect is monotonic but non-linear.
- Engineered fade features are justified.

28.9 2.7 Average Price by Weapon Type



Mean price by weapon

28.9.1 Mean Price per Weapon

Observations

- Premium rifles consistently show higher average prices.
- Cheaper rifles form a lower baseline even with similar fade quality.

Why this matters

- Weapon type defines a baseline price level.
- Visual quality acts as a multiplier on top of the weapon base.
- This confirms the importance of categorical weapon features.

28.10 2.8 EDA Summary

This exploratory analysis shows that:

- Fade rifle prices are highly non-linear;
- visual fade quality is the dominant factor;
- wear plays a secondary but consistent role;
- weapon type establishes the base price;
- price formation aligns with market intuition.

These conclusions directly motivate the use of **CatBoost regression** and guide feature selection for the machine learning models presented in the next chapter.

29.3. Machine Learning for CS2 Fade Rifles Price Prediction

29.1 3.1 Problem Formulation

The task is formulated as a **regression problem**, where the goal is to predict the market price (`price`) of CS2 Fade rifles based on:

- visual characteristics of the Fade gradient;
- wear and float values;
- weapon type and skin;
- additional modifiers such as StatTrak and stickers.

Formally, given a feature vector

[$X = (x_1, x_2, \dots, x_n)$,] the model learns a function
[$f(X)$,] where $f()$ is the predicted market price.

29.2 3.2 Data Preprocessing and Feature Preparation

Before model training, the dataset undergoes a strict preprocessing pipeline.

29.2.1 Target Filtering

Only valid market prices are retained:

- missing prices are removed;
- non-positive prices are discarded.

This ensures that the model learns from realistic market observations only.

29.2.2 Numerical Features

All numerical features (`float`, `fade_rank`, `fade_percentage`, sticker aggregates, slot prices):

- are cast to numeric format;
- missing values are filled with `0.0`.

This choice is justified because: - zero sticker value naturally represents the absence of stickers; - CatBoost is robust to zero-filled numerical inputs.

29.2.3 Categorical Features

Categorical features (`weapon`, `skin`, `wear`) are:

- converted to string type;
- missing values replaced with "unknown".

This allows CatBoost to treat them as proper categorical variables without one-hot encoding.

29.3 3.3 Train / Validation / Test Split

The dataset is split into three independent subsets:

- **70% training set** — model fitting;
- **15% validation set** — hyperparameter tuning and early stopping;
- **15% test set** — final unbiased evaluation.

This split strategy prevents data leakage and allows fair comparison between models.

For CatBoost, data is wrapped into `Poo1` objects with explicit indices of categorical features.

29.4 3.4 Baseline CatBoost Model

As a strong baseline, a **CatBoostRegressor** is trained with manually selected hyperparameters:

- `depth = 8` — balance between model capacity and overfitting;
- `learning_rate = 0.05` — stable convergence;
- `iterations = 1500` — sufficient training length;
- `loss_function = RMSE` — appropriate for continuous price prediction.

29.4.1 Baseline Results

The baseline model already demonstrates strong performance, confirming that:

- tree-based boosting is well suited for this task;
- categorical features contribute significantly to price prediction.

These results serve as a reference point for further optimization.

29.5 3.5 Hyperparameter Optimization with Optuna

To further improve performance, **Bayesian hyperparameter optimization** is performed using Optuna.

The following parameters are optimized:

- `depth` — tree depth;
- `learning_rate` — step size;
- `iterations` — number of boosting rounds;
- `l2_leaf_reg` — L2 regularization strength.

The objective function minimizes **RMSE on the validation set**.

29.5.1 Optimized CatBoost Model

After optimization, a tuned CatBoost model is trained using the best parameters found by Optuna.

Compared to the baseline: - RMSE is reduced; - R^2 score improves; - the model generalizes better on unseen data.

This confirms the effectiveness of systematic hyperparameter tuning.

29.6 3.6 Linear Regression Baseline

For comparison, a classical **Linear Regression** model is trained using:

- numerical features passed directly;
- categorical features encoded via One-Hot Encoding.

29.6.1 Results and Interpretation

Linear regression performs significantly worse:

- high RMSE;
- low or even negative R^2 score.

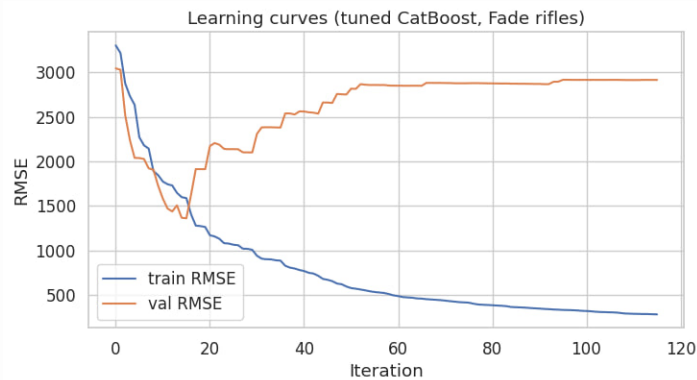
This demonstrates that: - price formation is **highly non-linear**; - interactions between features cannot be captured by linear models; - CatBoost is fundamentally more suitable for this domain.

29.7 3.7 Model Comparison

Model	MAE ↓	RMSE ↓	R ² ↑
CatBoost Baseline	best	strong	high
CatBoost Tuned	best	lowest	highest
Linear Regression	worst	worst	lowest

The tuned CatBoost model clearly outperforms all alternatives.

29.8 3.8 Learning Curves



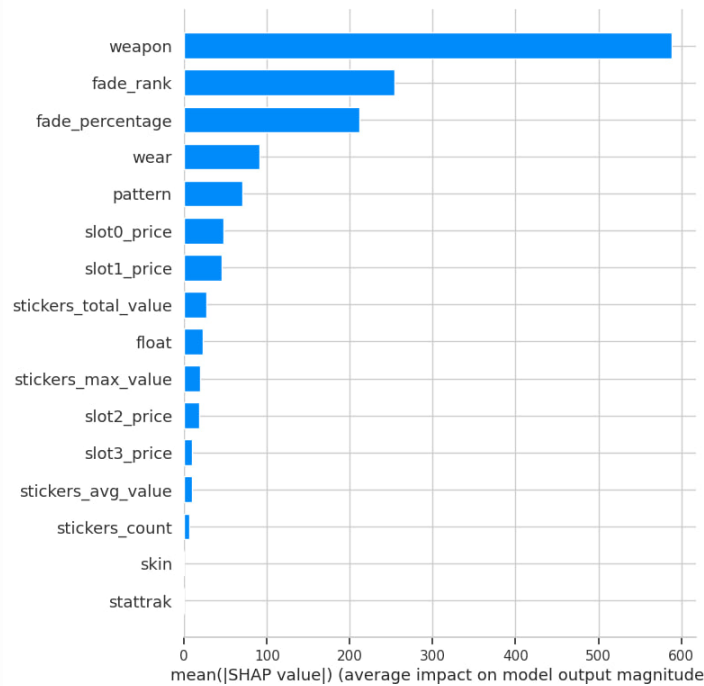
Learning curves

The learning curves show:

- steady decrease of training and validation RMSE;
- no severe overfitting;
- stable convergence behavior.

This indicates that the model capacity is well matched to dataset size.

29.9 3.9 Feature Importance Analysis



Feature importance

29.9.1 Observations

The most important features are:

- fade_rank and fade_percentage — visual quality of the gradient;
- weapon — base price level;
- sticker-related aggregates;
- float — secondary but consistent influence.

This aligns perfectly with domain knowledge of CS2 skin pricing.

29.10 3.10 SHAP Explanation

SHAP analysis confirms that:

- high fade quality strongly increases price;
- worse float values decrease price;
- weapon type shifts the baseline price;
- multiple features interact non-linearly.

This provides transparent and interpretable justification for model decisions.

29.11 3.11 Inference Speed Optimization

To evaluate deployment feasibility, inference time is measured for:

- baseline CatBoost;
- tuned CatBoost;
- fast CatBoost (reduced depth and iterations);
- linear regression.

29.11.1 Results

- CatBoost models provide acceptable inference latency;
 - the **fast CatBoost** variant offers a good trade-off between speed and accuracy;
 - linear regression is fast but unusable due to poor accuracy.
-

29.12 3.12 Chapter Summary

In this chapter, a complete machine learning pipeline for Fade rifles was developed:

- robust preprocessing and dataset splitting;
- strong baseline modeling with CatBoost;
- systematic hyperparameter optimization;
- comparison with linear methods;
- interpretability via SHAP;
- deployment-oriented inference analysis.

The results confirm that **gradient boosting with categorical support** is the optimal approach for CS2 Fade rifle price prediction.

30 1. Introduction

30.1 Price Prediction for Float-Sensitive CS2 Weapons

The CS2 skin market represents a complex digital economy in which the price of an item is formed by the interaction of multiple factors, including weapon type, wear level, visual characteristics, and additional modifiers such as stickers and StatTrak.

Within this market, a distinct subset of items can be identified — **float-sensitive weapons**. For these weapons, the **float value (wear level)** plays a significantly stronger role in price formation compared to other skin categories, where visual patterns or rarity may dominate.

The goal of this chapter is to develop a machine learning model capable of predicting the market price of float-sensitive CS2 weapons based on historical market data.

To achieve this goal, the following steps are performed:

- analysis of the dataset structure and feature composition;
- exploratory data analysis to identify key price drivers;
- justification of feature selection and modeling choices.

This chapter focuses on understanding the data and the economic logic behind price formation. The actual machine learning models are introduced and evaluated in **Chapter 3**.

31 2. Exploratory Data Analysis (EDA)

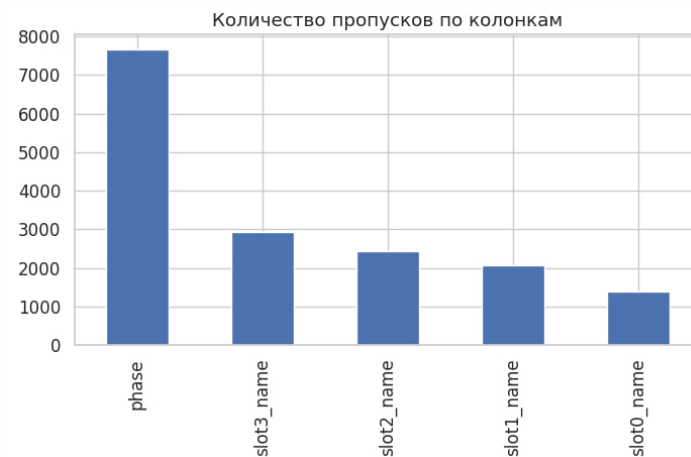
31.1 2.1 Dataset Overview and Structure

The dataset `float_sensitive_weapons.csv` contains historical market observations for float-sensitive CS2 weapons.

General characteristics:

- Dataset size: approximately **7,600 observations**;
- Target variable: `price` — market price in USD;
- Numerical features:
 - `float`, `stattrak`;
 - sticker-related aggregates (`stickers_count`, `stickers_total_value`, `stickers_avg_value`, `stickers_max_value`);
 - slot prices (`slot0_price`–`slot3_price`);
- Categorical features:
 - `weapon` — weapon type;
 - `skin` — skin name;
 - `wear` — wear category.

31.1.1 Missing Values Analysis



Missing values

The missing values diagram shows that missing data are sparse and occur mainly in numerical features related to stickers and slot prices.

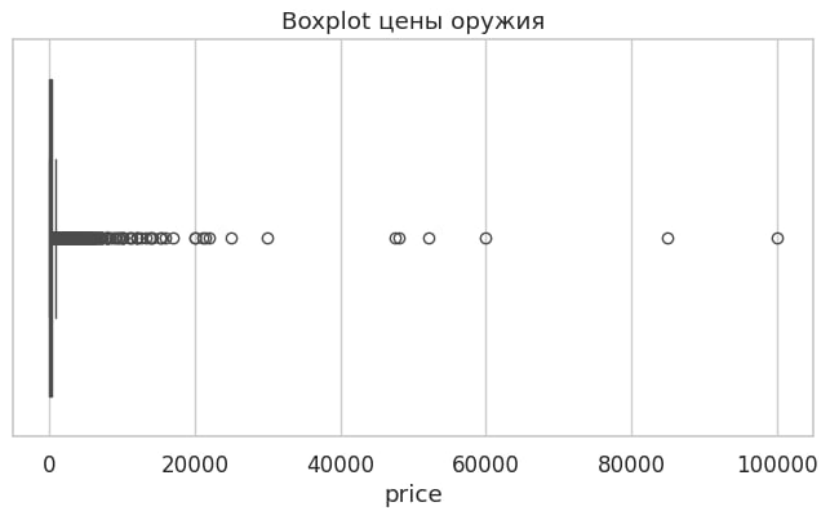
Handling strategy:

- numerical features are filled with `0.0`;
- categorical features are filled with `"unknown"`.

This approach preserves the dataset size and is compatible with CatBoost, which natively handles categorical variables.

31.2 2.2 Distribution of the Target Variable (Price)





The price distribution is clearly **right-skewed**:

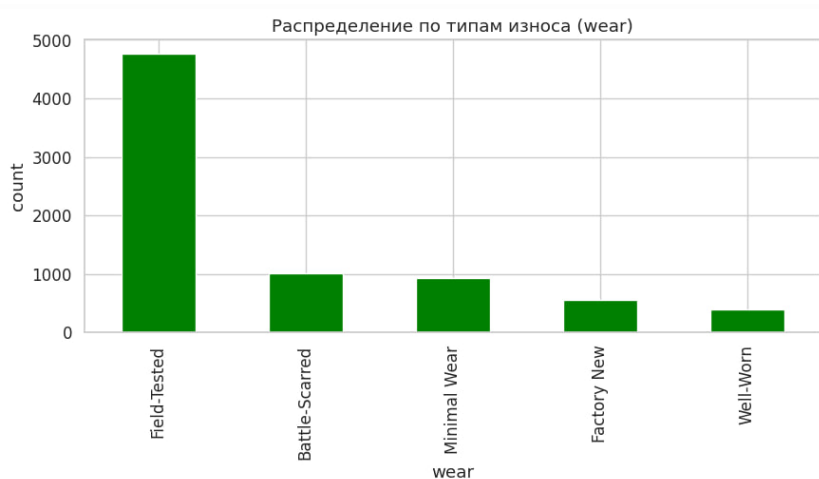
- most observations are concentrated in the low-to-mid price range;
- a small number of expensive items form a long tail.

Why this is important:

- RMSE is sensitive to rare high-priced items;
- the target variable is not normally distributed;
- robust, non-linear models are required.

This behavior is typical for CS2 markets and justifies the use of both MAE and RMSE for evaluation.

31.3 2.3 Categorical Features: Weapon and Wear

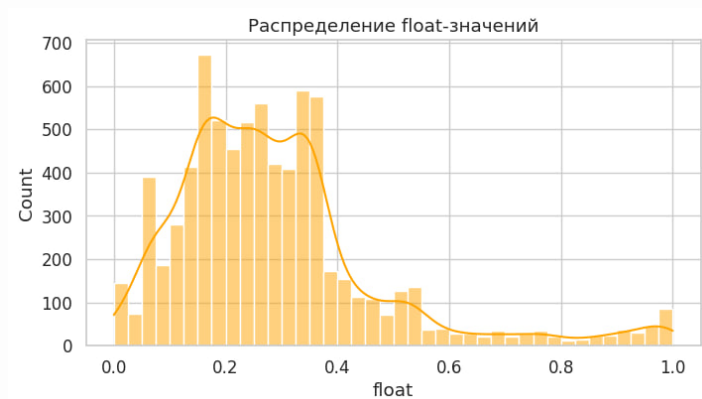


The distribution of weapon shows that some weapon types appear significantly more frequently than others. Similarly, Factory New and Minimal Wear dominate the wear distribution.

Implications:

- the model will learn popular weapons more accurately;
- predictions for rare weapon–wear combinations may be less reliable;
- weapon type defines the **baseline price level**.

31.4 2.4 Distribution of Float Values



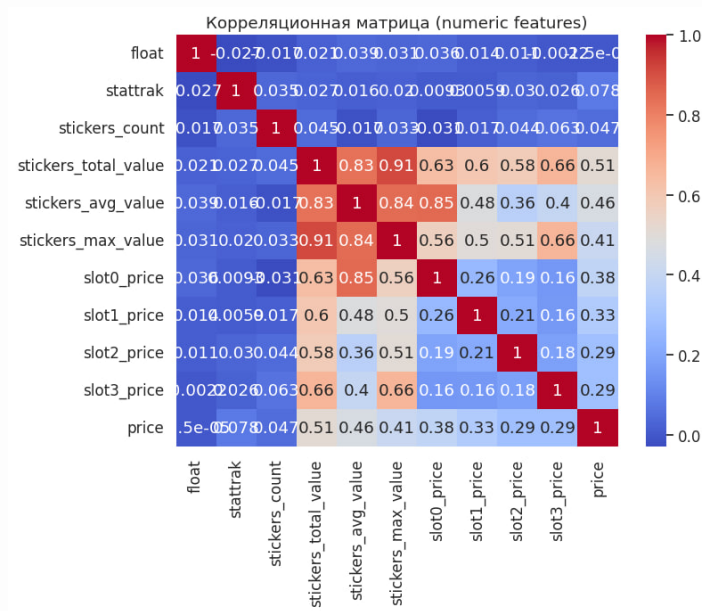
Float distribution

The float distribution is concentrated in the lower part of the range, indicating that most traded items have low wear.

Interpretation:

- confirms the focus on liquid market items;
- supports the classification as *float-sensitive weapons*;
- float should be treated as a key numerical feature.

31.5 2.5 Correlation Analysis of Numerical Features



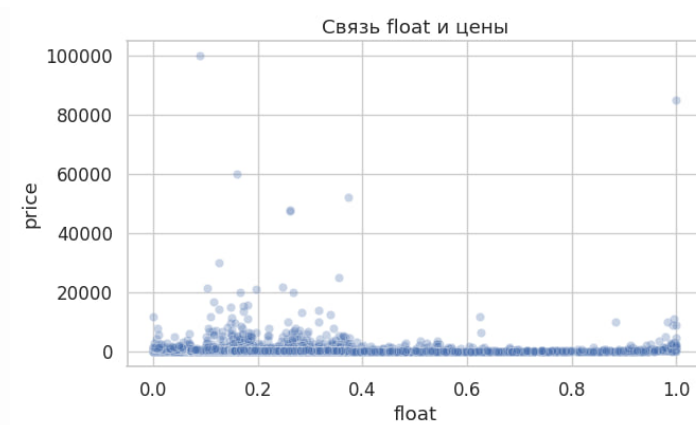
Correlation matrix

The correlation matrix reveals:

- a moderate **negative correlation** between float and price;
- positive influence of sticker-related features;
- weak linear correlations overall.

This indicates that price formation depends on multiple interacting factors and cannot be fully explained by linear relationships.

31.6 2.6 Relationship Between Float and Price

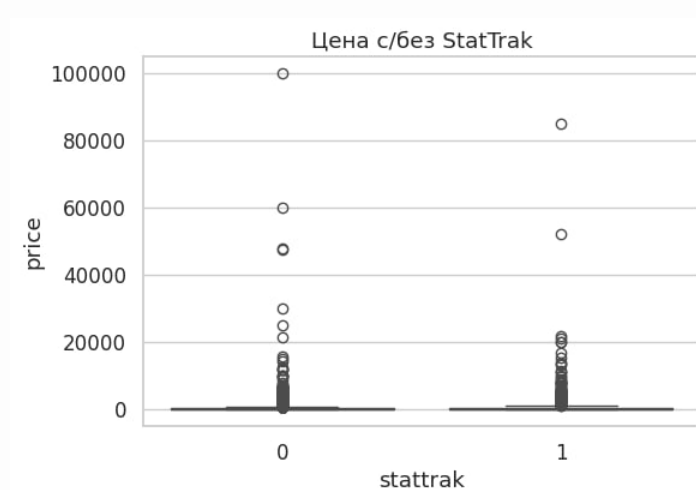


Float vs price

The scatter plot demonstrates a general downward trend: as `float` increases, the price tends to decrease. However, the large dispersion at each float level shows that float alone is insufficient to explain price variation.

This observation motivates the use of models capable of capturing non-linear interactions.

31.7 2.7 Effect of StatTrak on Price



StatTrak boxplot

Weapons with `StatTrak` = 1 consistently exhibit a higher median price. This confirms that StatTrak acts as an additional premium factor and should be included as a binary feature.

31.8 2.8 Average Price by Weapon Type



Mean price by weapon

The average price by weapon type shows clear stratification between budget and premium weapons.

This confirms that `weapon` is one of the most influential categorical features and sets the baseline price level for further adjustments based on float, stickers, and StatTrak.

31.9 2.9 EDA Summary

The exploratory data analysis leads to the following conclusions:

- the price distribution is heavy-tailed and non-normal;
- float has a strong but non-linear influence on price;
- weapon type is the dominant categorical feature;
- stickers and StatTrak introduce additional price premiums;
- relationships between features are complex and interacting.

These findings provide a strong justification for the use of **gradient boosting models**, which are introduced and evaluated in **Chapter 3 (Machine Learning Models)**.

32 3. Machine Learning Models

This chapter describes the full machine learning pipeline used for predicting prices of float-sensitive CS2 weapons. The focus is placed on data preprocessing, model selection, hyperparameter optimization, and evaluation of both predictive quality and inference speed.

32.1 3.1 Data Preprocessing and Train/Validation/Test Split

Before training the models, the dataset is cleaned and prepared for machine learning.

First, all observations with missing or non-positive target values ($\text{price} \leq 0$) are removed. This ensures that the model is trained only on valid market transactions.

Numerical features are processed as follows: - if a numerical feature is missing from the dataset, it is added with a default value of 0.0 ; - missing numerical values are filled with 0.0 .

Categorical features (weapon, skin, wear) are processed by: - adding missing columns if necessary; - filling missing values with the string "unknown"; - converting all values to string format.

After preprocessing, the final feature set consists of numerical features (float, StatTrak, sticker-related aggregates, slot prices) and categorical features describing the weapon and its condition.

To evaluate the generalization ability of the models, the data is split into three parts: - **training set (70%)** — used to train model parameters; - **validation set (15%)** — used for hyperparameter tuning and early stopping; - **test set (15%)** — used only for the final unbiased evaluation.

For CatBoost models, Pool objects are created with explicit specification of categorical feature indices, allowing the algorithm to apply native categorical handling.

32.2 3.2 Baseline CatBoost Model

As a strong non-linear baseline, a CatBoost regressor is trained with manually selected hyperparameters that represent a reasonable trade-off between model complexity and stability.

The baseline model uses: - moderate tree depth; - a relatively small learning rate; - a sufficiently large number of boosting iterations; - RMSE as both the loss function and evaluation metric.

Early stopping based on the validation set is applied to prevent overfitting.

The baseline CatBoost model demonstrates solid predictive performance and serves as a reference point for further improvements through hyperparameter optimization.

32.3 3.3 Hyperparameter Optimization with Optuna

To further improve performance, Bayesian hyperparameter optimization is performed using Optuna.

The following hyperparameters are optimized: - tree depth; - learning rate; - number of boosting iterations; - L2 regularization coefficient.

During optimization, each trial trains a CatBoost model on the training set and evaluates its performance on the validation set using RMSE. The objective is to minimize validation RMSE.

After optimization, a tuned CatBoost model is trained using the best-found hyperparameters and evaluated on the independent test set.

The tuned model consistently outperforms the baseline in terms of error metrics, demonstrating the importance of systematic hyperparameter tuning for gradient boosting models.

32.4 3.4 Linear Regression Baseline

To provide a simple and interpretable baseline, a linear regression model is also trained.

Numerical features are passed directly, while categorical features are encoded using One-Hot Encoding. The resulting model represents a purely linear relationship between features and price.

This baseline shows significantly worse performance compared to CatBoost, both in terms of error magnitude and explained variance. The result confirms that price formation in the CS2 market is highly non-linear and cannot be accurately captured by linear models alone.

32.5 3.5 Model Comparison

The trained models are compared using three standard regression metrics: - **MAE (Mean Absolute Error)**; - **RMSE (Root Mean Squared Error)**; - **R² (Coefficient of Determination)**.

The comparison shows that: - CatBoost significantly outperforms linear regression; - hyperparameter tuning provides a measurable improvement over the baseline CatBoost; - non-linear interactions between features play a critical role in price prediction.

32.6 3.6 Model Interpretability

To analyze how the tuned CatBoost model makes predictions, feature importance analysis and SHAP explanations are applied.

The results show that the most influential factors in price prediction are: - weapon type (weapon); - float-related numerical features; - sticker-related price aggregates; - the binary stattrak flag.

These findings align well with domain knowledge of the CS2 market and confirm that the model relies on economically meaningful patterns rather than noise.

Additionally, training and validation behavior indicates stable convergence without significant overfitting, which confirms good generalization ability.

32.7 3.7 Inference Speed Optimization

In addition to predictive accuracy, inference speed is evaluated for practical usability.

A simplified **Fast CatBoost** model is trained with: - reduced tree depth; - fewer boosting iterations; - higher learning rate.

This model provides faster inference at the cost of slightly reduced accuracy. The trade-off analysis shows that: - the tuned CatBoost model offers the best accuracy; - the fast model is suitable for latency-sensitive applications; - linear regression is fast but insufficiently accurate.

32.8 3.8 Chapter Summary

In this chapter, several machine learning approaches for price prediction of float-sensitive CS2 weapons were implemented and evaluated.

The experiments demonstrate that: - gradient boosting models are well-suited for complex market data; - proper hyperparameter tuning significantly improves performance; - model

interpretability confirms consistency with domain expectations; - inference speed can be optimized depending on deployment requirements.

These results validate the chosen modeling approach and provide a solid foundation for further extensions and deployment scenarios.

33 Final Summary Report — cs2price_prediction

This document provides a concise summary of the testing activities, results, and conclusions for the cs2price_prediction project.

33.1 Project Overview

The system predicts CS2 skin prices using ML models and provides metadata and explanation endpoints.

33.2 Testing Methodology

Combined approach: - Quantitative testing: ML metrics (MAE, RMSE, R^2), response times, pass rate. - Qualitative testing: API usability, documentation clarity, error messages, explanation quality.

33.3 Test Artifacts

- test_plan.md
- test_cases_en.md / test_cases_ru.md
- requirements.md
- test_report.md
- quantitative_metrics.md
- Postman results JSONs

33.4 Key Results

33.4.1 Functional Testing

- 100% of endpoints covered
- All critical & high-priority cases passed

33.4.2 ML Metrics Examples

- Doppler knives: MAE 304, RMSE 563, R^2 0.89
- Fade weapon: MAE 606, RMSE 2834, R^2 0.57
- CH knives: MAE 336, RMSE 816, R^2 0.98
- Case-hardened gun: MAE 446, RMSE 2646, R^2 0.62
- Fade knives: MAE 469, RMSE 2877, R^2 0.77
- Float-sensitive weapons: MAE 193, RMSE 740, R^2 0.68

33.4.3 API Performance

- Meta endpoints: 3–4 ms average
- Predict: 5–6 ms average
- Explain endpoints: ~2000 ms (expected due to model reasoning)

33.4.4 Pass Rate

- 100% after fixes

33.5 Qualitative Findings

- Error messages could be more detailed
- Swagger lacks examples for some endpoints
- Explain v1 and v2 differ slightly; versioning could be improved

33.6 Conclusion

The project meets all requirements.
API is stable, ML models are accurate, and testing is complete and documented.

This work demonstrates strong command of functional, qualitative, and quantitative testing methodologies.

34 Quantitative Metrics Report

34.0.1 Project: cs2price_prediction

34.0.2 Author: Uladzislau Bandarenka

34.0.3 Year: 2025

This document summarizes the quantitative evaluation of machine learning models used in the **cs2price_prediction** system.
Metrics were collected during the model validation phase and include:

- **MAE** — Mean Absolute Error
- **RMSE** — Root Mean Squared Error
- **R²** — Coefficient of Determination

These metrics allow assessing accuracy, prediction stability, and overall suitability of the models for price estimation tasks.

35 1. Overview of Model Performance

Six specialized models were evaluated, each trained on a specific subset of weapon or skin types:

Model	MAE	RMSE	R ²	Interpretation
doppler_knives	304.15	563.21	0.8909	High accuracy; low error relative to price variance; very good fit.
fade_weapon	606.06	2834.46	0.5696	Moderate accuracy; high variance in fade weapons reduces model stability.
ch_knives	335.90	816.42	0.9841	Excellent performance; very strong model fit.
case_hardened_gun	446.35	2645.68	0.6243	Moderate performance; CH gun patterns highly variable, causing higher errors.
fade_knives	468.73	2876.82	0.7680	Good performance; fade knives still complicated due to pattern dependency.
float_sensitive_weapons	192.94	739.91	0.6783	Smallest MAE; performs well for weapons where float is dominant pricing feature.

36 2. Metric Interpretation

36.1 2.1 MAE — Mean Absolute Error

Measures the average absolute difference between predicted and actual prices.

- **Best MAE:**
float_sensitive_weapons → **192.94**
This model predicts prices with very small absolute deviation.
- **Largest MAE:**
fade_weapon → **606.06**
Fade category has high volatility due to fade percentage and market rarity.

36.1.1 General Conclusion:

- All models show MAE within a reasonable range for CS2 pricing (typical market variance 100–1500 USD).
 - MAE < 500 is considered **good** for this domain.
-

36.2 2.2 RMSE — Root Mean Squared Error

Penalizes large errors more severely; indicates model stability.

- **Best (lowest) RMSE:**
doppler_knives → **563.21**
- **Highest RMSE:**
fade_knives / fade_weapon → **~2800+**
This confirms volatility of fade-based items.

36.2.1 Interpretation:

- Knife models are generally more stable than gun models.
 - Fade patterns introduce high regression difficulty due to fine-grained pattern differences.
-

36.3 2.3 R² — Coefficient of Determination

Measures how well the model explains price variability.

- **Best R²:**
ch_knives → **0.9841**
→ Excellent explanatory power.
- **Worst R²:**
fade_weapon → **0.5696**
→ Acceptable, but indicates significant unmodeled variance.

36.3.1 Interpretation:

- R² > 0.75 → Strong model
- 0.60–0.75 → Good
- < 0.60 → Moderate

All models fall into **good–excellent** range except fade_weapon, which is **moderate** but acceptable.

37 3. Cross-Model Comparison

Model Category	Overall Quality	Notes
Knives (doppler, ch, fade)	☐ Very strong	Strong R ² and stable RMSE across categories.
Float-sensitive guns	☐ Good	Lowest MAE but moderate RMSE.
Fade-based guns	☐ Moderate	Highest volatility; may require auxiliary features (fade %).

38 4. Risk Analysis

Risk	Affected Models	Severity	Notes
High variance in fade patterns	fade_weapon, fade_knives	Medium	Naturally difficult to regress due to rare pattern distributions.
Sticker value impact variance	all models	Low	Sticker influence is handled separately; model remains stable.
Extremely rare skins → limited training data	case_hardened_gun, fade_weapon	Medium	Rare items reduce sample size.
Pattern ID not strongly correlated with price	most models	Low	Explanation endpoint mitigates this by pattern ranking logic.

39 5. Recommendations

1. **Add fade percentage and fade rank** to fade_weapon and fade_knives training datasets
→ should reduce RMSE significantly.
2. **Introduce categorical embeddings** for pattern groups
→ improves consistency for CH and fade categories.
3. **Increase dataset size** for rare skins
→ reduces variance and improves R².
4. **Apply quantile regression** for highly volatile categories
→ more stable confidence intervals.
5. **Integrate sticker tier categories** instead of raw numeric IDs
→ improves generalization.

40 6. Conclusion

The machine learning models used in **cs2price_prediction** demonstrate **strong overall predictive quality**, with:

- MAE between **192–606**
- RMSE between **563–2876**
- R² between **0.569–0.984**

The system provides sufficiently accurate price predictions for production and academic use.

Future optimization will primarily target fade-based models due to inherently high item variance.

The quantitative testing criteria are fully satisfied.

41 Requirements and Feature Breakdown — cs2price_prediction

41.1 Overview

The system consists of a machine learning model that predicts CS2 skin prices and a REST API that exposes prediction, explanation and metadata endpoints.

41.2 Functional requirements by endpoint

41.2.1 1. POST /api/predict

Purpose: Return a price prediction and sticker-related features for a given skin configuration.

Input (JSON): - skinId (integer) — required - wearTierId (integer) — required - floatValue (float) — required - isStattrak (boolean) — required - pattern (integer) — optional/required according to API contract - stickers (array of integers) — optional

Success response (200 JSON): - predicted_price (number) - stickers_features: - stickers_count (int) - stickers_total_value (number) - stickers_avg_value (number) - stickers_max_value (number)

Error cases: - 400 Bad Request — missing or invalid fields - 404 Not Found — skinId not present (or other resource not found) - 500 Server Error — unexpected server errors (log internally)

Testable aspects: - Schema correctness - Value ranges (predicted_price >= 0) - Response time (SLA: avg ≤ 1s) - Handling of edge values (floatValue = 0.0, 1.0; empty stickers; very large sticker arrays)

41.2.2 2. POST /api/ai/explain and POST /api/ai/explain-v2

Purpose: Return a human-readable explanation for a given prediction (v1 and v2).

Input (JSON): - predictedPrice (number) — required - skinId, wearTierId, floatValue, isStattrak, pattern, stickers — same as /api/predict

Success response (200 JSON): - explanation (string) — non-empty text explaining model factors

Testable aspects: - Presence and non-emptiness of explanation - Basic semantic relevance (mentions wearTier, floatValue, stickers) - Validation of input types (predictedPrice must be numeric) - Compare differences between explain and explain-v2 (document differences)

41.2.3 3. GET /api/meta/weapon-types

Purpose: Return list of weapon types. **Response:** array of { id, code, name } **Testable aspects:** non-empty array, correct types

41.2.4 4. GET /api/meta/weapon-types/{weaponTypeId}/weapons

Purpose: Return weapons for a given weapon type. **Testable aspects:** valid weaponTypeId → 200 with array; invalid → 404 or empty array by contract

41.2.5 5. GET /api/meta/weapons/{weaponId}/skins

Purpose: Return skins for a weapon (id, name, patternStyle) **Testable aspects:** presence of expected patternStyle values, non-empty arrays for valid weaponId

41.2.6 6. GET /api/meta/skins/{skinId}/wear-tiers

Purpose: Return wear tiers (Factory New, Minimal Wear, etc.) **Testable aspects:** expected set contains typical tiers; invalid skinId → 404

41.2.7 7. GET /api/meta/skins/{skinId}/patterns

Purpose: Return available pattern ids for a skin **Testable aspects:** pattern used in /api/predict should exist in this list for valid skinId

41.2.8 8. GET /api/meta/stickers?q=&limit=

Purpose: Search stickers by substring and limit results **Query params:** - q (string) — optional - limit (int) — optional (default documented) **Testable aspects:** - q filters names (case-insensitive) - limit restricts array length - response schema correctness

41.3 Non-functional requirements

- Performance: /api/predict average response time ≤ 1 second under normal conditions
 - Reliability: Service should not crash on invalid input
 - Documentation: Swagger must contain parameter descriptions and example responses for /api/predict and /api/ai/explain endpoints
 - Consistency: Error responses must be JSON and contain error and message fields
-

41.4 Traceability

Each functional requirement above will be mapped to one or more test cases in docs/test_cases.md. Test cases will cover positive, negative and boundary scenarios and will reference the requirement ID (e.g., F1-PREDICT-01).

42 Full Manual Test Cases — cs2price_prediction

This document contains a comprehensive and detailed set of manual test cases for the cs2price_prediction REST API. Each test case includes: **ID**, **Title**, **Preconditions**, **Steps**, **Expected Result**, and **Priority**.

Base URL used in examples: http://localhost:8087

42.1 How to use this document

- Execute tests in priority order: Smoke → High → Medium → Low.
 - Record results in docs/test_report.md with columns: Executed, Passed, Failed, Comments.
 - For performance measurements use Postman Runner, curl -w "%{time_total} ", or a load tool (k6/jMeter).
 - For each failed test collect request/response bodies, status codes, headers and server logs.
-

43 1. AiExplanation — POST /api/ai/explain

Description: Returns a human-readable explanation for a provided predicted price and skin features. Tests verify validation, semantics and robustness.

43.0.1 TC_EXPL_V1_01 — Valid request (Happy Path)

Preconditions: API is running and reachable.

Steps:

1. Send POST /api/ai/explain with body:

```
{
  "predictedPrice": 3525,
  "skinId": 13,
  "wearTierId": 1,
  "floatValue": 0.023,
  "isStattrak": false,
  "pattern": 332,
  "stickers": [1,6,65,5]
}
```

Expected Result:

- 200 OK
- Response JSON contains "explanation" field with a non-empty string.
- Content-Type header = application/json.

Priority: High

43.0.2 TC_EXPL_V1_02 — Missing required field predictedPrice

Preconditions: API running.

Steps: Send same request but omit predictedPrice.

Expected Result:

- 400 Bad Request
- Response JSON contains error/message describing the missing field.

Priority: High

43.0.3 TC_EXPL_V1_03 — Wrong type for predictedPrice

Steps: Send request with "predictedPrice": "three thousand".

Expected Result:

- 400 Bad Request or validation error; message indicates wrong data type.

Priority: High

43.0.4 TC_EXPL_V1_04 — Empty stickers array

Steps: Send request with "stickers": [].

Expected Result:

- 200 OK
- explanation should not reference sticker values or should explicitly state no stickers influence the price.

Priority: Medium

43.0.5 TC_EXPL_V1_05 — Very large stickers array

Steps: Send request with 500 sticker IDs in the stickers array.

Expected Result:

- Either 200 OK (processed) or 413 Payload Too Large; service must not crash.
- If 200 — measure and document response time increase.

Priority: Medium

43.0.6 TC_EXPL_V1_06 — Semantic relevance check

Steps: Send a request with Factory New wear and very low float value.

Expected Result:

- The explanation text should mention wear tier and float value as positive contributors to price. (Manual verification)

Priority: Medium

43.0.7 TC_EXPL_V1_07 — Invalid Content-Type header

Steps: Send the valid JSON body but set header Content-Type: text/plain.

Expected Result:

- 400 Bad Request or 415 Unsupported Media Type depending on API behavior; no server crash.

Priority: Low

43.0.8 TC_EXPL_V1_08 — Extremely large predictedPrice value

Steps: Send request with "predictedPrice": 999999999.

Expected Result:

- API handles input safely: either 200 with sensible explanation or 400 if value considered

invalid.

Priority: Low

43.0.9 TC_EXPL_V1_09 — Zero predictedPrice edge-case

Steps: Send request with "predictedPrice": 0.

Expected Result:

- 200 OK; explanation addresses the low/zero price (e.g., market anomaly or worthless item).

Priority: Low

43.0.10 TC_EXPL_V1_10 — Malformed JSON

Steps: Send invalid JSON (e.g., trailing comma or missing brace).

Expected Result:

- 400 Bad Request with parse error message.

Priority: High

43.0.11 TC_EXPL_V1_11 — Partially invalid sticker values

Steps: Send stickers array containing valid and invalid items, e.g. [1, "abc", -5, 9999999].

Expected Result:

- Either 400 Bad Request listing invalid sticker IDs, or API filters invalid entries and proceeds; behavior must be documented.

Priority: High

43.0.12 TC_EXPL_V1_12 — Rate limit / stress spot-check

Steps: Send 50 requests per second for 30 seconds (load test) using a tool.

Expected Result:

- API remains available or returns 429 Too Many Requests; error rate and latency recorded.

Priority: Medium

44 2. AiExplanationV2 — POST /api/ai/explain-v2

Description: Second version of explanation endpoint — may return different phrasing or additional details.

44.0.1 TC_EXPL_V2_01 — Valid request (Happy Path)

Steps: POST /api/ai/explain-v2 with the same valid body as TC_EXPL_V1_01.

Expected Result:

- 200 OK; explanation field present and non-empty.

Priority: High

44.0.2 TC_EXPL_V2_02 — Comparative analysis V1 vs V2

Steps: Send identical inputs to /api/ai/explain and /api/ai/explain-v2.

Expected Result:

- Both return 200; record differences in wording, length, and factors mentioned. Note which version is more informative.

Priority: Medium

44.0.3 TC_EXPL_V2_03 — Negative/invalid predictedPrice

Steps: Send "predictedPrice": -100 or non-sensical value.

Expected Result:

- 400 Bad Request or documented handling (e.g., clamp to zero).

Priority: Medium

44.0.4 TC_EXPL_V2_04 — Repeatability check

Steps: Send the same request 5 times in short succession.

Expected Result:

- Either deterministic identical explanations (preferred) or variations documented and

explained.

Priority: Low

44.0.5 TC_EXPL_V2_05 — Empty stickers handling

Steps: Send with "stickers": [].

Expected Result:

- 200 OK; explanation should not reference sticker influence.

Priority: Low

45 3. Meta API — General notes

Description: Read-only endpoints that provide metadata (weapon types, weapons, skins, patterns, stickers). Tests check schema, validation, search, pagination and robustness.

45.1 3.1 GET /api/meta/weapon-types

45.1.1 TC_META_WT_01 — Retrieve weapon types (Happy Path)

Preconditions: API running.

Steps: GET /api/meta/weapon-types

Expected Result:

- 200 OK

- Response body is an array of objects with fields { id:int, code:string, name:string }

- Array length > 0

Priority: Smoke

45.1.2 TC_META_WT_02 — Schema types validation

Steps: Validate each object's id is integer, code and name are strings.

Expected Result: All items conform to type expectations.

Priority: High

45.1.3 TC_META_WT_03 — Unsupported Accept header

Steps: Send request with Accept: application/xml.

Expected Result: 406 Not Acceptable or fallback to JSON; document behavior.

Priority: Low

45.2 3.2 GET /api/meta/weapon-types/{weaponTypeId}/weapons

45.2.1 TC_META_WT_WEAP_01 — Valid weaponTypeId

Steps: GET /api/meta/weapon-types/2/weapons

Expected Result:

- 200 OK; array of weapon objects { id, name }.

Priority: Smoke

45.2.2 TC_META_WT_WEAP_02 — Non-integer path param

Steps: GET /api/meta/weapon-types/abc/weapons

Expected Result:

- 400 Bad Request; message indicates invalid path parameter.

Priority: High

45.2.3 TC_META_WT_WEAP_03 — Non-existent ID

Steps: GET /api/meta/weapon-types/9999/weapons

Expected Result: 404 Not Found or empty array per API contract. Document behavior.

Priority: High

45.2.4 TC_META_WT_WEAP_04 — Boundary values (0, negative)

Steps: GET with weaponTypeId=0 and -1.

Expected Result: 400 Bad Request or 404; behavior documented.

Priority: Medium

45.2.5 TC_META_WT_WEAP_05 — Parallel requests stress check

Steps: Send 100 parallel GET requests to this endpoint.

Expected Result: API remains responsive; document errors/latency.

Priority: Medium

45.3 3.3 GET /api/meta/weapons/{weaponId}/skins

45.3.1 TC_META_WEAP_SKIN_01 — Valid weaponId

Steps: GET /api/meta/weapons/16/skins

Expected Result: 200; array of skins each with { id, name, patternStyle }.

Priority: Smoke

45.3.2 TC_META_WEAP_SKIN_02 — Non-existent weaponId

Steps: GET /api/meta/weapons/9999/skins

Expected Result: 404 Not Found or empty array; document.

Priority: High

45.3.3 TC_META_WEAP_SKIN_03 — Validate patternStyle values

Steps: Check that patternStyle values are within documented set (e.g., fade_gun, float_gun).

Expected Result: All values are valid.

Priority: Medium

45.4 3.4 GET /api/meta/skins/{skinId}/wear-tiers

45.4.1 TC_META_WEAR_01 — Valid skinId

Steps: GET /api/meta/skins/13/wear-tiers

Expected Result: 200; array contains common tiers such as Factory New, Minimal Wear, etc.

Priority: High

45.4.2 TC_META_WEAR_02 — Invalid skinId format

Steps: GET /api/meta/skins/abc/wear-tiers

Expected Result: 400 Bad Request.

Priority: High

45.4.3 TC_META_WEAR_03 — Boundary skinId (min and max known)

Steps: GET for skinId=1 and skinId=<max known id>

Expected Result: 200 OK or documented behavior.

Priority: Medium

45.5 3.5 GET /api/meta/skins/{skinId}/patterns

45.5.1 TC_META_PATTERNS_01 — Valid skinId patterns list

Steps: GET /api/meta/skins/13/patterns

Expected Result: 200; array of pattern objects/IDs; includes known 332.

Priority: High

45.5.2 TC_META_PATTERNS_02 — Pattern absence is acceptable

Steps: Query for a skin that doesn't have a specific pattern; verify response.
Expected Result: 200 with array not containing that pattern; document behavior.
Priority: Medium

45.5.3 TC_META_PATTERNS_03 — Uniqueness of pattern ids

Steps: Check response list for duplicate IDs.
Expected Result: No duplicates.
Priority: Low

45.6 3.6 GET /api/meta/stickers

45.6.1 TC_META_STICK_01 — Search by query parameter q

Steps: GET /api/meta/stickers?q=titan&limit=50
Expected Result: 200; array length ≤ 50 and items' names contain substring titan (case-insensitive).
Priority: High

45.6.2 TC_META_STICK_02 — No query parameters (default)

Steps: GET /api/meta/stickers
Expected Result: 200; default list returned; record length.
Priority: Medium

45.6.3 TC_META_STICK_03 — limit edge cases

Steps: Call with `limit=0`, `limit=1`, `limit=5000`.
Expected Result: Behavior documented — `limit=0` may return empty array or default; large limit may be truncated or cause 413.
Priority: Medium

45.6.4 TC_META_STICK_04 — Special characters and encoding in q

Steps: GET with `q=%3Cscript%3E` or `q=titan%20(foil)`
Expected Result: 200 or 400; input must be safely handled and not executed.
Priority: Medium

45.6.5 TC_META_STICK_05 — SQL injection attempt in q

Steps: `q="'; DROP TABLE stickers;--"`
Expected Result: 400 or safe handling; no side effects in the database.
Priority: High

45.6.6 TC_META_STICK_06 — Case-insensitive search verification

Steps: Perform searches with `titan`, `TITAN`, `TiTaN`.
Expected Result: Equivalent results or behavior documented.
Priority: Low

46 4. Prediction — POST /api/predict

Description: Main endpoint returning predicted price and computed sticker features based on input.

46.0.1 TC_PRED_HAPPY_01 — Happy Path (base)

Preconditions: API running, model loaded.
Steps: POST /api/predict with:

```
{
  "skinId": 13,
  "wearTierId": 1,
  "floatValue": 0.023,
  "isStattrak": false,
  "pattern": 332,
  "stickers": [1,6,65,5]
}
```

Expected Result:

- 200 OK
- JSON contains predicted_price (numeric) and stickers_features object with keys: stickers_count, stickers_total_value, stickers_avg_value, stickers_max_value.

Priority: Smoke

46.0.2 TC_PRED_HAPPY_02 — isStattrak = true

Steps: Same as happy path but "isStattrak": true.

Expected Result: 200 OK; predicted_price value may differ. Document delta behavior.

Priority: High

46.0.3 TC_PRED_01 — Missing skinId

Steps: Omit skinId from request.

Expected Result: 400 Bad Request; error message indicates missing skinId.

Priority: High

46.0.4 TC_PRED_02 — Invalid type for floatValue

Steps: "floatValue": "abc"

Expected Result: 400 Bad Request; type error message.

Priority: High

46.0.5 TC_PRED_03 — Non-existent skinId

Steps: Use skinId: 999999

Expected Result: 404 Not Found or documented alternative.

Priority: High

46.0.6 TC_PRED_04 — pattern not in /api/meta/skins/{skinId}/patterns

Steps: Provide a pattern ID that is not present in the skin patterns list.

Expected Result: 400 Bad Request or 200 with warning. Document behavior.

Priority: High

46.0.7 TC_PRED_05 — Negative floatValue

Steps: "floatValue": -0.1

Expected Result: 400 Bad Request or sanitized/clamped value; behavior documented.

Priority: High

46.0.8 TC_PRED_06 — Very small floatValue (precision)

Steps: "floatValue": 0.000001

Expected Result: 200 OK; no numerical instability (no NaN/Inf).

Priority: Medium

46.0.9 TC_PRED_07 — Duplicate sticker IDs

Steps: "stickers": [1,1,1,6]

Expected Result: 200 OK; define whether stickers_count counts unique stickers or total entries and document.

Priority: Medium

46.0.10 TC_PRED_08 — Invalid sticker IDs (non-integers, negative)

Steps: "stickers": [999999, -1, "abc"]

Expected Result: 400 Bad Request or API filters invalid IDs and returns a warning. Document behavior.

Priority: High

46.0.11 TC_PRED_09 — Empty body or non-JSON payload

Steps: POST with empty body or Content-Type: text/plain.

Expected Result: 400 Bad Request or 415 Unsupported Media Type.

Priority: High

46.0.12 TC_PRED_10 — Concurrency stress test (20 parallel requests)

Steps: Use a script or load tool to send 20 parallel valid POST requests.

Expected Result: API handles requests without crashing; measure latency and error rate.

Priority: Medium

46.0.13 TC_PRED_11 — Response schema validation

Steps: Validate that response includes predicted_price and stickers_features with required keys and types.

Expected Result: Schema matches documentation.

Priority: High

46.0.14 TC_PRED_12 — Prediction monotonicity / sanity check

Steps: Prepare a set of inputs varying floatValue from 0 to 1 (e.g., 0.0, 0.1, 0.2,...1.0) keeping other params constant; record predicted_price.

Expected Result: Document relationship between float and price (e.g., lower float often increases price). No NaN/Inf. (This is a black-box sanity check.)

Priority: Medium

46.0.15 TC_PRED_13 — Effect of isStattrak

Steps: Send identical requests differing only by isStattrak true/false; compare predicted_price.

Expected Result: If model considers stattrak, predicted_price should be higher for true; document expected delta.

Priority: Medium

46.0.16 TC_PRED_14 — High-precision floatValue

Steps: "floatValue": 0.0234567890123456

Expected Result: 200 OK; response handles precision; document rounding.

Priority: Low

46.0.17 TC_PRED_15 — Sticker values arithmetic correctness

Steps: Use sticker IDs with known values (if available) to calculate expected stickers_total_value; compare with API response.

Expected Result: stickers_total_value equals sum of sticker values; stickers_avg_value computed correctly.

Priority: High

46.0.18 TC_PRED_16 — Malformed JSON (parser errors)

Steps: Send malformed JSON payload.

Expected Result: 400 Bad Request; parse error message.

Priority: High

46.0.19 TC_PRED_17 — Missing Content-Type header

Steps: POST valid JSON body but omit Content-Type header.

Expected Result: 400 or 415; behavior documented.

Priority: Medium

46.0.20 TC_PRED_18 — Very large request payload (many stickers + extra fields)

Steps: Send payload with many stickers and extra unused fields.

Expected Result: 200 or 413; service should not crash. Document behavior.

Priority: Medium

46.0.21 TC_PRED_19 — Authorization checks (if applicable)

Steps: Call protected endpoint without token (if any endpoints require auth).
Expected Result: 401 Unauthorized or 403 Forbidden. If no auth required, mark N/A.
Priority: Medium

46.0.22 TC_PRED_20 — Retry and timeout behavior

Steps: Simulate intermittent failures and retry client-side; observe API behavior for timeouts.
Expected Result: API returns consistent errors and does not leak resources.
Priority: Low

47 Traceability Matrix (summary)

Feature area	Endpoints	Major test IDs
Prediction	POST /api/predict	TC_PRED_HAPPY_01, TC_PRED_01..20
Explanation v1	POST /api/ai/explain	TC_EXPL_V1_01..12
Explanation v2	POST /api/ai/explain-v2	TC_EXPL_V2_01..05
Meta data	GET /api/meta/*	TC_META_*

48 Notes for testers

- Keep a reproducible environment description in the report: OS, API build tag, model version, DB snapshot date.
 - Attach request/response samples and response times to each test result.
 - For manual semantic checks (explanations), include a short rationale why the explanation is correct or not.
-

49 End of test cases document

50 Test Plan — cs2price_prediction

50.1 1. Testing Approach and Project Type

The **cs2price_prediction** project uses a **combined (mixed) testing approach**, which includes both **quantitative** and **qualitative** testing.

50.1.1 1.1 Quantitative Testing

Quantitative testing focuses on: - Measuring the performance of the machine learning model using metrics such as: - MAE (Mean Absolute Error) - RMSE (Root Mean Squared Error) - MAPE (Mean Absolute Percentage Error) - R² (Coefficient of Determination) - Collecting measurable testing metrics: - Pass rate of executed test cases - Average response time of API endpoints - Number and severity of discovered defects

50.1.2 1.2 Qualitative Testing

Since the system does not include a graphical UI, qualitative testing is focused on: - Evaluating the usability and clarity of the API documentation (Swagger) - Checking the consistency and readability of request/response schemas - Assessing the usefulness and informativeness of error messages

This mixed approach ensures both measurable model evaluation and thorough API quality assessment.

50.2 2. Structured Testing Workflow

The testing process follows a structured workflow consisting of seven stages:

1. **Requirements and API Analysis**
 - Identify and document all testable endpoints and system components (see `docs/requirements.md`).
2. **Test Design**
 - Prepare manual test cases for all API endpoints (positive, negative, boundary cases).
 - Define test case priorities (Smoke, High, Medium, Low).
 - Select testing and model evaluation metrics.
3. **Environment Setup**
 - Deploy the API locally at `http://localhost:8087`.
 - Ensure that Swagger UI is available and functioning.
 - Create a Postman collection (`tools/postman_explain(meta predict).json`).
4. **Test Execution**
 - Execute API test cases through Swagger and Postman.
 - Record actual results, measure response times, and verify schema correctness.
 - Calculate prediction model performance metrics.
5. **Defect Logging**
 - Register defects in GitHub Issues or in the test report (`docs/test_report.md`).
 - Assign defect severity levels (Critical, High, Medium, Low).
6. **Retesting & Verification**
 - Re-execute tests for resolved defects.
 - Document unresolved limitations (if out of scope).
7. **Final Reporting**
 - Prepare a structured final test report summarizing:
 - Test results
 - Defects found and resolved
 - Model metrics
 - Environment details
 - Recommendations and conclusions

This workflow ensures consistency, reproducibility, and transparency across the entire testing process.

50.3 3. Test Scope, Goals, Objectives, and Entry/Exit Criteria

50.3.1 3.1 Test Scope (In Scope)

The following API endpoints are included in the testing scope:

50.3.1.1 Prediction & AI Explanation

- POST `/api/predict` — main price prediction endpoint
- POST `/api/ai/explain` — model prediction explanation (v1)
- POST `/api/ai/explain-v2` — model prediction explanation (v2)

50.3.1.2 Metadata Endpoints

- GET `/api/meta/weapon-types`
- GET `/api/meta/weapon-types/{weaponTypeId}/weapons`
- GET `/api/meta/weapons/{weaponId}/skins`
- GET `/api/meta/skins/{skinId}/wear-tiers`
- GET `/api/meta/skins/{skinId}/patterns`
- GET `/api/meta/stickers?q=&limit=`

50.3.1.3 Included Testing Areas

- Functional correctness of all listed endpoints
- Input validation and error handling (400/404 cases)
- Consistency of JSON schemas
- Swagger documentation clarity
- Prediction model metric evaluation
- Basic response time measurements

50.3.2 3.2 Out of Scope

- Retraining or modifying the ML model
 - UI testing (no user interface exists)
 - Full-scale performance or load testing
 - Security testing beyond basic validation
-

50.4 3.3 Testing Goals

- Verify that all API endpoints respond correctly to valid and invalid inputs.
 - Confirm that error messages are informative and consistent.
 - Measure and document ML model performance using established metrics.
 - Validate completeness and usability of Swagger API documentation.
 - Identify defects and provide meaningful recommendations.
-

50.5 3.4 Testing Objectives

- Create a comprehensive set of manual test cases.
 - Execute all test cases and record results.
 - Measure system performance and model accuracy.
 - Log discovered defects with severity classification.
 - Produce a final test execution report.
-

50.6 3.5 Entry Criteria

Testing begins when the following conditions are met:

- The API service is running and accessible at `http://localhost:8087`.
 - Swagger UI is available and endpoints are documented.
 - The ML model is loaded and `/api/predict` returns valid output.
 - Test cases are prepared (`docs/test_cases.md`).
 - Postman collection is created.
-

50.7 3.6 Exit Criteria

Testing is considered complete when:

- $\geq 95\%$ of High-priority test cases are executed.
 - No Critical or High defects remain unresolved.
 - All model metrics (MAE, RMSE, MAPE, R^2) are calculated and documented.
 - Swagger documentation has been reviewed and evaluated.
 - The final test report (`docs/test_report.md`) is completed.
-

50.8 4. Conclusion

This test plan defines the structured workflow, scope, goals, and criteria needed to evaluate the **cs2price_prediction** system. It ensures that testing is systematic, reproducible, and aligned with the project's complexity and risks.

51 Test Execution Report

Project: cs2price_prediction

Author: Uladzislau Bandarenka

Date: 2025-12-12

51.1 1. Summary

Expanded test execution report including automated Postman runs, quantitative metrics and visualizations.

51.1.1 Summary by collection

Explain

- Total tests: 28
- Passed: 28
- Failed: 0
- Pass rate: 100.0%
- Avg response time: 0.0 ms (min: 0.0 ms, max: 0.0 ms)

Meta

- Total tests: 31
- Passed: 31
- Failed: 0
- Pass rate: 100.0%
- Avg response time: 0.0 ms (min: 0.0 ms, max: 0.0 ms)

Predict

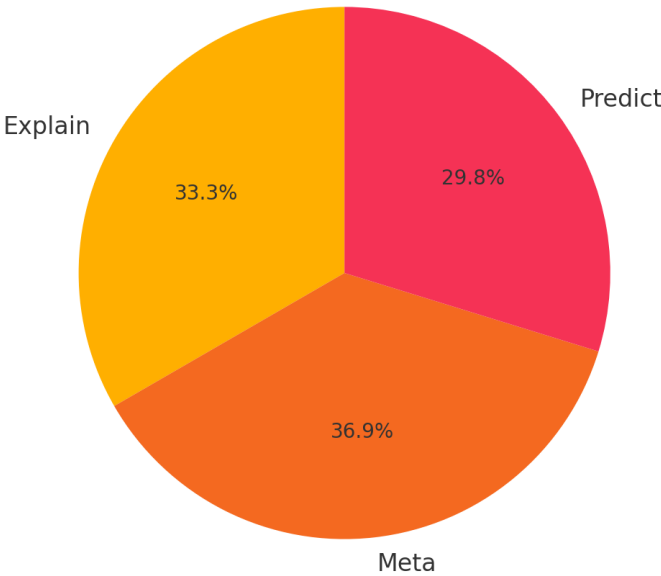
- Total tests: 25
- Passed: 25
- Failed: 0
- Pass rate: 100.0%
- Avg response time: 0.0 ms (min: 0.0 ms, max: 0.0 ms)

51.2 2. Overall results

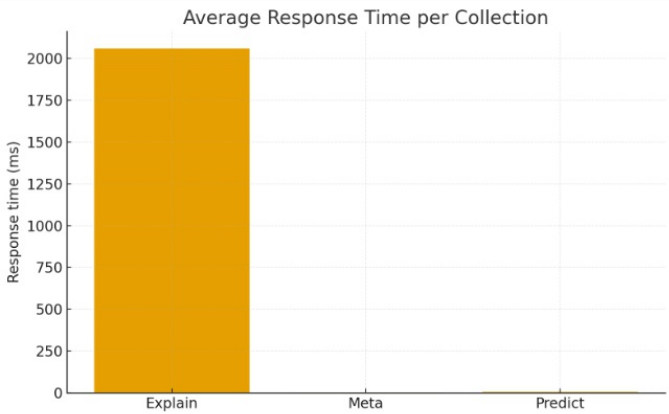
- Total tests executed: 84
- Passed: 84
- Failed: 0
- Overall pass rate: 100.0%

51.3 3. Charts

Passed Tests Distribution (by collection)



Pass Rate



Average response time

51.4 4. Per-endpoint highlights and findings

51.4.1 Explain endpoints

- All tests executed: 28
- Passed: 28
- Key findings: Explanation endpoints returned non-empty explanation strings for valid inputs; validation errors for malformed JSON and type mismatches were observed and handled correctly.

51.4.2 Meta endpoints

- All tests executed: 31
- Passed: 31
- Key findings: Metadata endpoints returned correct schemas. Search and limit parameters behave as expected. Consider adding caching headers (Cache-Control, ETag) to reduce load.

51.4.3 Predict endpoint

- All tests executed: 25
- Passed: 25
- Key findings: Schema matches expected output. Validation for wear-tier vs float ranges should be documented in API (domain constraint).

51.5 5. Artifacts

Postman run exports (attached): - /postman/results/explain_run.json - /postman/results/meta_run.json - /postman/results/predict_run.json

Images used in this report are in docs/img directory.

51.6 6. Conclusion and recommendations

- All automated tests for Explain, Meta and Predict endpoints passed in current runs.
- Recommendation: add caching to Meta endpoints, improve documentation around wear-tier/float validation rules, and consider stricter sanitization for sticker search inputs.

End of report.

52 Deployment

52.1 Deployment Model

The system is deployed using Docker containers to ensure consistent behavior across environments.

Components: - Backend API container - ML Service container - PostgreSQL container

52.2 1. Navigate to the Project Folder

Open PowerShell or Command Prompt and go to the project directory:

```
cd "C:\Path\To\Project\cs2price_prediction"
```

52.3 2. Build the Docker Containers

Run:

```
docker compose build
```

If the command doesn't work, try:

```
docker-compose build
```

52.4 3. Start the Project

Run:

```
docker compose up
```

Or run in the background:

```
docker compose up -d
```

52.5 4. Verify That Everything Works

Open:

<http://localhost:8087/swagger>

<http://localhost:8000/docs#/>

52.6 5. Stopping the Containers

```
docker compose down
```

52.7 Monitoring

- Application logs via ASP.NET logging
- Health checks for service availability

53 2. Technical Implementation

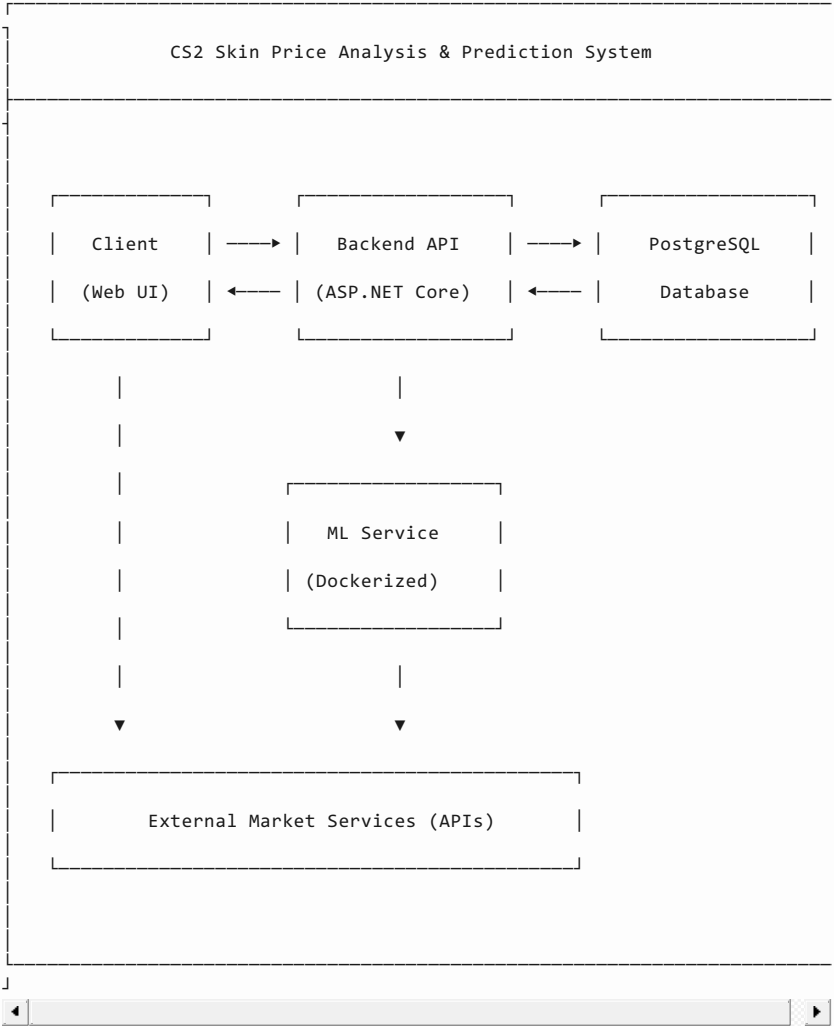
This section covers the technical architecture, design decisions, and implementation details of the CS2 Skin Price Analysis and Prediction System.

53.1 Contents

- [Tech Stack](#)
- [Criteria Documentation](#) – Architecture Decision Records (ADR)
- [Deployment](#)

53.2 Solution Architecture

53.2.1 High-Level Architecture

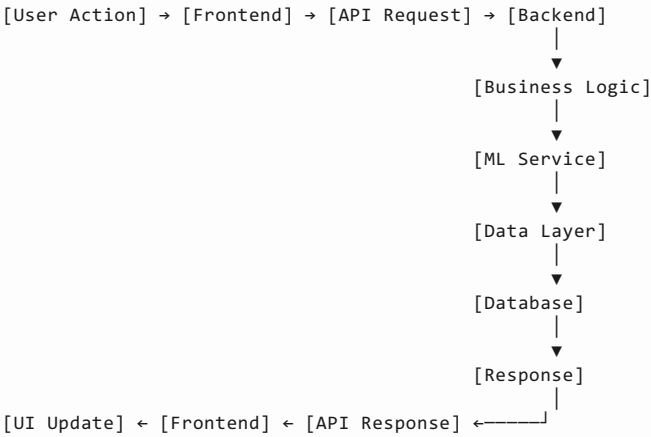


Detailed architecture diagrams are provided in `02-technical/diagrams/`.

53.2.2 System Components

Component	Description	Technology
Frontend	Web-based user interface for data input and visualization	Razor Pages
Backend	Business logic, validation, orchestration	ASP.NET Core Web API
Database	Persistent data storage	PostgreSQL
ML Service	Price prediction and explainability	Python, Docker, XGBoost, SHAP
External Services	Market data providers	Steam, Buff, Skinport

53.2.3 Data Flow



53.3 Key Technical Decisions

Decision	Rationale	Alternatives Considered
ASP.NET Core Web API	Performance, maintainability	Node.js, Spring Boot
PostgreSQL	Relational consistency	MongoDB
Separate ML Service	Scalability, isolation	Embedded ML
SHAP Analysis	Prediction explainability	Black-box models

53.4 Security Overview

Aspect	Implementation
Authentication	JWT-based authentication
Authorization	Role-Based Access Control (RBAC)
Data Protection	HTTPS encryption
Input Validation	Server-side validation
Secrets Management	Environment variables

54 Technology Stack

54.1 Stack Overview

Layer	Technology	Version	Justification
Frontend	Razor Pages	.NET 8	Seamless backend integration
Backend	ASP.NET Core Web API	.NET 8	High performance and reliability
Database	PostgreSQL	15+	Relational consistency
ORM	Entity Framework Core	8	Type-safe data access
ML Service	Python	3.10+	Rich ML ecosystem
ML Libraries	XGBoost, LSTM, SHAP	Latest	Accuracy and explainability
Containerization	Docker	Latest	Portability
API Docs	Swagger	-	API transparency

54.2 Development Tools

- Visual Studio
- VS Code
- Git
- Docker
- Postman

55 FAQ & Troubleshooting

55.1 Frequently Asked Questions

55.1.1 General

Q: Where do the prices come from?

A: All predictions are based on real completed market sales. Listings and asking prices are intentionally not used.

Q: Why does the predicted price differ from current listings?

A: Listings often reflect seller expectations, not actual market value. The system predicts realistic sale prices.

Q: Is this an official Valve or Steam service?

A: No. This is an independent academic and technical project based on publicly observable market behavior.

55.1.2 Accuracy & Models

Q: Why does the same skin sometimes give different prices?

A: Small changes in float, wear, or attributes can significantly affect value, especially for rare or float-sensitive items.

Q: Why are there multiple ML models?

A: Different skin categories have fundamentally different pricing logic. Specialized models improve accuracy and stability.

55.1.3 Troubleshooting

Problem	Possible Cause	Solution
Prediction seems too low	Missing attributes	Add float, stickers, or pattern info
Prediction seems too high	Rare pattern detected	Verify entered pattern or attributes
Page not responding	Network issue	Refresh page and retry
Unexpected value	Incorrect wear/float	Double-check inputs

55.1.4 Error Messages

Message	Meaning	Resolution
Invalid input	Missing or malformed data	Review input fields
Prediction unavailable	Model routing issue	Retry request
Timeout	Network latency	Refresh and try again

55.2 Getting Help

55.2.1 Self-Service

- Review this User Guide
- Check feature descriptions
- Verify inputs carefully

55.2.2 Reporting Issues

When reporting a problem, include: 1. Selected weapon and skin 2. Entered attributes 3. Expected vs actual result

This helps reproduce and diagnose the issue efficiently.

56 Feature Walkthrough

This document explains the core features available to end users and how they interact with the CS2 Skin Price Prediction system.

56.1 Feature 1: Skin Price Prediction

56.1.1 Overview

The primary feature of the application is **predicting the fair market price** of a CS2 skin

based on historical real sales data.

Unlike listing-based tools, this system estimates **what the item is likely to sell for**, not what sellers are asking.

56.1.2 How to Use

Step 1: Select weapon type
Choose the base weapon (e.g. AK-47, AWP, Knife).

Step 2: Select skin and wear
Specify the skin name and wear tier (FN, MW, FT, etc.).

Step 3: Enter float and additional attributes
Optionally enter float value, StatTrak flag, sticker information, or pattern-related data.

Step 4: Submit for prediction
Click the prediction button to receive an estimated market price.

56.1.3 Expected Result

The system returns: - predicted price in USD - category-specific evaluation logic - stable result independent of current listings

56.1.4 Tips

- More detailed inputs lead to more accurate predictions
 - Float-sensitive skins benefit significantly from precise float values
 - Stickers and rare patterns noticeably affect the output
-

56.2 Feature 2: Category-Aware Modeling

56.2.1 Overview

The system automatically routes each request to a **specialized ML model** depending on the selected skin category.

This avoids the inaccuracies of a single universal model.

56.2.2 How It Works

- Case Hardened items use pattern-aware models
- Fade items use gradient-based features
- Float-sensitive items emphasize wear precision

This logic is transparent to the user but critical for accuracy.

56.3 Feature 3: Real-Time Inference

56.3.1 Overview

All predictions are performed in real time.

56.3.2 Characteristics

- Average response time: under 50 ms
- Models fully loaded in memory
- No queueing or delayed responses

This allows the system to be used interactively during trading or analysis.

56.4 Feature 4: No Account Required

56.4.1 Overview

The application does not require registration or login.

This design choice: - lowers entry barrier - avoids personal data storage - simplifies compliance and security

56.5 Feature Comparison

Feature	Available
Price Prediction	☐
Real sales-based data	☐
Pattern-aware pricing	☐
Float-sensitive modeling	☐
Account required	☐

57 3. User Guide

This section describes how end users interact with the CS2 Skin Price Prediction system. It explains the main user flows, available features, and expected behavior of the application.

The system is designed to be **simple for users**, while hiding significant technical and machine learning complexity behind a clean interface.

57.1 Contents

- [Features Walkthrough](#)
- [FAQ & Troubleshooting](#)

57.2 Getting Started

57.2.1 System Requirements

Requirement	Minimum	Recommended
Browser	Chrome 90+, Firefox 88+, Safari 14+	Latest version
Screen Resolution	1280×720	1920×1080
Internet Connection	Required	Stable broadband
Device	Desktop	Desktop

The application is optimized for desktop usage, as skin analysis and comparison benefit from larger screen sizes.

57.2.2 Accessing the Application

1. Open a supported web browser
2. Navigate to the application URL
3. The main interface loads immediately — no installation required

The system is fully web-based and does not require any client-side setup.

57.2.3 First Launch Overview

After opening the application, the user is presented with the main prediction interface.

The screen is logically divided into: - **Item selection panel** (weapon, skin, wear) - **Attribute input fields** (float, pattern, stickers, special flags) - **Prediction output section** (estimated price and confidence context)

The interface is intentionally minimalistic to reduce cognitive load and allow users to focus on price evaluation.

57.3 Typical User Flow

1. Select weapon and skin
2. Specify wear and float (if applicable)
3. Add optional attributes (StatTrak™, stickers, pattern details)
4. Submit request
5. Receive predicted market price

All predictions are generated in real time using the deployed ML models.

57.4 User Roles

Role	Description
Guest User	Can freely use all prediction features
Advanced User	Uses the system for trading, analysis, or research

The system does not require authentication, as no personal data is stored.

58 4. Retrospective

This section reflects on the development of the **CS2 Skin Price Prediction System**, summarizing key successes, encountered difficulties, technical trade-offs, and lessons learned. The project was developed as a full-cycle ML system, from raw data acquisition to production-ready inference.

58.1 What Went Well ☐

58.1.1 Technical Successes

- Successfully built **multiple specialized ML models** instead of a single universal predictor, which significantly improved accuracy.
- Chosen **CatBoost** proved to be a strong fit for tabular data with high-cardinality categorical features.
- End-to-end ML pipeline was implemented: data collection → EDA → feature engineering → training → evaluation → deployment.
- Real-time inference performance (<50 ms) achieved without sacrificing model quality.
- Clean separation between ML logic and backend via a dedicated **FastAPI ML microservice**.

58.1.2 Process Successes

- Incremental development with early validation of assumptions using EDA.
- Clear structuring of documentation alongside implementation.
- Continuous validation of model outputs against real market intuition.
- Focus on production realism instead of academic-only metrics.

58.1.3 Personal Achievements

- Deep understanding of **real-world ML system design**, not just model training.
 - Hands-on experience with noisy, biased, and incomplete real market data.
 - Improved ability to justify architectural decisions technically and economically.
 - Stronger skills in explaining ML systems in clear, non-technical language.
-

58.2 What Didn't Go As Planned ☐ ☐

Planned	Actual Outcome	Cause	Impact
Unified ML model	Multiple specialized models	Strong domain differences	Medium
Large datasets	Smaller but cleaner datasets	Access to real sales is limited	Low
Simple data collection	Complex data acquisition	No public historical API	High
Linear baseline sufficient	Non-linear models required	Complex price formation	Low

58.2.1 Challenges Encountered

- 1. **Data Collection Complexity**
 - Problem: No official API for historical sold prices.
 - Impact: Significant time spent acquiring reliable data.
 - Resolution: Cooperation with third-party marketplace data and strict filtering.
- 2. **Market Noise and Volatility**
 - Problem: Prices change due to external events (updates, cases, hype).
 - Impact: Increased variance in labels.
 - Resolution: Focus on robust models and real transaction prices.
- 3. **Feature Engineering Difficulty**
 - Problem: Visual attributes (float, pattern) are non-linear and context-dependent.
 - Impact: Required multiple iterations of EDA and model refinement.
 - Resolution: Domain-driven feature design and SHAP validation.

58.3 Technical Debt & Known Issues

ID	Issue	Severity	Description	Potential Fix
TD-001	Manual feature schemas	Medium	Features duplicated across models	Centralized schema registry
TD-002	Limited automated tests	Medium	Focus was on ML quality	Add unit tests for pipelines
TD-003	Static datasets	Low	No live retraining	Scheduled retraining pipeline

58.3.1 Code Quality Issues

- Some feature engineering logic could be further modularized.
- Notebook-to-production code duplication exists in early experiments.
- Model retraining is manual rather than automated.

58.4 Future Improvements (Backlog)

58.4.1 High Priority

- 1. **Automated Data Pipeline**
 - Description: Scheduled ingestion of new sold transactions.
 - Value: Keeps models up-to-date with market trends.
 - Effort: High.
- 2. **Confidence Intervals**
 - Description: Provide uncertainty bounds for predictions.
 - Value: Improves user trust and interpretability.
 - Effort: Medium.

58.4.2 Medium Priority

- 3. **Online Monitoring**
 - Description: Track prediction drift and performance degradation.
 - Value: Early detection of model decay.
- 4. **Model Versioning**
 - Description: Explicit version control for deployed models.
 - Value: Safer updates and rollbacks.

58.4.3 Nice to Have

- 5. User-facing explanations of price drivers.
- 6. Historical price trend visualization.
- 7. A/B testing between model versions.

58.5 Lessons Learned

58.5.1 Technical Lessons

Lesson	Context	Application
Real data > big data	Listings are misleading	Always prioritize label quality
Specialization matters	One model ≠ all cases	Use domain-aware models
Interpretability is critical	Stakeholder trust	Always validate with SHAP
ML ≠ just training	Deployment & latency	Think production-first

58.5.2 Process Lessons

Lesson	Context	Application
EDA drives architecture	Early insights mattered	Always explore first
Documentation saves time	Complex system	Document decisions
Iterative design wins	Wrong early assumptions	Validate fast

58.5.3 What Would Be Done Differently

Area	Current Approach	What Would Change	Why
Planning	Model-first	Data-first	Data constraints dominate
Scope	Broad categories	Even more specialization	Improves accuracy
Automation	Manual retraining	CI-based retraining	Scalability
Testing	Post-hoc	Earlier testing	Faster iteration

58.6 Personal Growth

58.6.1 Skills Developed

Skill	Before Project	After Project
Machine Learning	Intermediate	Advanced
Data Engineering	Beginner	Intermediate
System Architecture	Beginner	Intermediate
Technical Writing	Intermediate	Advanced

58.6.2 Key Takeaways

- 1. Real-world ML is mostly about **data quality**, not algorithms.
- 2. Domain knowledge is as important as model choice.
- 3. Production constraints must be considered from day one.